# Risk Management and Financial Institutions

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# Preface

## Achieve Mastery Over the Art of Financial Risk Management

Ever wondered how financial institutions remain stable even in the stormiest of economic climates? The answer lies in effective risk management. Our course, "Risk Management and Financial Institutions", is a detailed, practical, and comprehensive exploration of this critical field.

Through this course, you will acquire a deep understanding of the sources of risk, and how financial institutions identify, measure, and manage these risks. It provides a clear and complete overview of risk management's significant aspects, spanning interest rate risk, market risk, credit risk, operational risk, and climate risk, among others.

## Here's what we'll explore in our journey:

1. **Introduction to Risk Management:** We'll delve into various sources of risk, setting a strong foundation for the rest of the course.
2. **Revision of the Basic Concepts:** Understand the necessary background from financial derivatibes, banking, and financial markets.
3. **Forwards and Futures Strategies:** We'll dissect forward and futures contracts and their use in protecting against potential losses.
4. **Options Strategies:** Understand the mechanism behind options and how they are used as effective hedging tools in risk management.
5. **Volatility, Correlations, and Copulas:** We will explore how volatility, correlations, and copulas are used in risk managment.
6. **Market Risk I:** Delve deep into the complexities of market risk, its components, measurement techniques, and management strategies.
7. **Reading Week:** Take some rest and revise the first six topics.
8. **Market Risk II:** Delve even deeper into the complexities of market risk, its components, measurement techniques, and management strategies.
9. **Credit Risk:** Learn to evaluate and manage credit risk, an essential aspect of risk management.
10. **Operational Risk, Liquidity Risk, and Model Risk:** Understand the different operational risks financial institutions face, liquidity risk management, and the potential pitfalls of financial modeling.

11. **Climate Risk and Enterprise Risk Management:** Explore the emerging field of climate risk management and understand the holistic approach of enterprise risk management.
12. **Regulation, Economic Capital, and RAROC:** Get an in-depth understanding of the regulatory environment, how economic capital is calculated, and the importance of Risk-Adjusted Return on Capital.
13. **Financial Innovation and Risk Management Mistakes:** Learn about the role of innovation in risk management and explore common risk management mistakes.

Finally, we will review the entire course and ensure you're prepared to apply your new skills and knowledge in the real world.

## Why Enroll in this Course?

- Learn from the best: Our curriculum is designed based on Hull (2023) and Pirie and Kritzman (2017), offering you top-notch content.
- Practical focus: We emphasize real-world applications of theoretical concepts, arming you with practical skills to handle real-life scenarios.
- Dynamic and engaging: Our course is far from dry textbook material. We make learning enjoyable and engaging through interactive exercises and relatable examples.
- End-to-end understanding: From the basics of risk management to advanced strategies, our course covers everything you need to master risk management in financial institutions.
- Empowering knowledge: Upon completion, you'll be equipped to make sound financial decisions, manage various risks effectively, and drive success in your or any financial institution.

**Take the step and transform your understanding of risk management!**

## References

The whole course is based on the following textbooks:


- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.


The course was created using a support of the ChatGPT.

# 1 Introduction to Risk Management

**References:**

- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 6 - Risk Management (Subchapters 1-4)
- CFA Program Curriculum, Level I, Volume 6 - Portfolio Management and Ethical and Professional Standards
  - Reading 53 - Introduction to Risk Management

**Learning Outcomes:**

- Define risk and risk management.
- Describe features of a risk management framework.
- Identify financial and non-financial sources of risk and describe how they may interact.
- Describe methods for measuring and modifying risk exposures and factors to consider in choosing among the methods.

## 1.1 Introduction

In financial institutions, risk is an inherent aspect of all investment decisions, making effective risk management vital for long-term success. The process of risk management involves identifying, measuring, and controlling risks to ensure alignment with the institution's financial objectives and regulatory requirements. Portfolio managers must be proficient in handling risks at both the portfolio level—focusing on specific assets and their correlations—and the enterprise level, where systemic and institutional risks are addressed. Continuous assessment is essential to classify risks into those that offer potential rewards, those that require mitigation, and those that should be avoided to preserve institutional stability and performance.

## 1.2 The Risk Management Process

Definition of risk

***Risk*** → exposure to uncertainty.
Risk is the chance of a loss or adverse outcome as a result of an action, inaction, or external event.

- Managing risk doesn't mean avoiding it but controlling it effectively.
- **Risk and return**: These are interrelated forces in finance; only risk can be controlled, not return.
- **Risk exposure**: The state of being vulnerable to potential risks.

Example

Consider an announcement in Japan that could lead to a 1% appreciation or depreciation of the yen. The potential risk reflects a range of outcomes, typically described by a probability distribution of future returns. For an underlying amount of ¥1,000,000, the risk exposure is ¥10,000 (1% of ¥1,000,000). The risk management process helps decide whether to invest in this risky currency.

- **Risk driver**: The underlying uncertainty (e.g., +1% or -1% yen movement).
- **Risk position**: The action or quantity at risk (e.g., ¥1,000,000 worth of yen).
- **Risk exposure**: The potential valuation change (e.g., ±¥10,000).

Definition of risk management

Risk management is the process of defining, measuring, and adjusting the level of risk to optimize a portfolio's or organization's value. It involves balancing risk to achieve goals with an acceptable chance of failure.
*Alternative definition*: Risk management involves decisions and actions that maximize *personal or organizational objectives within tolerable risk levels.

- Effective risk management quantifies, monitors, and adjusts risks to balance goals with failure chances.
- It provides a framework for decision-making before, during, and after risk events, rather than simply preventing losses.
- **Dynamic nature**: Risks evolve, making risk management a continuous process.

## 1.3 The Risk Management Framework

A risk management framework provides the infrastructure, processes, and analytics necessary for effective risk management. It integrates risk and return into decision-making to achieve the organization's goals within its risk tolerance. While organizations face numerous risks, they are typically driven by a small number of key risk factors.

### 1.3.1 Key Components of the Risk Management Framework

1. **Risk Governance**
   This top-down process guides risk management activities. Enterprise risk management (ERM) ensures alignment between risk practices and the organization's overall objectives, health, and value.

2. **Risk Identification and Measurement**
   Identifying relevant risk drivers and tracking exposures within the business or portfolio. This includes calculating risk metrics under various scenarios to size potential risks.

3. **Risk Infrastructure**
   Consists of the personnel and systems required for risk tracking and quantitative analysis, such as databases and analytic models.

4. **Defined Policies and Processes**
   Operationalizes governance by setting daily guidelines, limits, and processes that align with the organization's risk tolerance and regulatory requirements.

5. **Risk Monitoring, Mitigation, and Management**
   This involves ongoing reviews and adjustments to align risk exposure with the organization's tolerance levels. It is a dynamic process that integrates all aspects of the framework.

6. **Communication**
   Ensures clarity in risk governance, metrics, and exposure changes throughout the organization. Effective communication enables timely and appropriate responses to risks.

7. **Strategic Integration**
   Proactive use of risk management to enhance performance, optimize decision-making, and increase overall organizational value.

## 1.4 Risk Tolerance and Risk Budgeting

**Risk tolerance** defines the level of risk an organization is willing to accept, *including potential losses, opportunity costs, and the risk of failing to meet* objectives. It helps identify critical shortfalls in goal achievement and exposure to *uncertain factors. Risk tolerance must be formally established and communicated in* advance, guiding strategic risk selection.

**Risk budgeting** is the implementation of risk tolerance decisions approved at the *strategic or governance level. It quantifies risk using specific metrics and allocates* it across the organization. Common single-dimensional measures include standard *deviation, beta, value at risk (VaR), and scenario analysis. Multi-dimensional* approaches categorize risks by underlying classes (e.g., equity, fixed income, *commodities) and allocate investments accordingly through factor analysis.

Risk budgeting fosters intentional risk trade-offs and encourages a decision-making culture where risk is integral. Its goal is to invest where the return per unit of risk is highest, influencing decisions such as hedging or choosing lower-risk investments.

## 1.5 Identification of Risk

### 1.5.1 Financial Risks

Definition

**Financial risks** stem from financial markets, such as fluctuations in prices or interest rates.

- **Market risk**: Risk due to movements in interest rates, stock prices, exchange rates, and commodity prices.
- **Credit risk** (also known as default or counterparty risk): The risk of loss if one party fails to meet its obligations on bonds, loans, or derivatives.
- **Liquidity risk**: The risk of a significant downward price adjustment when selling a financial asset, often influenced by market transaction costs (bid-ask spread), asset liquidity, and position size.

### 1.5.2 Non-Financial Risks

> Definition
>
> **Non-financial risks** originate from sources outside financial markets, including internal organizational actions or external factors like relationships with regulators, governments, suppliers, or customers.

- **Settlement risk**: Risk of payment failure before default (e.g., sending funds to a counterparty that defaults).
- **Legal risk**: Risk of litigation or the failure of contracts to be upheld.
- **Compliance risk**: Encompasses regulatory, accounting, and tax risks.
- **Model risk**: Risk of errors from using incorrect models or misapplying correct ones.
- **Tail risk**: Risk of extreme events that occur more frequently than anticipated.
- **Operational risk**: Risk arising from internal failures (e.g., inadequate personnel, systems, or policies) or external events (e.g., fraud, natural disasters, cyber-attacks).
- **Solvency risk**: Risk of running out of cash, despite being otherwise solvent (distinct from liquidity risk in financial markets).

## 1.6 Interactions Between Risks

Risks often overlap across categories. For instance, market risk can lead to credit risk, which may, in turn, cause operational risk. Similarly, credit risk is linked to settlement risk, and legal risks can arise from market or credit risk when parties seek contract loopholes.

The interaction of risks can compound and amplify costs and reduce expected returns, particularly under stressed conditions, as seen during the 2008 financial crisis. **Combined risks are usually worse than the sum of individual risks**, following a **non-linear pattern**. Scenario planning helps identify these interactions early. Risk models often fail to capture these interactions, worsening the consequences. Governance bodies, management, and analysts should avoid treating risks as isolated and unrelated.

## 1.7 Measuring Risk

All risks stem from the uncertainty of the future. Financial markets reflect a blend of global and domestic economic conditions, industry dynamics, and company-specific factors. Government policies, regulations, and broader economic factors influence this environment, impacting company activities.

- **Systematic risk** affects entire economies or industries.

- **Idiosyncratic (unsystematic) risk** is specific to individual companies but can sometimes evolve into systematic risk, as with poor credit risk management triggering a global crisis.
- Diversified portfolios aim to eliminate unsystematic risks, though these risks still matter to company managers, poorly diversified investors, and financial analysts.

### 1.7.1 Metrics for Measuring Risk

Risk metrics are used to quantify exposure. Common measures include:

- **Probability**: Frequency of outcomes.
- **Standard deviation**: Measures the range of outcomes but has limitations for non-normal distributions.
- **Beta**: Sensitivity to market returns.
- **Greeks**: Measures specific to derivatives (e.g., delta, gamma).
- **Duration**: Applies to fixed-income assets.
- **Value at Risk (VaR)**: Estimates extreme loss potential but is subject to model limitations. Variants include Conditional VaR, extreme value theory, scenario analysis, and stress testing.

For credit risk, assessments include factors like **credit ratings**, liquidity, solvency, profitability, and leverage, along with macroeconomic indicators. Credit risk is difficult to measure due to the rarity of credit events, but default probabilities can be estimated using similar company characteristics or analyzing derivative instruments like credit default swaps. Operational risk, such as from cyberattacks, is also challenging to quantify, requiring assessment of rare but impactful events. Subjective evaluation may be needed for risks related to changes in accounting or regulation.

## 1.8 Methods of Risk Modification

Risk modification aims to align actual risk with acceptable levels and strategic objectives. It involves reducing or hedging risks and varies based on organizational goals. The primary methods include prevention and avoidance, acceptance, transfer, and shifting.

### 1.8.1 Risk Prevention and Avoidance

Risk avoidance is not always possible or desirable due to trade-offs between cost and benefits. Strategic decisions at the governance level help fine-tune exposure to desired risk levels. If risk exceeds acceptable thresholds, mitigation options include self-insuring, transferring, or shifting the risk.

### 1.8.2 Risk Acceptance: Self-Insurance and Diversification

Self-insurance involves accepting a risk that is too costly to eliminate through external means. This can either mean bearing the risk as is or setting aside reserves to cover potential losses. It must align with the enterprise's risk tolerance. Diversification, a key risk-mitigation technique, spreads investments across assets or sectors, and is central to modern portfolio theory.

### 1.8.3 Risk Transfer

Risk transfer shifts risk to another party, often through insurance. Insurers manage diversified risks by charging premiums and offering specialized coverage for unique risks.

### 1.8.4 Risk Shifting

Risk shifting alters the distribution of risk outcomes, typically using derivatives. Forward commitments and contingent claims efficiently shift risk across probability distributions and between parties. Derivatives can be traded on exchanges or created privately.

### 1.8.5 Choosing a Method for Modifying Risk

Organizations often **combine multiple risk-modification strategies** based on costs, benefits, and risk tolerance. For example, companies with foreign operations commonly hedge exchange rate risks using derivatives. Risk prevention and avoidance are preferred for low-benefit risks, while self-insurance may suit firms with strong cash flows. However, insurance may not cover risks affecting multiple parties simultaneously. Derivatives are often used to shift financial risks that exceed an organization's risk appetite. Each method affects risk profiles differently and requires careful evaluation.

# 2 Revision of the Basic Concepts

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 1 - Introduction: Risk-Return Trade-offs
  - Chapter 2 - Banks
  - Chapter 3 - Insurance Companies and Pension Plans
  - Chapter 4 - Fund Managers
  - Chapter 5 - Financial Instruments
  - Chapter 6 - The OTC Derivatives Market
  - Chapter 7 - Securitization and the Global Financial Crisis

**Learning Outcomes:**

- Quantify risk and return, and understand diversification and the efficient frontier.
- Analyze CAPM, systematic/nonsystematic risk, and $\alpha$ and $\beta$.
- Evaluate leverage, capital structure, and Modigliani-Miller propositions.
- Examine derivatives and the impact of the 2007-2008 crisis on OTC markets.
- Assess financial institutions' risks and differences between funds.
- Explore systemic risk, the 2007-2008 crisis, and regulatory frameworks.

## 2.1 Risk vs. Return for Investors

- How do investors quantify risk and return, both individually and in portfolios?
- What is the relationship between risk and expected return, and how does it influence investment decisions?
- How does diversification impact portfolio risk, and what is the efficient frontier's role in portfolio construction?
- How does adding a risk-free asset alter the efficient frontier, and how is the concept of the "market portfolio" derived?
- How do changes in market conditions, such as interest rate fluctuations or economic downturns, impact the risk-return trade-off in an investment portfolio?

## 2.2 The Capital Asset Pricing Model

- What is the CAPM model, and how is it defined?
- What is the systematic and nonsystematic risk, and where can you find it in the CAPM model?
- What is represented by $\alpha$ and $\beta$ in the CAPM model?
- What are the theoretical assumptions of the CAPM model? Is this model still useful in the real world?
- How is it possible to achieve positive $\alpha$, and how much is the weighted average $\alpha$ of all investors?

## 2.3 Capital Structure and Leverage

- How does leverage impact a firm's risk and return profile?
- What is the optimal capital structure for a company in terms of risk management?
- How does debt affect the company's risk of financial distress or bankruptcy?
- How do Modigliani-Miller propositions relate to risk management in firms with different capital structures?
- What role does equity financing play in reducing risk, and how does it compare to debt in terms of cost and financial flexibility?

## 2.4 Risk Management Tools and Derivatives

- How do exchange-traded and OTC derivatives differ in terms of risk exposure and regulation?
- What are the main challenges of using derivatives for risk management?
- What is meant by netting, and how is it related to events of default?
- How does financial crisis 2007-2008 changed OTC derivatives market?
- What is "rehypothecation", and what issues it causes on the OTC market?

## 2.5 Financial Institutions

- What are the roles of credit rating agencies in assessing the creditworthiness of institutions and securities?
- What are the potential conflicts of interest between commercial banking, securities services, and investment banking when they are all conducted under the same corporate umbrella?
- What is the definition of moral hazard and adverse selection?

- What are the main risks for insurance companies?
- What are the similarities and differences between mutual funds, ETFs, and hedge funds?

## 2.6 Regulatory and Systemic Risk

- What is systemic risk, and why is it crucial in the context of financial institutions?
- What were the origins and causes of the 2007 - 2008 financial crisis?
- What role do regulatory frameworks play in managing systemic risk?
- What went wrong during the crisis?
- What lessons can be learned from the crisis?

# 3 Forwards and Futures Strategies

**References:**

- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 7 - Risk Management Applications of Forward and Futures Strategies

**Learning outcomes:**

- Show how to use equity futures contracts for achieving a desired portfolio beta and calculate the necessary number of futures contracts.
- Create a synthetic stock index fund using cash and stock index futures.
- Describe the conversion of a long stock position into synthetic cash using stock index futures.
- Use equity and bond futures to rebalance a portfolio between equity and debt.
- Use futures to reallocate a portfolio across equity sectors.
- Define exchange rate risk and demonstrate the use of forward contracts to mitigate risk associated with future foreign currency transactions.
- Discuss the challenges and possible strategies for hedging the exchange rate risk of a foreign market portfolio.

## 3.1 Introduction

Definition of forward an futures

- Forward and futures contracts are commitments for one party to buy and the other to sell an underlying instrument at a future date at a pre-agreed price.
- Forward contracts are customized agreements, while futures contracts are standardized, created on a futures exchange, and subject to federal regulation.

- The material provides scenarios where these contracts are used for risk management.
- Businesses manage a variety of risks, often through hedging, which involves taking a market position to protect against unfavorable outcomes.
- Instead of just hedging, the broader concept of "risk management" will be emphasized.

## 3.2 Managing Interest Rate Risk

- Financial institutions face interest rate risk, affecting profitability and stability due to lending and borrowing activities.
- Interest rate risk impacts net income, asset values, and balance sheets, requiring effective risk management.
- Other companies also face interest rate risk in planning borrowing, including with fixed-rate loans, necessitating careful management to avoid unexpected cash demands.

### 3.2.1 Managing the Interest Rate Risk of a Loan Using a FRA

Definition of FRA

- A FRA or Forward Rate Agreement is a financial contract between two parties to exchange interest payments for a notional principal amount on a future, pre-agreed date.
- In a "6x18" FRA, "6" denotes the time (in months) until the loan's start, and "18" denotes the time until its end.
- So a "6x18" FRA means a one-year loan starting in six months.

- Lenders can lock in future loan rates by shorting a FRA, but this is risky due to borrowers' unpredictable needs and the possibility of losses if they decide not to take the loan.
- Banks offering floating-rate credit lines may use FRAs, but to mitigate risks, some choose options where the worst-case scenario is losing the option premium.

#### 3.2.1.1 Single-Payment Loan

- XYZ Corporation, planning to borrow $10 million in six months for a year to finance business expansion, is concerned about potential rising interest rates.
- They enter into a 6x18 FRA with a bank at a rate of 5.5% to hedge the risk.
- Outcome scenarios:
  1. If rates rise to 7%, the corporation borrows at 7% but the bank pays the difference, offsetting the increased cost (hedge was beneficial).
  2. If rates fall to 4%, the corporation pays the bank the difference, resulting in a cost (hedge wasn't beneficial).
- A FRA allows XYZ Corporation to lock in a maximum interest rate, hedging against rising interest rates.
- It provides certainty but can result in missed opportunities if rates fall.

- The payoff of a FRA at maturity can be calculated as follows:

$$\text{FRA Payoff} = \text{Notional Principal} \times \frac{(\text{Rate at Maturity} - \text{FRA Rate}) \times \frac{\text{Days}}{360}}{1 + \text{Rate at Maturity} \times \frac{\text{Days}}{360}}$$

- Where:
  - **Notional Principal** is the hypothetical amount of money that the FRA is based on.
  - **Rate at Maturity** is the actual market interest rate when the FRA matures.
  - **FRA Rate** is the interest rate agreed upon in the FRA contract.
  - **Days** is the number of days for which the contract is in effect.
- Note that this payoff is calculated assuming a 360-day year, which is a common convention for FRAs.
- The formula is based on the difference between the **FRA Rate** and the **Rate at Maturity**.
- Plugging these numbers into the FRA payoff formula, we get:

$$\text{FRA Payoff} = \$10,000,000 \times \frac{(0.07 - 0.055) \times \frac{360}{360}}{1 + 0.07 \times \frac{360}{360}} = \$140,187$$

$$\text{Pay back} = (\$10,000,000 - \$140,187) \times (1 + 0.07 \times \frac{360}{360}) = \$10,550,000$$

$$\text{Effective rate} = \left(\frac{\$10,550,000}{\$10,000,000} - 1\right) \times \frac{360}{360} \approx 5.5\%$$

#### 3.2.1.2 Floating-Rate Loan

- Floating-rate loans, with fluctuating interest rates, pose risk for borrowers.
- Borrowers can manage this through a series of FRAs aligned with the loan's rate reset dates, or futures contracts, though the latter may not match the reset periods.
- Financial institutions may use sophisticated methods to manage futures volatility, but these might not suit regular borrowers.
- Swaps could be a more accessible solution, and understanding FRAs helps in applying swaps for interest rate risk management.

### 3.2.2 Strategies and Applications for Managing Bond Portfolio Risk

- Longer-term loans pose bond market risk for borrowers and lenders.
- Managing government bond portfolios helps firms mitigate interest rate risk.
- Using futures on government bonds is a strategy for managing interest rate risk.
- The approach is similar to managing $\beta$ in the equity portfolio, only the bond duration is used instead of $\beta$.

## 3.3 Managing Equity Market Risk

- Stock market volatility poses greater risk compared to bond market volatility.
- The stock market is generally more liquid than the bond market.
- Futures contracts can help manage the risk associated with stock market volatility.
- Futures contracts are typically based on stock market indices, rather than individual stocks.
- Diversified investors manage risk at the portfolio level and prefer futures on broad-based indices.

### 3.3.1 Measuring and Managing the Risk of Equities

- Beta ($\beta$) is a measure of a stock's or a portfolio's volatility in relation to the market.
- It represents the tendency of a security's returns to respond to swings in the market.
- $\beta$ is used in the capital asset pricing model (CAPM), which calculates the expected return of an asset based on its beta and expected market returns.
- The formula for $\beta$ is:

$$\beta_i = \frac{Cov(R_i, R_m)}{Var(R_m)}$$

- Where:
  - $\beta_i$ is the beta of the stock or portfolio ($i$).
  - $Cov(R_i, R_m)$ is the covariance between the return of the stock/portfolio ($R_i$) and the return of the market ($R_m$).
  - $Var(R_m)$ is the variance of the return of the market.
- If $\beta$ is:
  - 1, the security's price will move with the market
  - less than 1, the security will be less volatile than the market
  - greater than 1, the security's price will be more volatile than the market.

- The number of futures contracts ($N$) needed to adjust $\beta$ is:

$$N = \frac{(\text{Target Beta} - \text{Current Beta}) \times \text{Portfolio Value}}{\text{Futures Price} \times \text{Futures Beta}}$$

- Let's break down the elements of this formula:
  - **Target Beta**: The desired beta of the portfolio after hedging.
  - **Current Beta**: The beta of the portfolio before the adjustment.
  - **Portfolio Value**: The total value of the portfolio that you want to hedge.
  - **Futures Price**: The price of one futures contract.
  - **Futures Beta**: The beta of the futures contract relative to the market.
- If the result of this formula is positive, it means you should buy futures contracts.
- Conversely, if the result is negative, it means you should sell futures contracts.
- The calculation assumes that the futures contract and the portfolio are based on closely correlated assets.

### 3.3.2 Numerical Example of Managing $\beta$

- Current Portfolio Beta: 1.2
- Target Beta: 0.8
- Portfolio Value: $\$1,000,000$
- Futures Price: $\$50,000$
- Futures Beta: 0.95

$$N = \frac{(0.8 - 1.2) \times \$1,000,000}{\$50,000 \times 0.95} \approx -8$$

- We need to short 8 futures contracts in order to adjust the portfolio beta to the target of 0.8.
- A negative value indicates shorting futures contracts because we want to decrease the portfolio's market risk.
- Later, the market increases by 3.7% and the portfolio increases by 4.4%, and futures price increases by 3.45%.
- Determine the overall value of the position and the effective $\beta$.

$$\text{Portfolio value} = \$1,000,000 \times 1.044 = \$1,044,000$$

$$\text{Futures value} = -8 \times (\$50,000 \times 1.0345 - \$50,000) = -\$13,800$$

$$\text{Total return} = \frac{\$1,044,000 + (-\$13,800)}{\$1,000,000} - 1 = 0.0302$$

$$\text{Effective beta} = \frac{0.0302}{0.037} = 0.816$$

- The effective $\beta = 0.816$ is pretty close from the target 0.8.

### 3.3.3 Creating Equity out of Cash

- Stock index futures help create synthetic equity positions, reducing transaction costs and conserving liquidity.
- A relationship between futures/forwards and the asset is established using a formula connecting the risk-free interest rate and dividends.
- Synthetic replication of the asset is beneficial in creating a synthetic stock index fund or converting cash into equity while retaining liquidity.

$$\text{Long Stock} + \text{Short Futures} = \text{Long Risk-Free Bond}$$

- Risk-free bonds and futures can be combined to "equitize" cash, converting it into an equity position while retaining liquidity.
- This strategy may be used by funds that wish to invest in equities but are constrained by the need for quick liquidation or regulatory restrictions.
- Despite these limitations, the fund can still gain equity exposure without significant risk of loss or difficulty in liquidation.

#### 3.3.3.1 Synthetic Index Fund

$$\text{Long Stock} = \text{Long Risk-Free Bond} + \text{Long Futures}$$

- It is created by combining risk-free bonds and futures on the desired index.
- To calculate the number of futures contracts, we use:

$$N = \frac{V(1+r)^t}{qf}$$

- Where:

- $V$ = amount of money to be invested.
- $f$ = futures price.
- $t$ = time to expiration of futures.
- $r$ = risk-free rate.
- $q$ = multiplier.
- $\delta$ = dividend yield on the index.

- On 1st January, an investment firm wants to construct a synthetic index fund of \$200 million invested in US stock.
- The index is the S&P 500, with a dividend yield of 1.5%.
- A futures contract on the S&P 500 is priced at \$3,000 with a multiplier of \$50.
- The position will be held for three months, after which it will be renewed.
- The US risk-free rate is 2%, and both the risk-free rate and dividend yield are compounded annually.

$$N = \frac{V(1+r)^t}{qf} = \frac{\$200,000,000(1.02)^{0.25}}{\$50 \times \$3,000} = 1,339.95$$

- We can only buy whole futures contracts, so we round up to $1,340$ contracts.

$$\frac{1,340 \times \$50 \times \$3,000}{1.02^{0.25}} = \$200,007,377$$

- We're effectively investing \$200,007,377 in the stock, and we put this much money into risk-free bonds.
- After three months the amount grows to $\$200,007,377 \times 0.02^{0.25} = \$201,000,000$.
- The number of units of stock that we have effectively purchased at the start is:

$$\frac{N \times q}{(1+\delta)^t} = \frac{1,340 \times \$50}{1.015^{0.25}} = 66,751.08$$

- If the stock were actually purchased, dividends would be reinvested into additional shares, so the number of shares would grow to $66,751,08 \times 1.015^{0.25} = 69,574.35$.
- When the futures contract expires at price $S_T$, the futures contracts will have a payoff:

$$\text{Futures payoff} = 1,340 \times \$50 \times (S_T - \$3,000) = 69,574.35 \times S_T - \$201,000,000$$

### 3.3.4 Creating Cash out of Equity

- Converting a long stock position into synthetic cash involves selling futures.
- This can minimize transaction costs and help avoid the sale of a large volume of stocks at once.

$$\text{Long Risk-Free Bond} = \text{Long Stock} + \text{Short Futures}$$

- The approach is simillar to the creation of synthetic index fund from the previous example.
- There are only two modifications in the equation:
  1. Minus at the beginning becasue we are selling futures.
  2. $V$ is the market value of our stock investment.

$$N = -\frac{V(1+r)^t}{qf}$$

- This approach is comparable to changing $\beta$ to zero.
- In this example the stock portfolio has to be indentical to the index (the same $\beta$) and it is a full hedge.
- The formula for managing $\beta$ is more general approach to eliminate the systematic risk on any portfolio.
- The number of futures contracts needed for transforming stocks into a risk-free asset might not match those required to nullify portfolio $\beta$.
- Eliminating systematic risk doesn't imply a complete absence of risk in the portfolio; diversifiable risk may persist.
- Both $\beta$ reduction and full hedging formulas align when applied to a portfolio identical to the index on which the futures are based.
- Derivatives provide a cost-effective and flexible solution for $\beta$ adjustments, especially for temporary adjustments.

## 3.4 Asset Allocation with Futures

### 3.4.1 Adjusting the Allocation among Asset Classes

- Suppose that our current portfolio is worth \$300 million allocated to 80% stock (\$240 million) and 20% bonds (\$60 million).
- Our goal is to change allocation to 50% stock (\$150 million) and 50% bonds (\$150 million).

- We need to reduce stock allocation by \$90 million and increase bond allocation by \$90 million.
- Solution:
  - Sell stock index futures contracts to eliminate beta on \$90 million of stock, effectively converting it to cash.
  - Purchase bond futures contracts to increase the duration on the cash to desired level.

### 3.4.2 Pre-Investing in an Asset Class

- Pre-investing via futures contracts allows investors to enter the market without immediate cash.
- It is effectively borrowing against the cash that will be received in the future and investing in the underlying.
- The strategy involves taking long positions in futures with correct beta and duration, aligning the results closely with direct investment.

$$\text{Long Stock} = \text{Long Futures} + \text{Long Risk-Free Bond}$$

$$\text{Long Stock} + \text{Short Risk-Free Bond} = \text{Long Futures}$$

- Short Risk-Free Bond is equivalent to a loan.
- Upon receipt of the anticipated cash, the futures are closed, and cash is invested, equating to paying off the implicit loan.
- This speculative strategy involves potential risks if the market doesn't perform adequately to cover borrowing costs.

## 3.5 Managing Foreign Currency Risk

- Companies engaging in international business face exchange rate risks that impact planning, income targets, and balance sheets.
- **Transaction exposure** arises when future cash flows require conversion from foreign to domestic currency.
- **Translation exposure** occurs when foreign subsidiary balance sheets are consolidated into the company's domestic currency balance sheet.
- **Economic exposure** refers to the impact of exchange rate changes on a company's competitiveness. Even companies not dealing in foreign markets may be affected, for instance, if a strong domestic currency lowers inbound tourism.

- The management of these risks requires different approaches: forward contracts for transaction exposure, accounting knowledge for translation exposure, and demand forecasting for economic exposure.
- Companies often use forward contracts to manage single cash flow conversions from one currency to another as futures contracts may be too standardized for their needs.

### 3.5.1 Managing the Risk of a Foreign Currency Receipt/Payment

- When a company expects to receive cash flows in a foreign currency, it is considered "long" on that currency.
- If the domestic currency increases in value while waiting for the cash flow, the company will receive fewer units of the domestic currency for the given amount of foreign currency.
- To manage this risk, the company should consider selling the foreign currency in the forward market by going "short" on a currency forward contract.
- The process for hedging a foreign currency cash outflow using a forward contract is similar to hedging a cash inflow but in reverse.

### 3.5.2 Managing the Risk of a Foreign-Market Asset Portfolio

- Foreign currency risk is a crucial consideration when investing abroad.
- It is challenging to hedge both the foreign market risk and the foreign currency risk simultaneously.
- Estimating the future portfolio value helps in determining hedging strategies.
- It is generally not possible to hedge the currency risk while leaving the local equity market exposed.
- Companies can choose to hedge the local market return or leave both the local market and currency risks unhedged.
- If only the foreign market returns is hedged, the gain is equal to foreign risk-free rate.
- If foreign market returns and exchange rate are hedged, the gain is equal to domestic risk-free rate.
- Neither strategy makes much sense for the long run.
- In the short run, however, this strategy can be a good tactic for investors who are already in foreign markets and who wish to temporarily take a more defensive position without liquidating the portfolio and converting it to cash.

## 3.6 Final Comments

- Betas and durations are unstable and difficult to measure, making it difficult to accurately assess risk sensitivities in stocks and bonds.

- Derivatives shouldn't be blamed if they don't yield the exact hedging results expected, and they should not be maligned for their speculative use due to their valuable hedging uses.
- Futures and forwards typically offer much lower transaction costs compared to stock indices, a major reason for their existence and survival as risk management products.
- Futures and forwards are less disruptive to portfolio managers and can be used to effectively change the risk profile or asset allocation of a portfolio.
- Despite requiring less capital to trade, futures and forwards are not immune to liquidity issues, particularly for contracts with long-term expirations.
- Organizations may have varying policies regarding the use of futures, forwards, and options, often based on leverage and credit-risk considerations.
- Futures and forward contracts have zero value at the start and offer linear payoffs, unlike options which require a cash investment at the start and offer non-linear payoffs.

## 3.7 Practice Questions and Problems

1. An investment management firm wishes to increase the beta for one of its portfolios under management from 0.95 to 1.20 for a three-month period. The portfolio has a market value of $175,000,000. The investment firm plans to use a futures contract priced at $105,790 in order to adjust the portfolio beta. The futures contract has a beta of 0.98.

    - A. Calculate the number of futures contracts that should be bought or sold to achieve an increase in the portfolio beta.
    - B. At the end of three months, the overall equity market is up 5.5%. The stock portfolio under management is up 5.1%. The futures contract is priced at $111,500. Calculate the value of the overall position and the effective beta of the portfolio.

2. Consider an asset manager who wishes to create a fund with exposure to the Russell 2000 stock index. The initial amount to be invested is $300,000,000. The fund will be constructed using the Russell 2000 Index futures contract, priced at 498.30 with a $500 multiplier. The contract expires in three months. The underlying index has a dividend yield of 0.75%, and the risk-free rate is 2.35% per year.

    - A. Indicate how the money manager would go about constructing this synthetic index using futures.
    - B. Assume that at expiration, the Russell 2000 is at 594.65. Show how the synthetic position produces the same result as investment in the actual stock index.

3. An investment management firm has a client who would like to temporarily reduce her exposure to equities by converting a $25 million equity position to cash for a period of four months. The client would like this reduction to take place without liquidating her equity position. The investment management firm plans to create a synthetic cash position using an equity futures contract. This futures contract is priced at 1170.10, has

a multiplier of $250, and expires in four months. The dividend yield on the underlying index is 1.25%, and the risk-free rate is 2.75%.

- A. Calculate the number of futures contracts required to create synthetic cash.
- B. Determine the effective amount of money committed to this risk-free transaction and the effective number of units of the stock index that are converted to cash.
- C. Assume that the stock index is at 1031 when the futures contract expires. Show how this strategy is equivalent to investing in the risk-free asset, cash.

4. Consider a portfolio with a 65% allocation to stocks and 35% to bonds. The portfolio has a market value of $200 million. The beta of the stock position is 1.15, and the modified duration of the bond position is 6.75. The portfolio manager wishes to increase the stock allocation to 85% and reduce the bond allocation to 15% for a period of six months. In addition to altering asset allocations, the manager would also like to increase the beta on the stock position to 1.20 and increase the modified duration of the bonds to 8.25. A stock index futures contract that expires in six months is priced at $157,500 and has a beta of 0.95. A bond futures contract that expires in six months is priced at $109,000 and has an implied modified duration of 5.25. The stock futures contract has a multiplier of one.

   - A. Show how the portfolio manager can achieve her goals by using stock index and bond futures. Indicate the number of contracts and whether the manager should go long or short.
   - B. After six months, the stock portfolio is up 5% and bonds are up 1.35%. The stock futures price is $164,005 and the bond futures price is $110,145. Compare the market value of the portfolio in which the allocation is adjusted using futures to the market value of the portfolio in which the allocation is adjusted by directly trading stocks and bonds.

5. A pension fund manager expects to receive a cash inflow of $50,000,000 in three months and wants to use futures contracts to take a $17,500,000 synthetic position in stocks and $32,500,000 in bonds today. The stock would have a beta of 1.15 and the bonds a modified duration of 7.65. A stock index futures contract with a beta of 0.93 is priced at $175,210. A bond futures contract with a modified duration of 5.65 is priced at $95,750.

   - A. Calculate the number of stock and bond futures contracts the fund manager would have to trade in order to synthetically take the desired position in stocks and bonds today. Indicate whether the futures positions are long or short.
   - B. When the futures contracts expire in three months, stocks have declined by 5.4% and bonds have declined by 3.06%. Stock index futures are priced at $167,559, and bond futures are priced at $93,586. Show that profits on the futures positions are essentially the same as the change in the value of stocks and bonds during the three-month period.

6. Consider a US company, GateCorp, that exports products to the United Kingdom. GateCorp has just closed a sale worth 200,000,000 GBP. The amount will be received in two months. Because it will be paid in pounds, the us company bears the exchange risk. In order to hedge this risk, GateCorp intends to use a forward contract that is priced at $1.4272 per pound. Indicate how the company would go about constructing the hedge. Explain what happens when the forward contract expires in two months.

7. ABCorp is a US-based company that frequently imports raw materials from Australia. It has just entered into a contract to purchase 175,000,000 AUD worth of raw wool, to be paid in one month. ABCorp fears that the australian dollar will strengthen, thereby raising the US dollar cost. A forward contract is available and is priced at $0.5249 per australian dollar. Indicate how ABCorp would go about constructing a hedge. Explain what happens when the forward contract expires in one month.

# 4 Options Strategies

**References:**

- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 8 - Risk Management Applications of Option Strategies

**Learning outcomes:**

- Compare the use of covered calls and protective puts for risk management.
- Calculate/interpret values for option strategies including bull spread, bear spread, butterfly spread, collar, straddle, box spread.
- Determine the effective annual rate for an anticipated loan managed with interest rate options.
- Calculate payoffs for a floating rate loan combined with an interest rate cap, floor, or collar.
- Explain delta hedging for dealers, including adjustments and why delta changes.
- Interpret the gamma of a delta-hedged portfolio and explain how gamma changes with options moving toward expiration.

## 4.1 Introduction

Definition of options

Options are contracts that give the buyer the right, but not the obligation, to buy (in the case of a call option) or sell (in the case of a put option) an underlying asset at a predetermined price, known as the strike price, within a specified time frame. The seller of the option, in turn, has the obligation to fulfill the transaction if the buyer decides to exercise the option.

- Options have non-linear payoffs and require upfront payment.
- Options allow benefiting from favorable movements and minimizing harm from unfavorable movements.
- Sellers of options face potential large losses and must be adequately compensated.

### 4.1.1 Notation and Assumptions

- Assumptions:
  - Option user has a view on underlying movements or volatility.
  - Options are customized and European-style.
  - No early termination of strategies.
- Time notation:
  - Time 0 is the time at which the strategy is initiated.
  - Time $t$ is the time the option expires, stated as a fraction of a year.
- Time until expiration: $t - 0 = t$, which is $\frac{\text{Days to expiration}}{365}$.
- Symbol definitions:
  - $c_0, c_t$ ... Price of the call option at time 0 and time $t$.
  - $p_0, p_t$ ... Price of the put option at time 0 and time $t$.
  - $X$ ... Exercise price.
  - $S_0, S_t$ ... Price of the underlying at time 0 and time $t$.
  - $V_0, V_t$ ... Value of the position at time 0 and time $t$.
  - $\Pi$ ... Profit from the transaction, calculated as $V_t - V_0$.
  - $r$ ... Risk-free rate.

## 4.2 Option Strategies for Equity Portfolios

- The following strategies can be applied to variety of asset classes.
- However, they are primarily explained and understood within the context of investing in equities or equity indices.

### 4.2.1 Standard Long and Short Positions

- Based on the course "Financial Derivatives" you should be able to define the following:
  - Long position in a stock and short position in a stock.
  - Long call and short call options.
  - Long put and short put options.
  - Calculate profit and value of the option.
  - Determine the maximum profit or loss for each position.

### 4.2.2 Risk Management Strategies with Options and the Underlying

#### 4.2.2.1 Covered Call

- Covered call strategy involves owning the underlying asset (like stocks) and then selling call options on that same asset.
- The goal is to generate additional income from the option premium, which the investor gets to keep regardless of whether the option is exercised or not.
- Covered calls reduce the risk of a position, contradicting beliefs that options inherently increase risk.
- While they decrease risk, covered calls also reduce expected returns compared to only holding the stock.
- This strategy may limit gains in bull markets but provides a cushion against losses in bear markets through the option premium.
- To summarize the covered call, we have the following:

$$V_t = S_t - \max(0, S_t - X)$$

$$\Pi = V_t - S_0 + c_0$$

$$\text{Maximum profit} = X - S_0 + c_0$$

$$\text{Maximum loss} = S_0 - c_0$$

$$\text{Breakeven} = S_t = S_0 - c_0$$

#### 4.2.2.2 Covered Call Example

- This example illustrates how a covered call strategy can generate income and potentially increase profits, but it also shows that the strategy can limit upside potential if the underlying asset's price increases significantly.
- Suppose you own 100 shares of Company XYZ, which are currently trading at \$50 per share.
- You decide to implement a covered call strategy by selling one call option (equivalent to 100 shares) with a strike price of \$55 for a premium of \$2 per share.
- If the price of the shares rises to \$60, the option will be exercised:

$$\text{Profit} = \text{Profit from shares} + \text{Profit from option}$$

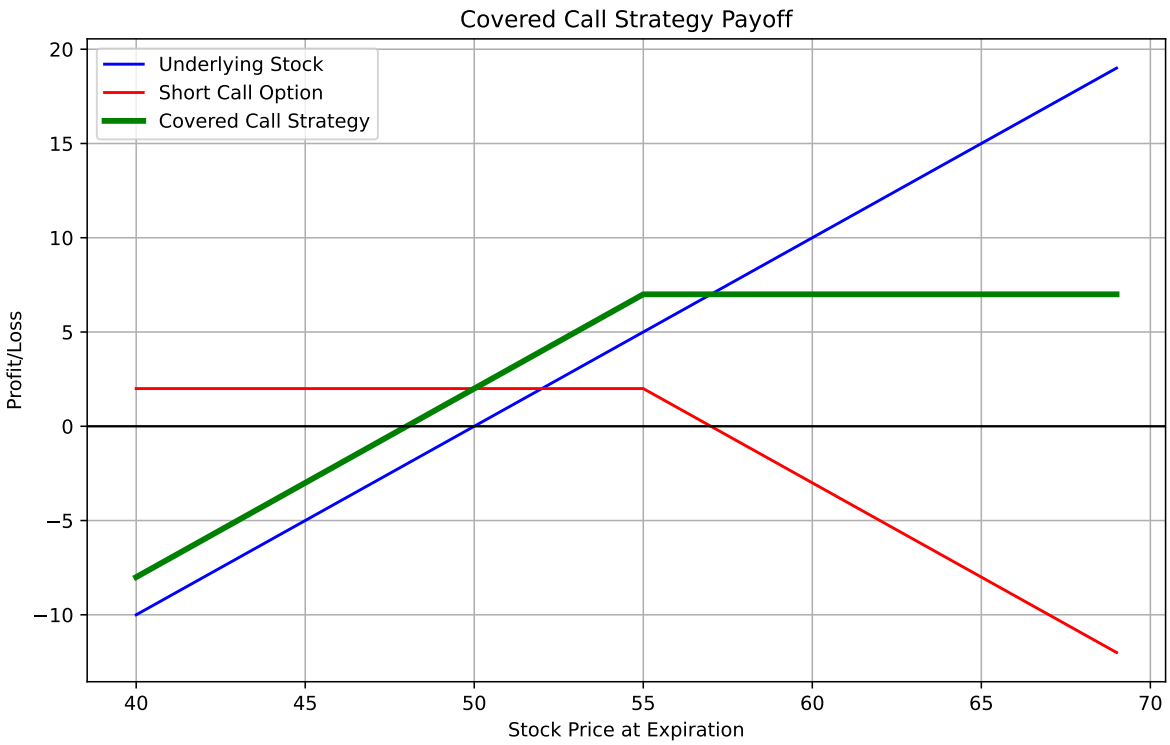
$$\text{Profit} = (\$60 - \$50) \times 100 + (-\$300) = \$1000 - \$300 = \$700$$

- If the price of the shares stays below \$55, the option will not be exercised:

$$\text{Profit} = (\$2 \times 100) + ((\$55 - \$50) \times 100) = \$200 + \$0 = \$200$$

```
import numpy as np
import matplotlib.pyplot as plt

# Variables
stock_price = 50
strike_price = 55
premium = 2
shares = 100

# Stock price range for the graph
stock_price_range = np.arange(40, 70, 1)

# Underlying stock payoff calculation
stock_payoff = stock_price_range - stock_price

# Call option payoff calculation (for short position)
call_payoff = np.where(
    stock_price_range <= strike_price,
    premium,
    premium - (stock_price_range - strike_price),
)

# Covered call payoff calculation
covered_call_payoff = stock_payoff + call_payoff

# Plotting
plt.figure(figsize=(10, 6))
plt.plot(stock_price_range, stock_payoff, label="Underlying Stock", color="blue")
plt.plot(stock_price_range, call_payoff, label="Short Call Option", color="red")
plt.plot(
    stock_price_range,
    covered_call_payoff,
```

```
    label="Covered Call Strategy",
    color="green",
    linewidth=3,
)
plt.axhline(y=0, color="black", linewidth=1.2)
plt.xlabel("Stock Price at Expiration")
plt.ylabel("Profit/Loss")
plt.title("Covered Call Strategy Payoff")
plt.legend()
plt.grid(True)
plt.show()
```




#### 4.2.2.3 Protective Put

- A protective put strategy involves buying an asset (like stocks) and buying a put option for the same asset.
- The goal of this strategy is to limit the downside risk of owning the asset.
- If the asset's price falls, the put option increases in value, offsetting the loss from the asset.

- A protective put strategy offers downside protection with upside potential, but it's important to consider the costs and trade-offs.
- Protection against downside loss is expensive and the buyer of a protective put must give up considerable upside potential.
- The protective put strategy is often viewed as a form of insurance.
- To summarize the protective put, we have the following:

$$V_t = S_t + \max(0, X - S_t)$$

$$\Pi = V_t - S_0 - p_0$$

$$\text{Maximum profit} = \infty$$

$$\text{Maximum loss} = S_0 + p_0 - X$$

$$\text{Breakeven} = S_t = S_0 + p_0$$

#### 4.2.2.4 Protective Put Example

- Suppose you own 100 shares of Company XYZ, which are currently trading at \$50 per share.
- You decide to implement a protective put strategy by buying one put option (equivalent to 100 shares) with a strike price of \$45 for a premium of \$2 per share.
- If the price of the shares rises to \$60, the option will not be exercised:

$$\text{Profit} = \text{Profit from shares} + \text{Profit from option}$$

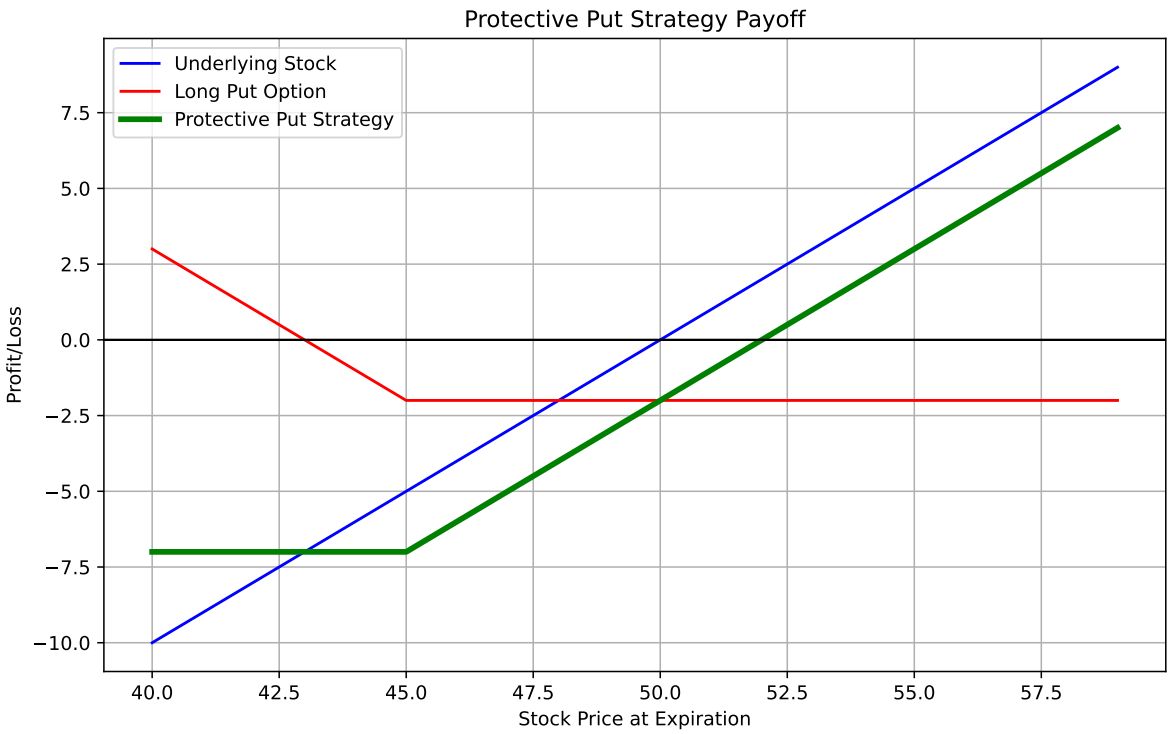
$$\text{Profit} = (\$60 - \$50) \times 100 + (-\$2) \times 100 = \$1000 - \$200 = \$800$$

- If the price of the shares drops to \$40, the option will be exercised:

$$\text{Profit} = (\$40 - \$50) \times 100 + (-\$2) \times 100 + (\$45 - \$40) \times 100$$

$$\text{Profit} = -\$1000 - \$200 + \$500 = -\$700$$

```python
import numpy as np
import matplotlib.pyplot as plt

# Variables
stock_price = 50
strike_price = 45
premium = 2
shares = 100

# Stock price range for the graph
stock_price_range = np.arange(40, 60, 1)

# Underlying stock payoff calculation
stock_payoff = stock_price_range - stock_price

# Put option payoff calculation (for long position)
put_payoff = (
    np.where(stock_price_range <= strike_price, strike_price - stock_price_range, 0)
    - premium
)

# Protective put payoff calculation
protective_put_payoff = stock_payoff + put_payoff

# Plotting
plt.figure(figsize=(10, 6))
plt.plot(stock_price_range, stock_payoff, label="Underlying Stock", color="blue")
plt.plot(stock_price_range, put_payoff, label="Long Put Option", color="red")
plt.plot(
    stock_price_range,
    protective_put_payoff,
    label="Protective Put Strategy",
    color="green",
    linewidth=3,
)
plt.axhline(y=0, color="black", linewidth=1.2)
plt.xlabel("Stock Price at Expiration")
plt.ylabel("Profit/Loss")
plt.title("Protective Put Strategy Payoff")
plt.legend()
plt.grid(True)
plt.show()
```




### 4.2.3 Money Spreads

- In options trading, "money spreads" or "spreads" refer to strategies that involve simultaneously buying and selling options of the same class (e.g., calls or puts) on the same underlying asset but with different strike prices and/or expiration dates.
- These strategies are often used to reduce risk and can also be a way to reduce the initial cost of entering a position.
- Spreads are categorized based on how they make or lose money, and below are some common types:

1. **Vertical Spreads**:
   - **Bull Call Spread**: Buying a call option with a lower strike price and selling another call option with a higher strike price, both with the same expiration date.
   - **Bull Put Spread**: Selling a put option with a higher strike price and buying a put option with a lower strike price.
   - **Bear Put Spread**: Buying a put option with a higher strike price and selling
   - **Bear Call Spread**: Selling a call option with a lower strike price and buying a call option with a higher strike price.
2. **Horizontal Spreads** (also known as **Calendar Spreads**):

- Involves options with the same strike price but different expiration dates.

3. **Diagonal Spreads**:
   - Combines elements of both vertical and horizontal spreads, involving different strike prices and different expiration dates.
4. **Iron Condor**:
   - A combination of two vertical spreads (one using calls, the other using puts) with the same expiration date but different strike prices.
5. **Butterfly Spread**:
   - A combination of three strike prices, usually involving both calls and puts. It can be constructed using all calls (call butterfly) or all puts (put butterfly).

- The profit and loss of these strategies depend on the underlying asset's price at expiration, relative to the strike prices of the options used in the spread.
- Money spreads help traders target a specific range of potential price movement, and they often involve less risk than outright buying or selling single options.
- The tradeoff, generally speaking, is that they also offer more limited profit potential.

### 4.2.4 Combinations of Calls and Puts

#### 4.2.4.1 Collars

- A collar option strategy is a defensive strategy that investors use when they have a long position in the underlying asset and want to protect against potential losses.
- It consists of three components:
  1. **Owning the Underlying Asset**: The investor holds shares of a stock or another underlying asset.
  2. **Buying a Put Option**: The investor buys a put option with a strike price below the current price of the underlying asset. This put option serves as insurance, providing the right to sell the underlying asset at the strike price if the asset's price falls significantly. It defines a floor on potential losses.
  3. **Selling a Call Option**: At the same time, the investor sells (or "writes") a call option with a strike price above the current price of the underlying asset. This generates premium income, which can offset the cost of buying the put option. However, it also caps the potential profit if the underlying asset's price rises above the call's strike price, as the call buyer will have the right to buy the asset at that price.
- The combination of these three elements forms a "collar" around the potential profit and loss from the underlying asset:

  - **Protection from Loss**: If the underlying asset's price falls, the put option compensates for the loss, limiting the downside risk.
  - **Limitation on Profit**: If the underlying asset's price rises above the strike price of the sold call, the profit is capped, as the asset may be called away at that price.
  - **Cost Mitigation**: The premium received from selling the call option can offset the cost of buying the put, potentially creating a cost-neutral or low-cost hedge.

- The collar strategy is often used by investors who have significant gains in the underlying asset and want to protect those gains without selling the asset.
- It can be tailored to the investor's outlook by choosing different strike prices and expiration dates for the options, balancing the level of protection, profit potential, and cost.

#### 4.2.4.2 Straddle

- A straddle option strategy is a neutral options strategy that involves simultaneously buying or selling a call option and a put option with the same strike price and expiration date on the same underlying asset.
- It's used when an investor expects a significant price movement but is uncertain about the direction of the movement (up or down). There are two main variations of the straddle strategy:
- **Long Straddle**: Suitable for investors who anticipate significant price volatility but are uncertain about the direction. The cost is the premium paid for both options, and the profit potential is unlimited.
- **Short Straddle**: Suitable for investors who anticipate little to no price movement in the underlying asset. The profit is limited to the premiums received, and the risk can be substantial if the asset price moves sharply.
- **Strangle**: Similar to a straddle but uses different strike prices for the call and put, usually with the call's strike price above and the put's strike price below the current asset price. It can be implemented as a long or short strategy.
- The straddle strategy is commonly used around significant news events, earnings announcements, or other situations where substantial price swings are possible. It allows investors to potentially profit from volatility itself, rather than having to correctly predict price direction.

## 4.3 Interest Rate Option Strategies

- Underlying asset is an interest rate, usually on a 90- or 180-day underlying instrument.
- The purpose is a protection against changes in interest rates.
- Call option is exercised if the underlying interest rate is above the exercise rate.

- Put option is exercised if the underlying interest rate is below the exercise rate.
- Notional principal determines the payoff when the option is exercised.
- Payoff is delayed by a period corresponding to the life of the underlying instrument.
- The rate is determined on one day, the option expiration, and payment is made $m$ days later.
- This practice is standard in floating-rate loans and thus is used with interest rate options, which are designed to manage the risk of floating-rate loans.
- **Payoff for Interest Rate Call Option**:

$$\text{Notional principal} \times \max(0, \text{Underlying rate at expiration} - \text{Exercise rate}) \times \frac{\text{Days in underlying rate}}{360}$$

- **Payoff for Interest Rate Put Option**:

$$\text{Notional principal} \times \max(0, \text{Exercise rate} - \text{Underlying rate at expiration}) \times \frac{\text{Days in underlying rate}}{360}$$

- Days in underlying refers to the maturity of the instrument from which the underlying rate is taken.

### 4.3.1 Using Interest Rate Calls with Borrowing - Example

| Loan Details | Value |
|---|---|
| Loan Amount | \$40,000,000 |
| Underlying | 180-day LIBOR |
| Spread | 200 basis points over LIBOR |
| Current LIBOR | 5.5% |
| Expiration | 128 days later |
| Exercise Rate | 5% |
| Call Premium | \$100,000 |
| LIBOR at expiration | 8% |

- **Call Payoff at Expiration** (received 180 days later):

$$\$40,000,000 \times \max(0, 0.08 - 0.05) \times \frac{180}{360} = \$600,000$$

- **Premium Compounded** from option initiation to option expiration (128 days) at original LIBOR of 5.5% plus 200 basis points:

$$\$100,000 \times \left[1 + (0.055 + 0.02) \times \frac{128}{360}\right] = \$102,667$$

- **Effective Loan Proceeds**:

$$\$40,000,000 - \$102,667 = \$39,897,333$$

- **Loan Interest**:

$$\$40,000,000 \times (0.08 + 0.02) \times \frac{180}{360} = \$2,000,000$$

- **Effective Interest**: Loan interest minus the call payoff
- **Effective Rate on the Loan**:

$$\left(\frac{\$40,000,000 + \$2,000,000 - \$600,000}{\$39,897,333}\right)^{365/180} - 1 = 7.79\%$$

- For various LIBOR rates at option expiration, the highest possible effective rate is 7.79% (if the option is exercised). If the option is not exercised, the effective rate decreases.
- The similar analysis can be used **for lending** by using interest rate **put options**.
- A lender can buy a put that pays off if the interest rate falls below a chosen level.
- The put payoff then compensates the lender for the lower interest rate on the loan.
- The equations have to be slightly adjusted for put options, everything else is similar.

### 4.3.2 Floating-Rate Cap, Floor, and Collar

- Many corporate loans are floating-rate loans that have periodic interest payments with regularly scheduled rate resets, causing multiple distinct risks.
- **Interest Rate Calls (Caplets)**:
  - Borrowers might use multiple call options (caplets) to set a ceiling on borrowing rates.
  - Each caplet has its own expiration date; exercise rates are typically the same.
  - A combination of these options is called **a cap**.
- **Interest Rate Puts (Floorlets)**:

  - Lenders like may be concerned about falling rates and can buy puts expiring on rate reset dates.
  - This combination is known as **a floor**, with individual options called floorlets.

- **Interest Rate Collars**:
  - Combines a long position in a cap with a short position in a floor.
  - Zero-Cost collar means that exercise rate on the floor matches the cap premium, seeming to have no "cost," but the borrower gives up gains from potential falling rates.
  - Borrowers often buy caps and sell floors, while lenders might buy floors and sell caps.
  - Most interest rate collars are initiated by borrowers, to protect against varying interest rates.

## 4.4 Option Portfolio Risk Management Strategies

- Dealers provide liquidity in options, taking on risk and hedging positions to earn the bid-ask spread without taking on the risk themselves.
- When a customer wants to buy a call option, the dealer quotes a price and sells it, but this creates a dangerous short call position.
- Ideally, the dealer would find someone to take the opposite position or use put-call parity to hedge the risk.
- However, this often isn't practical, so dealers often resort to delta hedging.

### 4.4.1 Delta Hedging

- Delta is a ratio expressing how the option price changes relative to the underlying price, typically between 0.0 and 1.0.
- A long position in the underlying asset can offset a short call position; the size of the long position depends on the option's delta.
- If the dealer sells a given number of calls, they need to own a number of shares equal to the number of underlying shares in the options times the delta.
- **Challenges with Delta Hedging**:
  - **Delta as an Approximation**: Delta is only an approximate measure, and it changes as the ***underlying price or time changes***, meaning the hedged position can quickly become unhedged.
  - **Rounding Off**: The number of underlying units per option has to be rounded, leading to imprecision in balancing the two positions.

- **Continuous Monitoring Required**: Constant adjustments are needed to maintain the delta-hedged position, especially when the underlying changes significantly.

### 4.4.2 Gamma Hedging

- Gamma measures the rate of change in the delta with respect to the change in the underlying asset's price.
- It reflects the deviation between the exact option price change and the price change approximated by delta, making it a measure of the delta's sensitivity.
- **Role in Delta-Hedging**:
  - If a delta-hedged position were risk-free, its gamma would be zero.
  - A large gamma indicates a greater deviation from being risk-free.
  - The larger the gamma, the more the position is affected by rapid movements in the delta.
- Gamma is the largest for at-the-money and near-expiration options.
- Deep in-the-money or out-of-the-money options have slow delta movements and pose fewer problems for hedgers.
- Gamma hedging requires adding another option and combining it with the underlying and existing options to make both delta and gamma zero.

### 4.4.3 Vega and Volatility Risk

- Vega measures the sensitivity of an option's price to changes in the volatility of the underlying asset.
- Volatility is the only unobservable variable required to value an option, making it a critical factor.
- An option's price is highly sensitive to changes in the underlying asset's volatility.
- Changes in volatility can have a significant impact on a dealer's position, even if it's delta-hedged with zero or insignificant gamma.
- An increase in volatility can raise the value of options, potentially leading to substantial losses for a dealer holding the underlying and selling options to delta hedge.
- An option price is more sensitive to volatility when it is at-the-money.
- Dealers attempt to measure and monitor vega, and in some cases, hedge it by taking a position in another option.
- The dealer must jointly monitor and manage the risks associated with delta, gamma, and vega.

## 4.5 Practice Questions and Problems

### 4.5.1 Case Study

Stanley Singh, CFA, is the risk manager at SS Asset Management. Singh works with individual clients to manage their investment portfolios. One client, Sherman Hopewell, is worried about how short-term market fluctuations over the next three months might impact his equity position in Walnut Corporation. While Hopewell is concerned about short-term downside price movements, he wants to remain invested in Walnut shares as he remains positive about its long-term performance. Hopewell has asked Singh to recommend an option strategy that will keep him invested in Walnut shares while protecting against a short-term price decline. Singh gathers the information in the following table to explore various strategies to address Hopewell's concerns.

- **Current stock Price**: $67.79
- **Walnut Corporation European options:**

| Exercise Price | Market Call Price | Call Delta | Market Put Price | Put Delta |
|---|---|---|---|---|
| $55.00 | $12.83 | 4.7 | $0.24 | –16.7 |
| $65.00 | $3.65 | 12.0 | $1.34 | –16.9 |
| $67.50 | $1.99 | 16.5 | $2.26 | –15.3 |
| $70.00 | $0.91 | 22.2 | $3.70 | –12.9 |
| $80.00 | $0.03 | 35.8 | $12.95 | –5.0 |

**Note:** Each option has 106 days remaining until expiration.

Another client, Nigel French, is a trader who does not currently own shares of Walnut Corporation. French has told Singh that he believes that Walnut shares will experience a large move in price after the upcoming quarterly earnings release in two weeks. However, French tells Singh he is unsure which direction the stock will move. French asks Singh to recommend an option strategy that would allow him to profit should the share price move in either direction.

A third client, Wanda Tills, does not currently own Walnut shares and has asked Singh to explain the profit potential of three strategies using options in Walnut: a bull call spread, a straddle, and a butterfly spread. In addition, Tills asks Singh to explain the gamma of a call option. In response, Singh prepares a memo to be shared with Tills that provides a discussion of gamma and presents his analysis on three option strategies:

- **Strategy 1:** A straddle position at the $67.50 strike option
- **Strategy 2:** A bull call spread using the $65 and $70 strike options
- **Strategy 3:** A butterfly spread using the $65, $67.50, and $70 strike call options

1. The option strategy Singh is most likely to recommend to Hopewell is a:

- a. collar.
- b. covered call.
- c. protective put.

2. The option strategy that Singh is most likely to recommend to French is a:

   - a. straddle.
   - b. butterfly.
   - c. box spread.

3. Based upon the table, strategy 1 is profitable when the share price at expiration is *closest* to:

   - a. $63.24.
   - b. $65.24.
   - c. $69.49.

4. Based upon the table, the maximum profit, on a per share basis, from investing in strategy 2, is *closest* to:

   - a. $2.26.
   - b. $2.74.
   - c. $5.00.

5. Based upon the table, and assuming the market price of Walnut's shares at expiration is $66, the profit or loss, on a per share basis, from investing in strategy 3, is *closest* to:

   - a. –$1.57.
   - b. $0.42.
   - c. $1.00.

6. Based on the data in the table, Singh would advise Tills that the call option with the largest gamma would have a strike price *closest* to:

   - a. $55.
   - b. $67.50.
   - c. $80.

# 5 Volatility, Correlations, and Copulas

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 8 - Volatility
  - Chapter 9 - Correlations and Copulas
  - Chapter 10 - Valuation and Scenario Analysis

**Learning outcomes:**

- Understand the concept of volatility and implied volatility.
- Become familiar with standard volatility models like EWMA and GARCH(1,1).
- Explore basic concept of correlation and copulas.
- Describe the funcioning of a simple copula.
- Be aware of copulas shortcomings and criticism.
- Describe the process of valuation and scenario analysis.

## 5.1 Volatility

- Monitoring the volatilities of market variables such as exchange rates, equity prices, and commodity prices is crucial for a financial institution to manage the value of its portfolio.
- The popular models are Exponentially Weighted Moving Average (EWMA), Autoregressive Conditional Heteroscedasticity (ARCH), and Generalized Autoregressive Conditional Heteroscedasticity (GARCH).
- These models acknowledge that volatility is not constant and varies over time; it can be low during some periods and high during others.

### 5.1.1 Definition of Volatility

- The volatility, $\sigma$, is usually calculated as a standard deviation of daily returns.
- When volatility is used for option pricing, the unit of time is usually one year.
- When volatility is used for risk management, the unit of time is usually one day.
- Volatility is usually much greater when the market is open than when it is closed.
- For this reason time is usually measured in **trading days** not calendar days.

- It is assumed that there are **252 trading days** in one year for most assets.

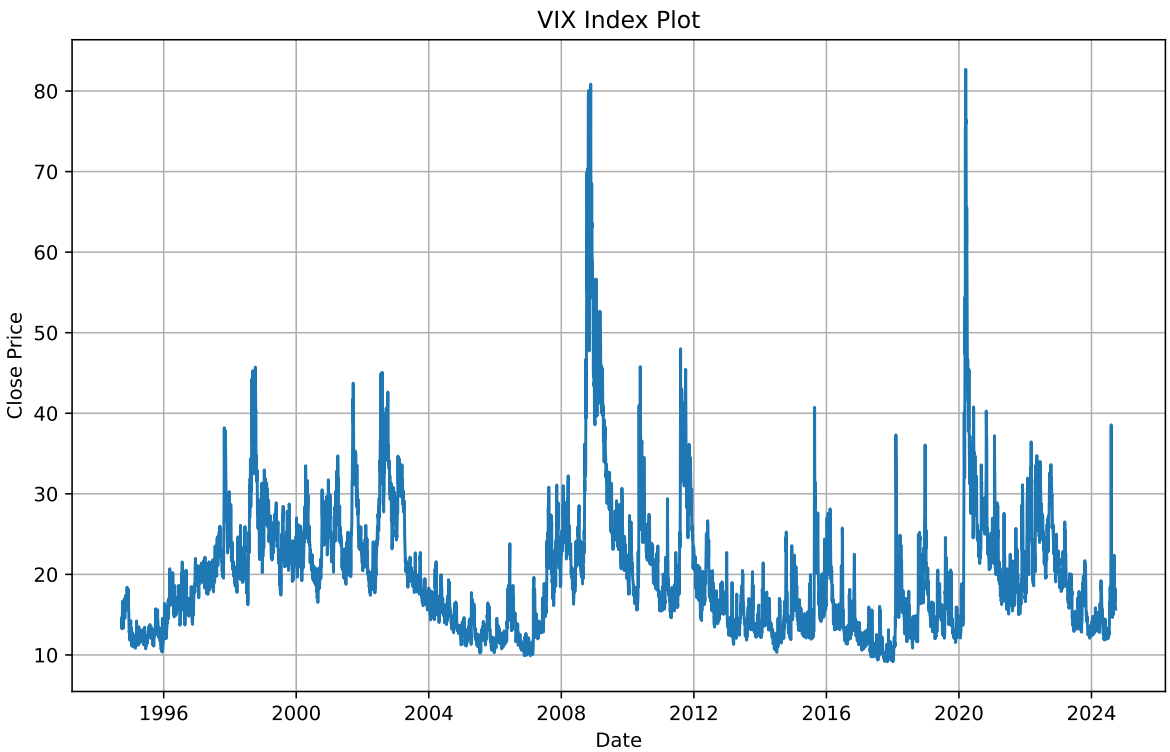
$$\sigma_{year} = \sigma_{day} \times \sqrt{252}$$

$$\sigma_{day} = \sigma_{year} \times \frac{1}{\sqrt{252}}$$

### 5.1.2 Implied Volatility

- Volatility inferred from option prices is called the **implied volatility**.
- The implied volatility of an option is the volatility for which the Black-Scholes-Merton price equals the market price.
- The key advantage in using implied volatility is that it provides information regarding the market's perception of uncertainty going forward.
- The VIX S&P 500 Volatility Index.

```
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt

# Download historical data for VIX
data = yf.download("^VIX", period="30y", progress=False)

# Plot the closing price
plt.figure(figsize=(10, 6))
plt.plot(data["Close"])
plt.title("VIX Index Plot")
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.grid(True)
plt.show()
```




### 5.1.3 Are Daily Percentage Changes in Financial Variables Normal?

- The following code tests if the daily returns AAPL are normally distributed.
- Download daily data from Yahoo Finance, calculate log returns, and show descriptive statistics.

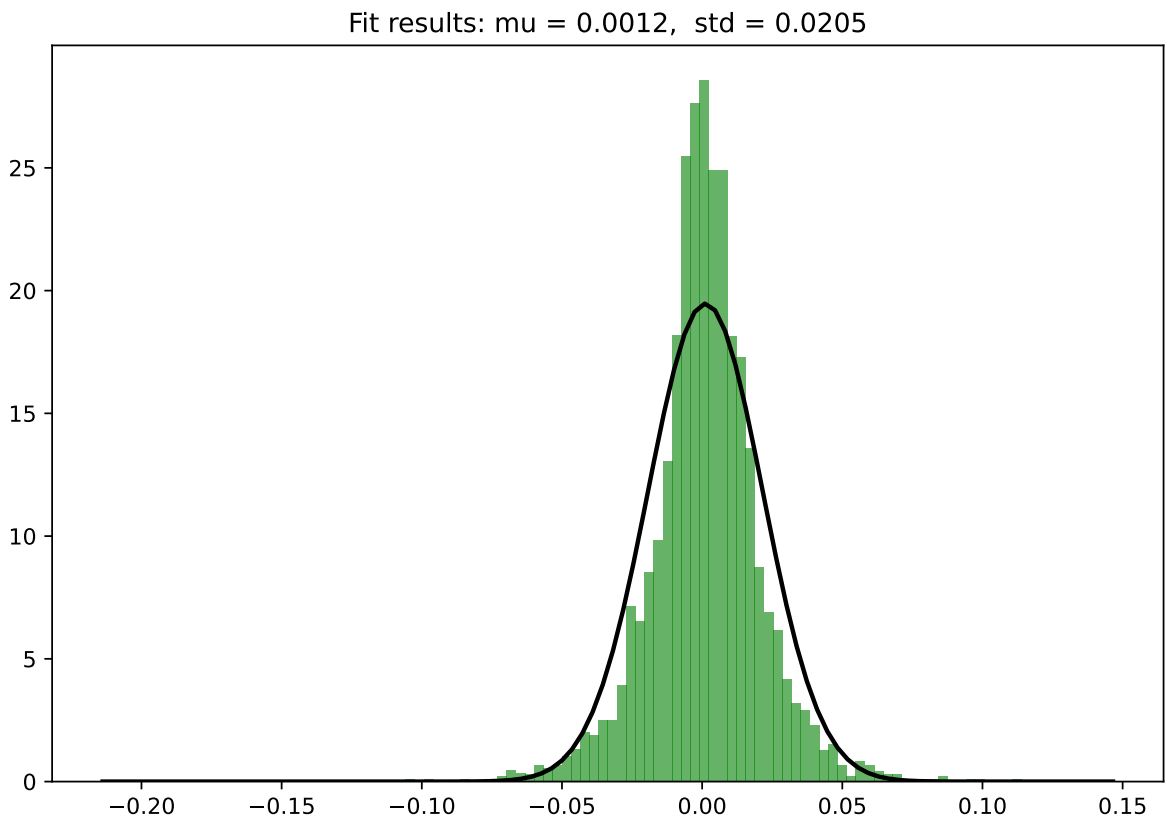
```
import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt
from scipy.stats import norm, kstest

# Download historical data for AAPL
data = yf.download("AAPL", period="20y", progress=False)

# Calculate the daily returns
data["Return"] = np.log(data["Close"] / data["Close"].shift(1)).dropna()

# Show descriptive statistics
```

```
desc = pd.DataFrame(data["Return"].describe())
print("Descriptive Statistics:")
print(desc)
```

```
Descriptive Statistics:
            Return
count  5033.000000
mean      0.001159
std       0.020491
min      -0.197470
25%      -0.008359
50%       0.001085
75%       0.011737
max       0.130194
```

- Plot the histogram of returns and fit a normal distribution to the data.
- It is visible that AAPL returns are not normally distributed.
- AAPL shows fat tails (heavy-tailed distribution) which means that large changes in the variable are more likely compared to normal distribution.

```
# Plot the histogram of returns
plt.figure(figsize=(9, 6))
plt.hist(data["Return"], bins=100, density=True, alpha=0.6, color="g")

# Fit a normal distribution to the data
mu, std = norm.fit(data["Return"].dropna())

# Plot the PDF
xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = norm.pdf(x, mu, std)
plt.plot(x, p, "k", linewidth=2)
title = "Fit results: mu = %.4f,  std = %.4f" % (mu, std)
plt.title(title)

plt.show()
```




- Use Kolmogorov-Smirnov test to check normality of the returns.
- The Kolmogorov-Smirnov (K-S) test is a non-parametric test used to determine if a sample data set follows a specific distribution, such as the normal distribution.
- $H_0$ : The sample data follows the normal distribution.
- $H_1$ : The sample data does not follow the normal distribution.
- p-value in this example is extremely small, therfore, $H_0$ could be rejected.

```
# Perform a Kolmogorov-Smirnov test
ks_statistic, p_value = kstest(data["Return"].dropna(), "norm", args=(mu, std))

print("KS Statistic:", ks_statistic)
print("p-value:", p_value)
```

```
KS Statistic: 0.0735566098061799
p-value: 3.974745020370105e-24
```

### 5.1.4 The Exponentially Weighted Moving Average Model

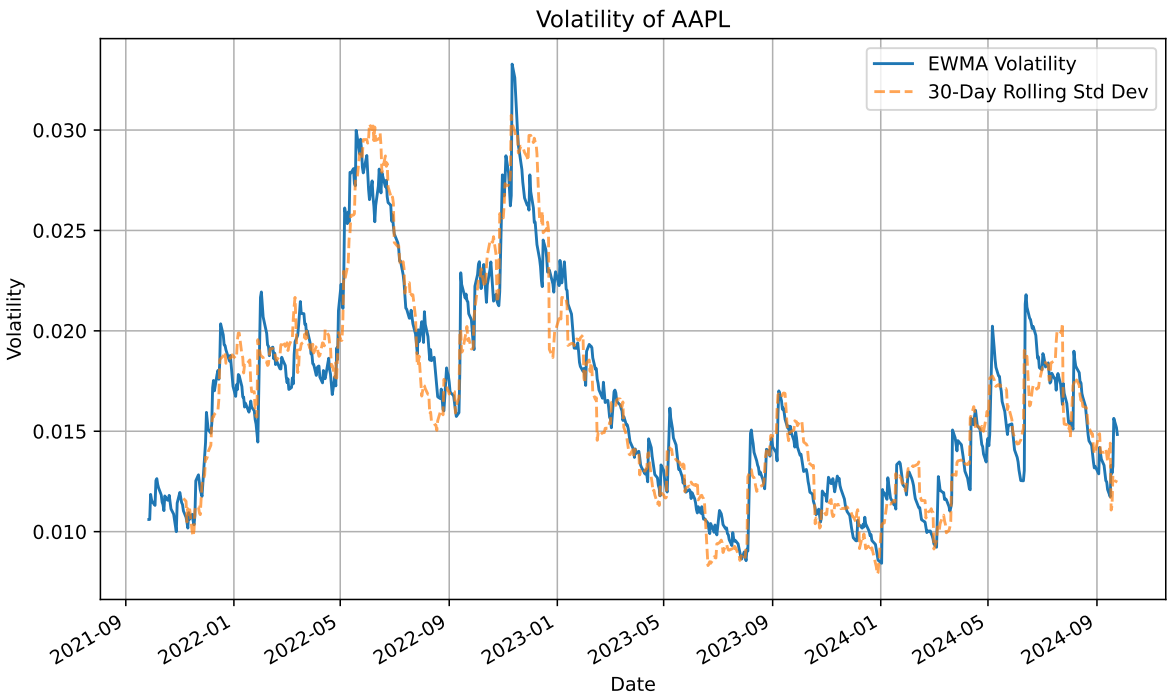
- The Exponentially Weighted Moving Average (EWMA) model for volatility is a method of measuring the volatility of financial returns where more weight is assigned to more recent returns.
- This is in contrast to the simple moving average model, which assigns equal weight to all observations in the sample.
- The idea behind giving more weight to recent returns is that they are more indicative of future volatility than older returns.
- The EWMA model can be defined by the following equation:

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

- Where:
  - $\sigma_t^2$ is the forecasted variance for time $t$,
  - $\sigma_{t-1}^2$ is the variance for time $t-1$,
  - $r_{t-1}^2$ is the square of the return at time $t-1$,
  - and $\lambda$ is the decay factor, which determines the weights.
- In the EWMA model, $\lambda$ lies between 0 and 1.
- A smaller $\lambda$ will assign more weight to recent returns.
- Now let's implement a simple numerical example in Python.
- In this example, we first download the historical closing prices for AAPL and compute the daily returns.
- We then calculate the volatility series using the EWMA model, with a decay factor of 0.94.
- Finally, we plot the volatility series.

```
import pandas as pd
import yfinance as yf
import numpy as np
import matplotlib.pyplot as plt

# Download historical data for AAPLpepe
data = yf.download("AAPL", period="3y", progress=False)

# Calculate the daily returns
data["Return"] = np.log(data["Close"] / data["Close"].shift(1))
```

```python
# Drop missing values
data = data.dropna().copy()

# Set the decay factor
lambda_factor = 0.94

# Initialize the variance
variance = data["Return"][0] ** 2

# Initialize the volatility series with the first variance
volatility_series = [np.sqrt(variance)]

# Calculate the volatility series
for t in range(1, len(data)):
    variance = (
        lambda_factor * variance + (1 - lambda_factor) * data["Return"].iloc[t - 1] ** 2
    )
    volatility_series.append(np.sqrt(variance))

# Convert volatility series to pandas Series
volatility_series = pd.Series(volatility_series, index=data.index)

# Calculate the rolling 30-day standard deviation as volatility
data["30D Std Dev"] = data["Return"].rolling(window=30).std()

# Plot the volatility
plt.figure(figsize=(10, 6))
volatility_series.plot(label="EWMA Volatility")
data["30D Std Dev"].plot(label="30-Day Rolling Std Dev", linestyle="--", alpha=0.7)
plt.title("Volatility of AAPL")
plt.xlabel("Date")
plt.ylabel("Volatility")
plt.legend()
plt.grid(True)
plt.show()
```




### 5.1.5 The GARCH(1,1) Model

- The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is a statistical model used to estimate the volatility of financial markets.
- The GARCH(1,1) model is the most widely used GARCH variant, where "1,1" indicates that the model uses one lag of the squared residuals (ARCH term) and one lag of the conditional variance (GARCH term).
- The model is specified as follows:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

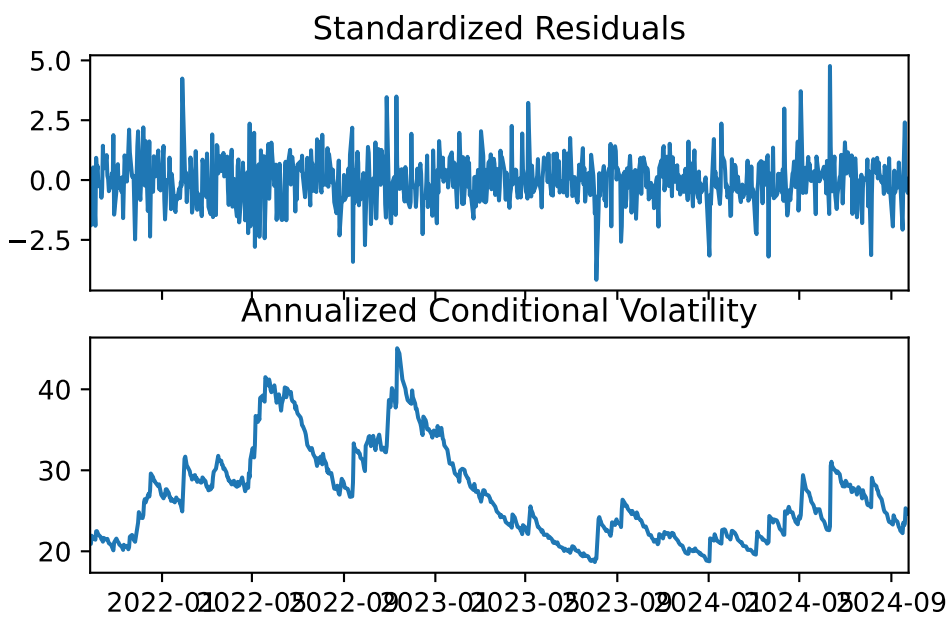
- Where:
  - $\sigma_t^2$ is the conditional variance at time $t$,
  - $r_{t-1}^2$ is the square of the return at time $t-1$,
  - $\sigma_{t-1}^2$ is the conditional variance at time $t-1$,
  - $\omega$, $\alpha$, and $\beta$ are parameters to be estimated. $\alpha$ is the ARCH parameter that determines the impact of the previous error, and $\beta$ is the GARCH parameter that determines the impact of the previous period's variance.

- In a GARCH(1,1) model, the variance today depends on the weighted average of long-term variance ($\omega$), the impact of the return from the previous day ($\alpha$), and the variance observed the previous day ($\beta$).
- We can estimate a GARCH(1,1) model in Python using the `arch` package.
- Here's an example with real historical data for the Apple Inc. (AAPL) stock.
- More information about the package available at https://bashtage.github.io/arch/univariate/introduction.html

```python
import pandas as pd
import numpy as np
import yfinance as yf
from arch import arch_model
import matplotlib.pyplot as plt

# Download historical data for AAPL
data = yf.download("AAPL", period="3y", progress=False)

# Calculate the daily returns
data["Return"] = np.log(data["Close"] / data["Close"].shift(1)) * 100

# Drop missing values
data = data.dropna()

# Define the GARCH model
model = arch_model(data["Return"], vol="Garch", p=1, q=1)

# Fit the model
model_fit = model.fit(disp=False)

# Print the summary of the model
print(model_fit.summary())

# Plot the conditional standard deviation
model_fit.plot(annualize="D")
plt.show()
```

```
                     Constant Mean - GARCH Model Results
==============================================================================
Dep. Variable:                 Return   R-squared:                       0.000
Mean Model:             Constant Mean   Adj. R-squared:                  0.000
```

```
Vol Model:                      GARCH   Log-Likelihood:               -1447.77
Distribution:                  Normal   AIC:                           2903.55
Method:            Maximum Likelihood   BIC:                           2922.05
                                        No. Observations:                  753
Date:                Tue, Sep 24 2024   Df Residuals:                      752
Time:                        20:29:00   Df Model:                            1
                                 Mean Model
============================================================================
                 coef    std err          t      P>|t|     95.0% Conf. Int.
----------------------------------------------------------------------------
mu             0.0887  5.864e-02      1.513      0.130 [-2.621e-02,  0.204]
                              Volatility Model
==============================================================================
                 coef    std err          t      P>|t|       95.0% Conf. Int.
------------------------------------------------------------------------------
omega          0.0387  2.221e-02      1.742  8.150e-02 [-4.839e-03,8.221e-02]
alpha[1]       0.0346  9.725e-03      3.555  3.778e-04  [1.551e-02,5.363e-02]
beta[1]        0.9526  1.217e-02     78.251      0.000      [  0.929,  0.976]
==============================================================================

Covariance estimator: robust
```




- In this script, we first download the historical closing prices for AAPL and compute the daily log returns (in percentage for model easier estimation).

- We then define a GARCH(1,1) model and fit the model to the returns data.
- The `arch_model` function is used to define the GARCH model, where `p=1` and `q=1` specify the GARCH(1,1) model.
- Finally, we print the summary of the model fit, which includes the estimated parameters, and plot the conditional standard deviation.
- The plot shows the evolution of the estimated volatility over time.
- Please note that the code uses daily returns to fit the model, which may lead to extreme values in the estimation of the variance because of daily market noise. A solution could be to use longer periods (weekly, monthly), depending on the purpose of the analysis.

### 5.1.6 Choosing between the Models

- GARCH(1,1) and EWMA models estimate financial market volatility.
- GARCH(1,1) incorporates mean reversion, unlike EWMA.
- Despite being complex, GARCH(1,1) is theoretically superior due to its consideration of mean reversion.
- In GARCH(1,1), best-fit parameter estimation is critical.
- If $\omega$ is zero in GARCH(1,1), it reduces to EWMA.
- If $\omega$ is negative, GARCH(1,1) is unstable and EWMA is preferred.

## 5.2 Correlations

- Risk managers need to estimate not just volatilities, but also correlations between changes in market variables when assessing risk exposures.
- Correlations can be monitored in a similar manner to volatilities.

### 5.2.1 Definition of Correlation

- Correlation between two variables, $V1$ and $V2$, is represented by the correlation coefficient $\rho$, calculated using the formula:

$$\rho = \frac{E(V_1V_2) - E(V_1)E(V_2)}{SD(V_1)SD(V_2)}$$

- where $E(.)$ stands for expected value and $SD(.)$ for standard deviation.
- No correlation exists when $E(V_1V_2) = E(V_1)E(V_2)$, making $\rho = 0$.
- If $V_1 = V_2$, $\rho$ equals 1.
- Covariance between $V_1$ and $V_2$ is defined as:

$$\text{cov}(V_1, V_2) = E(V_1V_2) - E(V_1)E(V_2)$$

- This allows us to write the correlation as:

$$\rho = \frac{\text{cov}(V_1, V_2)}{SD(V_1)SD(V_2)}$$

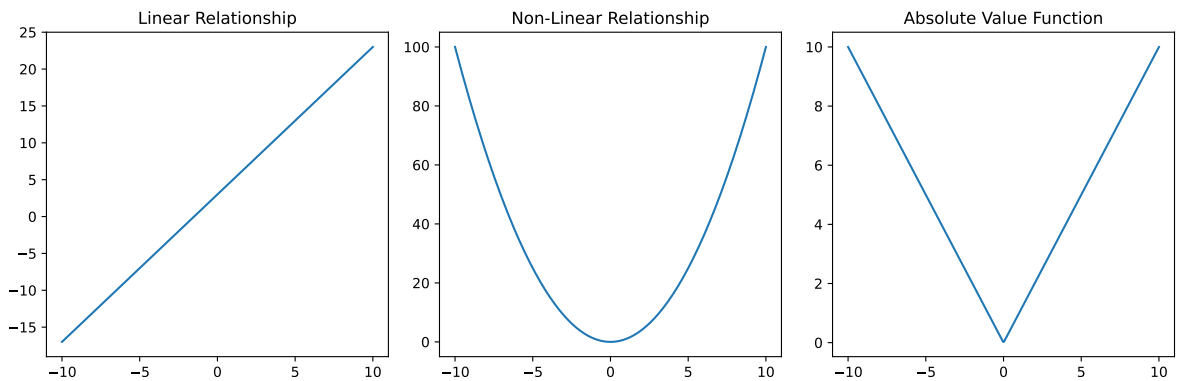
- Though understanding correlation may be more intuitive, covariances prove to be fundamental in the analysis.
- Two variables are statistically independent if knowing one doesn't affect the probability distribution for the other. Formal representation:

$$f(V_2|V_1 = x) = f(V_2)$$

- for all $x$, where $f(.)$ represents the probability density function.
- Zero correlation doesn't imply no dependence between variables.
- Correlation measures linear dependence, but there can be many other ways two variables can relate. These relationships can be visualized by plotting $E(V_2)$ against $V_1$.
- In certain situations, the standard deviation of $V_2$ depends on the value of $V_1$, indicating another aspect of how $V_2$ depends on $V_1$.

```
import numpy as np
import matplotlib.pyplot as plt

# Generate a range of x values
x = np.linspace(-10, 10, 400)

# Linear Relationship
y_linear = 2 * x + 3
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.plot(x, y_linear)
plt.title("Linear Relationship")

# Non-linear Relationship (Quadratic)
y_nonlinear = x**2
plt.subplot(1, 3, 2)
plt.plot(x, y_nonlinear)
plt.title("Non-Linear Relationship")

# Absolute Value Function
```

```
y_abs = np.abs(x)
plt.subplot(1, 3, 3)
plt.plot(x, y_abs)
plt.title("Absolute Value Function")

plt.tight_layout()
plt.show()
```




### 5.2.2 Monitoring Correlation

- It is common to work with variance-covariance matrices.
- Similar approaches as Exponentially Weighted Moving Average (EWMA) and GARCH models can be used to monitor the covariance rate between two variables.
- The returns on variables $X$ and $Y$ on day $i$ are:

$$x_i = \frac{X_i - X_{i-1}}{X_{i-1}}$$

$$y_i = \frac{Y_i - Y_{i-1}}{Y_{i-1}}$$

- The covariance rate between $X$ and $Y$ on day $n$ is:

$$\text{cov}_n = E(x_n y_n) - E(x_n)E(y_n)$$

- In practice, risk managers assume that expected daily returns are zero when calculating the covariance rate per day, hence the covariance rate per day between $X$ and $Y$ on day $n$ is simplified to:

$$\text{cov}_n = E(x_n y_n)$$

- Using equal weights for the last $m$ observations on $x_i$ and $y_i$ gives the following:

$$\text{cov}_n = \frac{1}{m}\sum_{i=1}^{m} x_{n-i} y_{n-i}$$

$$\text{var}_{x,n} = \frac{1}{m}\sum_{i=1}^{m} x_{n-i}^2$$

$$\text{var}_{y,n} = \frac{1}{m}\sum_{i=1}^{m} y_{n-i}^2$$

$$\rho_n = \frac{\text{cov}_n}{\sqrt{\text{var}_{x,n}\text{var}_{y,n}}}$$

- The Exponentially Weighted Moving Average (EWMA) model can be used to weight recent observations more heavily. The formula for updating a covariance estimate in the EWMA model is:

$$\text{cov}_n = \lambda \text{cov}_{n-1} + (1-\lambda) x_{n-1} y_{n-1}$$

- The lower the value of $\lambda$, the greater the weight that is given to recent observations.
- GARCH models can also be used for updating covariance rate estimates and forecasting the future level of covariance rates. The GARCH(1,1) model for updating a covariance rate between $X$ and $Y$ is:

$$\text{cov}_n = \omega + \alpha x_{n-1} y_{n-1} + \beta \text{cov}_{n-1}$$

- This formula gives some weight to a long-run average covariance, some to the most recent covariance estimate, and some to the most recent observation on covariance.
- The long-term average covariance rate is $\omega/(1-\alpha-\beta)$.
- Example of covariance between Apple and Microsoft from the past 5 years of daily returns:

```python
import yfinance as yf
import numpy as np

# Download daily closing prices of Apple and Microsoft from the past 5 years
df = yf.download(["AAPL", "MSFT"], period="5y", progress=False)["Adj Close"]

# Calculate daily returns
df = df.pct_change().dropna()

# Define decay factor and initialize covariance
decay_factor = 0.94
ewma_cov = df["AAPL"].iloc[0] * df["MSFT"].iloc[0]

# Calculate EWMA covariance
for i in range(1, len(df)):
    ewma_cov = (
        decay_factor * ewma_cov
        + (1 - decay_factor) * df["AAPL"].iloc[i] * df["MSFT"].iloc[i]
    )

print(f"The EWMA covariance is: {ewma_cov}")
```

```
The EWMA covariance is: 7.973599904442859e-05
```

## 5.3 Copulas

- This section introduces the concept of copulas, which define a correlation structure between two or more variables, regardless of the shapes of their probability distributions.
- It illustrates how a copula can model default correlation for a portfolio of loans, a model that forms part of the Basel II capital requirements.

### 5.3.1 Introduction to Copulas

- In risk management, copulas are statistical tools used to model and manage dependencies among different risk types or among different components within a risk type.
- A copula is a function that links univariate marginal distributions to their joint multivariate distribution.
- Copulas allow risk managers to separate the study of the behavior of individual risk factors (captured by the marginal distributions) from the study of the dependencies among these risk factors (captured by the copula).

- This separation property is particularly useful in risk management because it allows practitioners to fit marginal distributions and dependency structures separately.

Some key points:

1. **Multivariate distribution:** When dealing with multiple risk factors, it is often necessary to understand their joint behavior, which is described by their multivariate distribution. However, multivariate distributions can be difficult to estimate and work with directly.
2. **Marginal distributions and copulas:** The copula approach decomposes the multivariate distribution into marginal distributions (one for each risk factor) and a copula that captures the dependencies among the risk factors.
3. **Benefits:** The benefits of copulas include flexibility in modeling and the ability to capture a wide range of dependency structures. Different copula types can be used to model different types of dependencies (e.g., Gaussian copula, t-copula, etc.).
4. **Applications:** In risk management, copulas are used in various applications, such as estimating portfolio credit risk, modeling the spread of contagion in financial networks, pricing complex derivatives, and more.
5. **Limits and criticisms:** Despite their benefits, copulas also have limits. In particular, they have been criticized for their role in the financial crisis of 2007-2008, where the widespread use of Gaussian copulas for modeling dependencies among mortgage-backed securities led to underestimation of risk due to the fact that Gaussian copulas do not capture tail dependence well.

Applications in risk management:

1. **Portfolio Risk Management:** Copulas can model the dependence structure between different assets in a portfolio, separate from their individual behaviors. This allows for a more accurate computation of portfolio risk measures, such as Value-at-Risk (VaR) and Expected Shortfall (ES), by capturing the impact of extreme joint movements in asset prices.
2. **Credit Risk Management:** Copulas are extensively used in the pricing and risk management of credit derivatives, especially Collateralized Debt Obligations (CDOs). They can model the likelihood of joint default events (default correlations) across different entities. The Gaussian copula was famously used in the pricing of CDOs in the run-up to the 2007-2008 financial crisis.
3. **Operational Risk Management:** In operational risk, losses are usually due to a number of high impact, low frequency events (e.g., legal risks, risks of fraud, etc.). Copulas can be used to model the dependencies among these different types of operational risks, improving the accuracy of capital requirement calculations.
4. **Insurance Risk Management:** Copulas can also be used to model dependencies in insurance claims from different sectors or regions, helping insurers to better manage their overall risk.

5. **Stress Testing:** Copulas can be useful in stress testing scenarios. By defining the dependence structure between various risk factors, stress tests can be more accurately modeled to reflect extreme conditions.

### 5.3.2 Gaussian Copula

- A Gaussian copula is a specific type of copula used to model the dependency structure among various dimensions or variables.
- It assumes that these dependencies follow a multivariate normal distribution.
- To understand the Gaussian copula, consider a collection of random variables.
- Each of these variables has its own individual (marginal) distribution, which describes its behavior.
- The copula then describes how these variables interact or are correlated with each other.

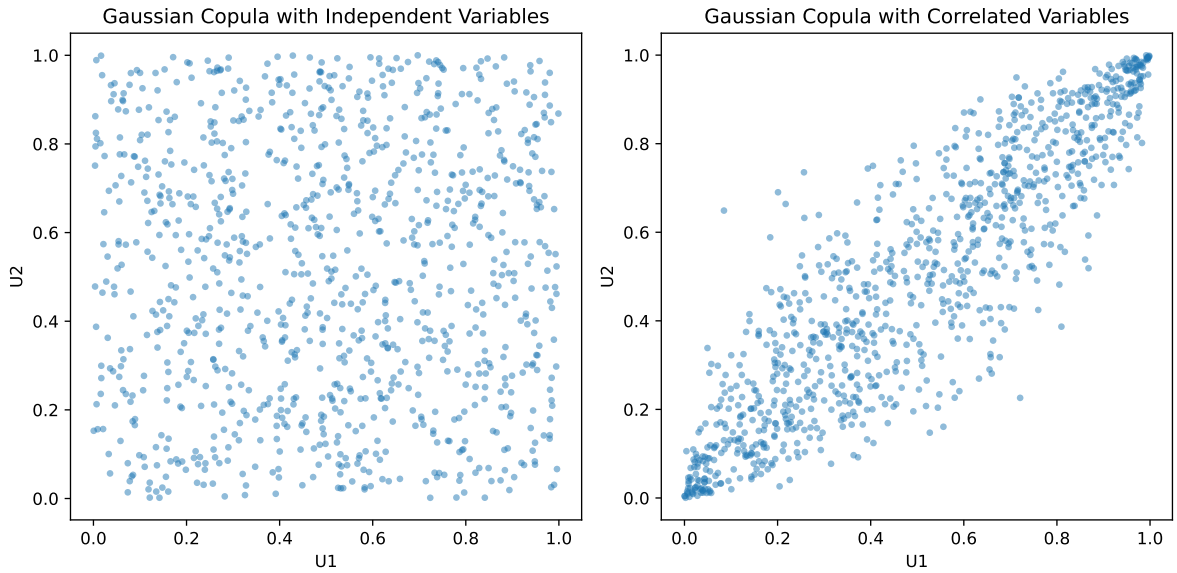
Simplified description of how a Gaussian copula is constructed:

1. **Start with random variables:** Let's say we have a set of random variables, each following its own arbitrary distribution.
2. **Transform to uniform:** Apply the cumulative distribution function (CDF) to each random variable. This transforms each variable into a uniform distribution on the interval [0, 1].
3. **Apply inverse Gaussian CDF:** Take these uniform variables and apply the inverse of the cumulative distribution function of the multivariate normal distribution. This step ensures that the new variables have a multivariate normal distribution.
4. **Correlation Matrix:** The Gaussian copula is parameterized by a correlation matrix, which captures the dependencies among the variables.

- The Gaussian copula is popular due to its simplicity and the tractability of the Gaussian distribution.
- However, it has limitations, particularly in modeling tail dependencies.
- It was famously used in the pricing of collateralized debt obligations (CDOs) and was implicated in the mispricing of risk that contributed to the 2008 financial crisis.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Set the random seed for reproducibility
np.random.seed(42)

# Generate independent data
data_independent = np.random.normal(size=(1000, 2))
```

```python
# Generate correlated data
correlated_mean = [0, 0]
correlated_cov = [[1, 0.9], [0.9, 1]]
data_correlated = np.random.multivariate_normal(
    correlated_mean, correlated_cov, size=1000
)

# Gaussian Copula for independent data
u_independent = norm.cdf(data_independent)

# Gaussian Copula for correlated data
u_correlated = norm.cdf(data_correlated)

# Plotting both scatter plots side by side
fig, axs = plt.subplots(1, 2, figsize=(10, 5))

# Scatter plot for independent data
axs[0].scatter(u_independent[:, 0], u_independent[:, 1], marker=".", alpha=0.5)
axs[0].set_xlabel("U1")
axs[0].set_ylabel("U2")
axs[0].set_title("Gaussian Copula with Independent Variables")

# Scatter plot for correlated data
axs[1].scatter(u_correlated[:, 0], u_correlated[:, 1], marker=".", alpha=0.5)
axs[1].set_xlabel("U1")
axs[1].set_ylabel("U2")
axs[1].set_title("Gaussian Copula with Correlated Variables")

plt.tight_layout()
plt.show()
```




### 5.3.3 Gaussian Copula Example

- In this example, we'll use historical data from two stocks to demonstrate the use of a Gaussian copula.
- First, we'll download the data and compute the daily returns for each stock.

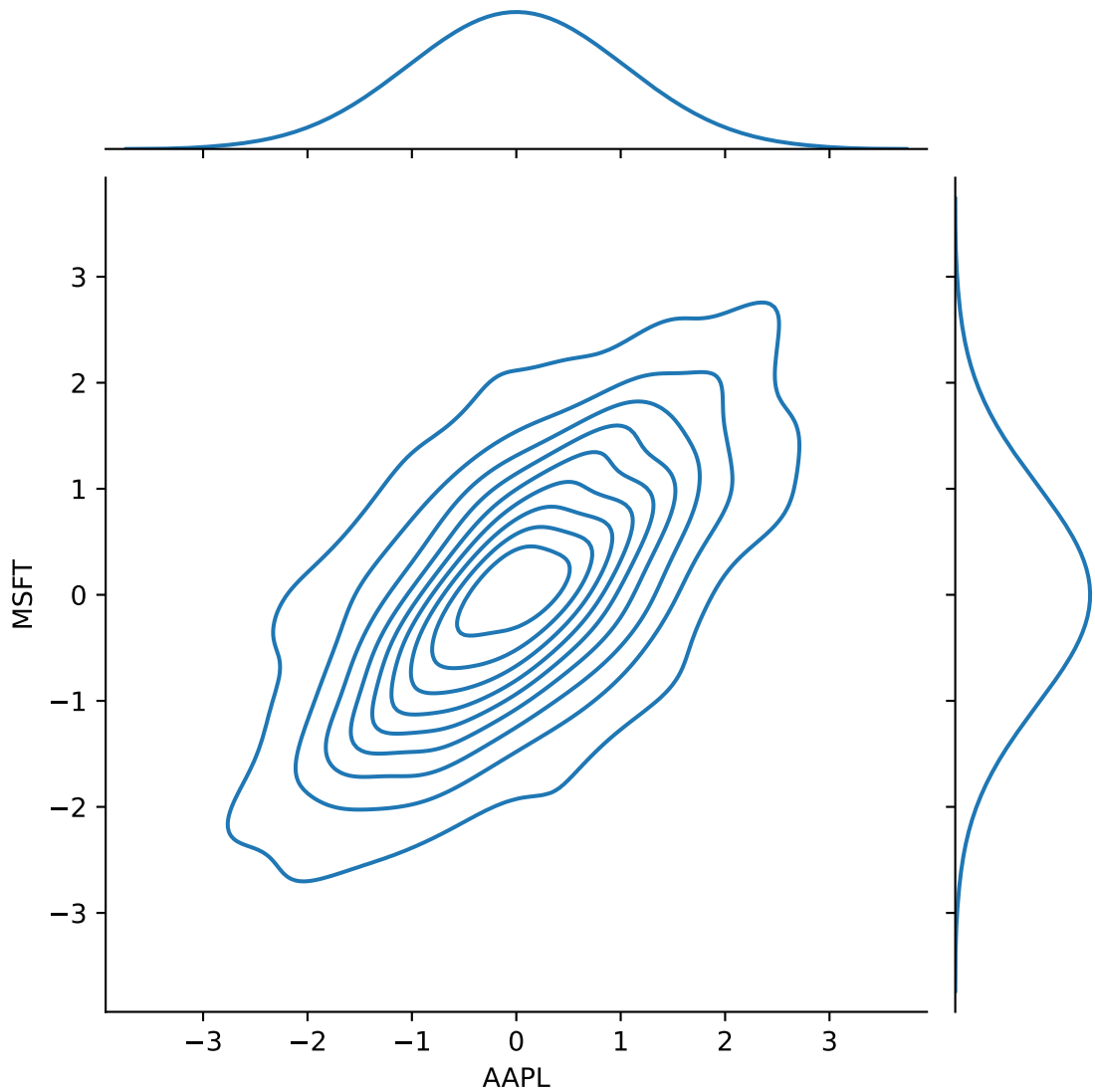
```
import yfinance as yf
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import norm, gaussian_kde

# Download historical data for desired stocks
tickers = ["AAPL", "MSFT"]

data = yf.download(tickers, period="2y", progress=False)["Adj Close"]

# Calculate the daily returns
returns = data.pct_change().dropna()
```

- Next, we'll transform these daily returns to be uniformly distributed by using the empirical cumulative distribution function.

```
# Calculate empirical cumulative distribution functions
ecdf_aapl = returns["AAPL"].rank() / len(returns["AAPL"])
ecdf_msft = returns["MSFT"].rank() / len(returns["MSFT"])
```

- Finally, we'll transform these uniform variables to be normally distributed and plot the joint distribution to see their dependence structure.

```
# Transform to standard normal distribution
norm_aapl = norm.ppf(ecdf_aapl)
norm_msft = norm.ppf(ecdf_msft)

# Create DataFrame for plotting
df_norm = pd.DataFrame({"AAPL": norm_aapl, "MSFT": norm_msft})

# Plot joint distribution
sns.jointplot(data=df_norm, x="AAPL", y="MSFT", kind="kde")
plt.show()
```




- The plot shows a smoothed estimate of the joint distribution of the transformed returns.
- There is a clear linear pattern suggesting a strong dependence between the two stocks' returns.
- Remember, in practice, you should check the appropriateness of the Gaussian copula for your data.
- While it's a simple and popular choice, it might not always capture the dependence structure correctly, especially in the tails of the distribution.

### 5.3.4 Clayton Copula Example

- **Tail Dependence**: The main difference between the Clayton and Gaussian copulas is how they model tail dependence. The Clayton copula is suitable for modeling lower tail dependence (correlation between extreme negative values), while the Gaussian copula can model both lower and upper tail dependence, but not as effectively for extreme values.
- **Parameterization**: The Clayton copula uses a single parameter (the Clayton parameter) to determine the strength and direction of the dependence. In contrast, the Gaussian copula uses a correlation matrix to model the linear dependence between variables.
- **Origins**: The Clayton copula is an Archimedean copula, while the Gaussian copula is derived from the multivariate normal distribution.
- This example fetches historical stock data for two companies (e.g., Apple and Microsoft) and shows their tail dependence using the Clayton copula.
- The plots show the lower tail dependence between the daily returns of Apple and Microsoft.
- This can be helpful for analyzing the risk of simultaneous extreme losses in these two stocks.

## 5.4 Valuation and Scenario Analysis

- **Valuation and Scenario Analysis**: Both activities used by financial institutions, concerned with estimating future cash flows, but with different objectives.
  - **Valuation**: Aims to estimate the present value of future cash flows. Calculates expected values of future cash flows across all outcomes and discounts them to today.
  - **Scenario Analysis**: Focuses on exploring the full range of future situations, often emphasizing adverse outcomes to answer "How bad can things get?"
- **Real-world vs Risk-Neutral Projections**:
  - **Real-World Projections**: Underlie scenario analysis.
  - **Risk-Neutral Projections**: Used for valuation, helpful for time-evolving variables or when an outcome depends on a particular discrete event, such as a company defaulting.

## 5.5 Practice Questions and Problems

### 5.5.1 Volatility

1. The volatility of an asset is 2% per day. What is the standard deviation of the percentage price change in three days?
2. The volatility of an asset is 25% per annum. What is the standard deviation of the percentage price change in one trading day? Assuming a normal distribution with zero mean, estimate 95% confidence limits for the percentage price change in one day.
3. What is implied volatility? What does it mean if different options on the same asset have different implied volatilities?
4. Calculate daily volatility using the last 30 observations for selected asset.
5. Test normality of daily returns for selected asset.
6. Explain the exponentially weighted moving average (EWMA) model for estimating volatility from historical data.
7. What is the difference between the exponentially weighted moving average model and the GARCH(1,1) model for updating volatilities?
8. The most recent estimate of the daily volatility of an asset is 1.5% and the price of the asset at the close of trading yesterday was \$30.00. The parameter $\lambda$ in the EWMA model is 0.94. Suppose that the price of the asset at the close of trading today is \$30.50. How will this cause the volatility to be updated by the EWMA model?
9. A company uses an EWMA model for forecasting volatility. It decides to change the parameter $\lambda$ from 0.95 to 0.85. Explain the likely impact on the forecasts.
10. A company uses the GARCH(1,1) model for updating volatility. The three param- eters are $\omega$, $\alpha$, and $\beta$. Describe the impact of making a small increase in each of the parameters while keeping the others fixed.
11. Assume that an index at close of trading yesterday was 1,040 and the daily volatility of the index was estimated as 1% per day at that time. The parameters in a GARCH(1,1) model are $\omega = 0.000002$, $\alpha = 0.06$, and $\beta = 0.92$. If the level of the index at close of trading today is 1,060, what is the new volatility estimate?

### 5.5.2 Correlations and Copulas

1. If you know the correlation between two variables, what extra information do you need to calculate the covariance?
2. What is the difference between correlation and dependence? Suppose that $y = x^2$ and $x$ is normally distributed with mean zero and standard deviation one. What is the correlation between $x$ and $y$?
3. Explain what is meant by tail dependence. How can you vary tail dependence by the choice of copula?
4. Calculate Clayton copula for selected asset and show appropriate graphs.

### 5.5.3 Valuation and Scenario Analysis

1. What is the difference between *valuation* and *scenario analysis*?
2. What is meant by risk-neutral valuation and the risk-neutral world?
3. A stock price has an expected return of 12% and a volatility of 20%. It is currently \$50. What is the probability that it will be greater than \$70 in two years?
4. An analyst calculates the expected future value of a stock index in (a) the real world and (b) the risk-neutral world. Which would you expect to be higher? Why?
5. The value of a derivative that pays off \$100 after one year if a company has defaulted during the year is \$3. The value of a derivative that pays off \$100 after one year if a company has not defaulted is \$95. What is the risk-free rate? What is the risk neutral probability of default?
6. Explain why it is sometimes necessary to work in both the real world and the risk-neutral world when carrying out a scenario analysis to determine a confi- dence interval for the value of a portfolio in one year.

# 6 Market Risk I

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 11 - Value at Risk and Expected Shortfall
  - Chapter 14 - Interest Rate Risk
  - Chapter 15 - Derivatives Risk
  - Chapter 16 - Scenario Analysis and Stress Testing
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 6 - Risk Management

**Learning outcomes:**

- Explain the use of Value at Risk (VaR) in measuring portfolio risk.
- Describe advantages and limitations of VaR.
- Explain the use of Expected Shortfall (ES) in measuring portfolio risk.
- Understand conditions for coherent risk measures and provide examples.
- Define how to choose parameters for VaR and ES estimation.

## 6.1 Introduction

- **Market risk** refers to the risk arising from changes in stock prices, interest rates, exchange rates, and commodity prices.
- Risk management involves identifying and measuring risk and ensuring that it aligns with desired risk levels.
- Effective risk management requires judgment, experience, and an understanding of model strengths and limitations.
- Market risk is easier to analyze compared to other financial risks due to the availability of data and collective experience.
- Estimating potential losses for market risk is challenging as they are unknown and based on historical data.
- Risk management models blend historical data with forward-looking judgment to create a framework for testing and assessing risk.

- **Value at Risk (VaR)** and **Expected Shortfall (ES)** are metrics that summarize total portfolio risk with a single number.
- VaR was first introduced by JPMorgan and is now widely used by corporate treasurers, fund managers, and financial institutions.
- Regulators have traditionally used VaR for capital requirement calculations, but have switched to ES for some market risk.

## 6.2 Understanding Value at Risk (VaR)

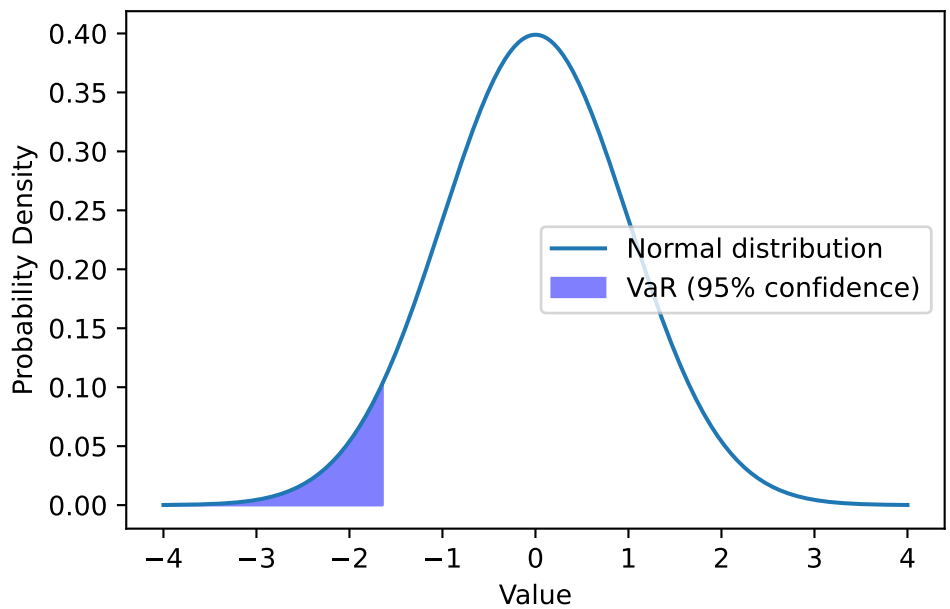
Definition of VaR

- Value at risk is the minimum loss that would be expected a certain percentage of the time over a certain period of time given the assumed market conditions.
- E.g.: The 5% VaR of a portfolio is $2.2 million over a one-day period.
- Using the example given, it is correct to say any of the following:
  - $2.2 million is the minimum loss we would expect 5% of the time.
  - 5% of the time, losses would be at least $2.2 million.
  - We would expect a loss of no more than $2.2 million 95% of the time.

- Value at Risk (VaR) is an important risk measure in global financial markets.
- VaR is defined as the minimum loss expected over a certain period of time with a specified level of confidence.
- VaR can be expressed in currency units or as a percentage of portfolio value.
- VaR is a minimum loss, not the total possible loss of the portfolio.
- A VaR statement includes the frequency of losses and the time horizon over which they are expected to occur.
- A 5% VaR implies a 95% level of confidence.
- VaR statements can be rephrased to indicate the minimum loss expected or the maximum loss not expected to occur.
- The concept of VaR is often illustrated using a probability distribution, such as the normal distribution, although other distributions may be more appropriate for financial market returns.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Parameters
mu = 0
sigma = 1
```

```python
x = np.linspace(-4, 4, 1000)
confidence_level = 0.95

# Calculate VaR
VaR = norm.ppf(1 - confidence_level, mu, sigma)

# Plot the normal distribution
plt.plot(x, norm.pdf(x, mu, sigma), label="Normal distribution")

# Highlight the VaR area
x_fill = np.linspace(-4, VaR, 1000)
plt.fill_between(
    x_fill,
    norm.pdf(x_fill, mu, sigma),
    color="blue",
    alpha=0.5,
    label=f"VaR (95% confidence)",
)

# Add labels and legend
plt.xlabel("Value")
plt.ylabel("Probability Density")
plt.legend()

# Show the plot
plt.show()
```




### 6.2.1 Simple VaR Calculations

**Example 1** Suppose that the gain from a portfolio during six months is normally distributed with a mean of \$2 million and a standard deviation of \$10 million. How would you calculate the VaR for the portfolio with a time horizon of six months and confidence level of 99%, given the properties of the normal distribution?

**Example 2** Suppose that for a one-year project all outcomes between a loss of \$50 million and a gain of \$50 million are considered equally likely, resulting in a uniform distribution extending from –\$50 million to +\$50 million. How would you determine the VaR with a one-year time horizon and a 99% confidence level for this project?

**Example 3** A one-year project has a 98% chance of leading to a gain of \$2 million, a 1.5% chance of leading to a loss of \$4 million, and a 0.5% chance of leading to a loss of \$10 million. How would you calculate the VaR with a confidence level of 99% and a one-year time horizon for this project, given the cumulative loss distribution?

## 6.3 Advantages and Limitations of VaR

### 6.3.1 Advantages of VaR

- **Simple concept:** VaR is easy to understand, making it accessible to decision-makers without technical backgrounds by asking the question, "How bad can things get?".

- **Easily communicated:** VaR consolidates a large amount of information into a single, significant number.
- **Provides a basis for risk comparison:** VaR allows for the comparison of risks across different asset classes, portfolios, and trading units.
- **Facilitates capital allocation decisions:** VaR provides a benchmark for capital allocation decisions across different trading units or portfolio positions.
- **Can be used for performance evaluation:** VaR can be used for risk-adjusted performance measurement, which provides a more comprehensive view of profitability.
- **Reliability can be verified:** VaR can be backtested, making it easier to check its reliability.
- **Widely accepted by regulators:** VaR is commonly used and encouraged by regulators like the SEC, making it a common sight in annual reports of financial firms.

### 6.3.2 Limitations of VaR

- **Subjectivity:** The estimation of VaR involves a lot of subjective decisions, such as choosing the VaR cutoff, time horizon, and estimation method.
- **Underestimating the frequency of extreme events:** VaR can underestimate the likelihood of extreme adverse events, often referred to as "left-tail events".
- **Failure to take into account liquidity:** VaR can be understated in the case of relatively illiquid assets, particularly during major market downturns.
- **Sensitivity to correlation risk:** VaR does not fully account for the fact that correlations among assets can rise significantly during times of market stress.
- **Vulnerability to trending or volatility regimes:** VaR can lead to an underestimation of losses during periods of low volatility and may not fully capture the risk in trending markets.
- **Misunderstanding the meaning of VaR:** VaR is not a worst-case scenario measure, and losses can exceed the VaR.
- **Oversimplification:** While VaR is easy to communicate, it can also oversimplify the risk picture.
- **Disregard of right-tail events:** VaR focuses on the left tail (losses), often ignoring the right tail (potential gains).
- VaR can be misleading when used to limit a trader's risks. A trader can appear to meet a set VaR limit while taking on unacceptable risks.
- For example, a trader can create a portfolio with a 99.1% chance of a daily loss under \$10 million, but a 0.9% chance of a \$500 million loss, meeting a bank's one-day 99% VaR limit of \$10 million.
- Such skewed probability distributions, where large losses are more likely, are not uncommon. Trading strategies like writing out-of-the-money options often yield good returns but carry a small chance of huge losses.
- Traders often seek high risks for high returns, and if they find ways to take high risks without violating risk limits, they will do so.

### 6.3.3 Extensions of VaR

- No single risk model can answer all questions a risk manager may have, hence VaR has numerous variations, each providing additional information.
- **Conditional VaR** (CVaR, Expected shortfall) estimates the average loss when VaR is exceeded.
- **Incremental VaR** (IVaR) measures the change in portfolio VaR due to a change in position size.
- **Marginal VaR** (MVaR) assesses the effect of a small change in position, useful for evaluating each asset's contribution to total VaR in diversified portfolios.
- IVaR and MVaR help evaluate a trade's potential effect before execution.
- **Relative VaR**, or ex ante tracking error, quantifies how much a portfolio's performance might deviate from its benchmark.

## 6.4 Expected Shortfall

- ES is also known as conditional value at risk, conditional tail expectation, or expected tail loss.
- Expected Shortfall (ES) is an alternative risk measure to Value at Risk (VaR) that provides better incentives for traders.
- It provides a more comprehensive view of risk than VaR by considering the magnitude of extreme losses.
- ES is particularly useful for understanding the risk of rare but significant losses.
- While VaR measures the maximum potential loss at a given confidence level, ES calculates the average loss beyond the VaR threshold.
- To calculate ES, it is necessary to calculate VaR first. ES is the expected loss over time horizon T, given that the loss exceeds the VaR.
- ES depends on two parameters: T (time horizon) and X (confidence level).
- An example: If $X = 99$, $T$ is 10 days, and the VaR is \$64 million, then ES is the average loss over 10 days when the loss is greater than \$64 million.
- Setting ES limits for traders reduces risky position-taking.
- ES has better properties than VaR as it always recognizes the benefits of diversification.
- However, ES is not as straightforward as VaR and is harder to understand.
- It is also more challenging to back-test a procedure for calculating ES compared to back-testing for VaR.
- Other risk measures can be defined with different weightings, such as the exponential spectral risk measure (ES use equal-weights).

## 6.5 Coherent Risk Measures

- Coherent risk measures provide a robust and consistent framework for assessing risk.
- They take into account the benefits of diversification and ensure that increasing the scale of a portfolio increases the risk measure proportionally.
- One example of a coherent risk measure is the Expected Shortfall (ES).
- Value at Risk (VaR) is not coherent as it does not always satisfy the subadditivity property.
- It is characterized by the following four axioms:

1. **Monotonicity**: If portfolio $A$ always has returns greater than or equal to portfolio $B$, the risk measure for $A$ should be less than or equal to the risk measure for $B$.
2. **Subadditivity**: The risk measure of the combined portfolio $A$ and $B$ should be less than or equal to the sum of the risk measures of $A$ and $B$ individually. This property ensures that diversification is rewarded.
3. **Positive Homogeneity**: If a portfolio's holdings are scaled up by a positive factor $\lambda$, the risk measure should also scale up by the same factor. Mathematically, if $A$ is a portfolio and $\lambda > 0$, then the risk measure of $\lambda A$ should be $\lambda$ times the risk measure of $A$.
4. **Translation Invariance**: If a risk-free asset with return $r$ is added to a portfolio, the risk measure should decrease by the amount $r$. Mathematically, if $A$ is a portfolio and $r$ is a risk-free return, the risk measure of $A + r$ should be the risk measure of $A$ minus $r$.

### 6.5.1 Subadditivity Example

- Each of two independent projects has a 2% probability of a \$10 million loss and a 98% probability of a \$1 million loss in one year.
- The one-year 97.5% **VaR (Value at Risk)** for each project is \$1 million.
- When combined into a portfolio, there is:
  - a 0.04% probability of a \$20 million loss,
  - 3.92% probability of an \$11 million loss, and
  - 96.04% probability of a \$2 million loss.
- The one-year 97.5% VaR for the portfolio is \$11 million, which is greater than the sum of the individual project VaRs (\$2 million) by \$9 million, **violating subadditivity**.
- The **ES (Expected Shortfall)** for a single project at 97.5% confidence level considers 2% at \$10 million loss and 0.5% at \$1 million loss. This results in an expected loss of \$8.2 million.
- When combined, the ES for the portfolio considers 0.04% at \$20 million loss and 2.46% at \$11 million loss, resulting in an expected loss of \$11.144 million.
- The sum of individual project ES (\$16.4 million) is greater than the portfolio ES (\$11.144 million), **satisfying the subadditivity condition**.

## 6.6 Choice of Parameters for VaR and ES

- For both Value at Risk (VaR) and Expected Shortfall (ES), users must choose two parameters: the time horizon and the confidence level.
- One assumption is that the change in the portfolio value at the time horizon is normally distributed, although this may not always be appropriate.
- When the loss in the portfolio value has a mean of $\mu$ and a standard deviation of $\sigma$, VaR is given by:

$$\text{VaR} = \mu + \sigma N^{-1}(X)$$

- where $X$ is the confidence level, and $N^{-1}(\cdot)$ is the inverse cumulative normal distribution.
- For short time horizons, $\mu$ is often assumed to be zero, making VaR for a particular confidence level proportional to $\sigma$.
- When the loss is assumed to be normally distributed with mean $\mu$ and standard deviation $\sigma$, ES with a confidence level of $X$ is given by:

$$\text{ES} = \mu + \sigma \frac{e^{-(Y^2/2)}}{\sqrt{2\pi}(1-X)}$$

- where $Y$ is the $X$th percentile point of the standard normal distribution.
- **Example:** The loss from a portfolio over a 10-day horizon is normally distributed with a mean of zero and a standard deviation of \$20 million. The 10-day 99% VaR and ES are:

$$\text{VaR} = 20N^{-1}(0.99) = \$46.5 \text{ million}$$

$$\text{ES} = 20\frac{e^{-(2.326^2/2)}}{\sqrt{2\pi} \times 0.01} = \$53.3 \text{ million}$$

### 6.6.1 The Time Horizon

- Depends on liquidity and trading frequency.
- Short time horizons for liquid, actively-traded positions; long horizons for less liquid portfolios.
- Portfolios with mixed liquidity can have tailored risk measures for each asset.
- Analysts often start with a 1-day time horizon.
- T-day VaR and T-day ES can be calculated from 1-day measures using the formulas:

$$\text{T-day VaR} = \text{1-day VaR} \times \sqrt{T}$$

$$\text{T-day ES} = \text{1-day ES} \times \sqrt{T}$$

- These formulas are exactly true when daily changes in portfolio value have independent identical normal distributions with mean zero.
- In other cases, these formulas are approximations.

### 6.6.2 Impact of Autocorrelation

- Changes in portfolio value aren't always independent.
- Higher autocorrelation of portfolio returns means higher underestimation of VaR and ES.
- It is possible to adjust standard deviation ($\sigma$) to reflect autocorrelation when calculating VaR and ES.
- The standard deviation of the sum of $T$ daily changes, considering autocorrelation, is given by:

$$\sigma\sqrt{T + 2(T-1)\rho^1 + 2(T-2)\rho^2 + ... + 2\rho^{T-1}}$$

### 6.6.3 Confidence Level

- The choice depends on factors such as maintaining credit ratings.
- A bank aiming for an AA rating might use a 99.98% confidence level for VaR to align with a 0.02% annual default chance.
- It's tough to estimate VaR directly with very high confidence levels.
- Extreme value theory can help to solve this issue.

### 6.6.4 Back-Testing

- Back-testing evaluates the performance of risk measure models, like VaR, using historical data.
- It is relatively simple for VaR models.
- The analysis focuses on "exceptions," where actual losses exceed the VaR prediction.
- A high or low percentage of exceptions may indicate model weaknesses.
- Statistical tests can assess whether a VaR model should be rejected based on exceptions.
- Regulators may increase market risk capital requirements if back-testing results over 250 days are unsatisfactory.

## 6.7 Additional Topics

### 6.7.1 Interest Rate Risk

- Financial institutions have exposures to different yield curves, requiring calculation of VaR and ES.
- Two approaches for this: considering each rate as a separate risk factor or using principal components analysis.
- Bank's net interest margin measures the excess of earned interest over paid interest; asset/liability management ensures constancy.
- Duration is key in interest rate markets, measuring portfolio sensitivity to shifts in the zero-coupon yield curve.
- Convexity of the portfolio adds precision for large parallel shifts, but not for nonparallel shifts.
- Several duration or delta measures quantify different ways the yield curve can change.
- Principal components analysis can be an alternative to multiple deltas, effectively hedging against standard shifts in yield curves.

### 6.7.2 Derivatives Risk

- A trader monitors Greek letters (such as delta, gamma, vega) to keep them within specified limits.
- Delta ($\Delta$) is the rate of change in portfolio value concerning the underlying asset's price; delta hedging aims to create a delta-neutral position.
- Hedging involves taking a position of $-\Delta$ in the underlying asset, and complex portfolios require periodic rebalancing.
- After achieving delta neutrality, traders often look at gamma (curvature of the relationship) and vega (rate of change with respect to volatility).
- Gamma and vega can be altered by trading options on the asset.
- Derivatives traders typically rebalance daily to maintain delta neutrality; maintaining gamma and vega neutrality regularly is usually infeasible.
- Corrective action or trading curtailment may occur if gamma and vega measures get too large.

### 6.7.3 Scenario Analysis and Stress Testing

- Stress testing is crucial in risk management, considering impacts of extreme scenarios usually ignored by VaR or ES analysis.
- It allows understanding and mitigation of risks and can uncover potential systemic problems.
- Scenarios generation methods include:

  - Examining extreme movements in one market variable.
  - Using historical market variables from extreme shock periods.
  - A committee of senior management and economists generating plausible extremes.
  - Reverse stress testing using algorithms to find large loss scenarios.

- Scenarios should consider knock-on effects; market turmoil may lead to significant consequences like flight to quality or liquidity shortage.
- Regulators in the US, EU, and UK develop stress scenarios for large financial institutions, possibly leading to capital raising requirements.
- Integrating stress testing with VaR through subjective probabilities is an idea not widely adopted by financial institutions or regulators.

## 6.8 Practice Questions and Problems

1. What is the difference between expected shortfall and VaR? What is the theoretical advantage of expected shortfall over VaR?
2. A fund manager announces that the fund's one-month 95% VaR is 6% of the size of the portfolio being managed. You have an investment of $100,000 in the fund. How do you interpret the portfolio manager's announcement?
3. A fund manager announces that the fund's one-month 95% expected shortfall is 6% of the size of the portfolio being managed. You have an investment of $100,000 in the fund. How do you interpret the portfolio manager's announcement?
4. Suppose that each of two investments has a 0.9% chance of a loss of $10 million and a 99.1% chance of a loss of $1 million. The investments are independent of each other.

   - What is the VaR for one of the investments when the confidence level is 99%? [1 mil.]
   - What is the expected shortfall for one of the investments when the confidence level is 99%? [9.1 mil]
   - What is the VaR for a portfolio consisting of the two investments when the confidence level is 99%? [11 mil.]
   - What is the expected shortfall for a portfolio consisting of the two investments when the confidence level is 99%? [11.07 mil.]
   - Show that in this example VaR does not satisfy the subadditivity condition, whereas expected shortfall does. [18.2 > 11.07]

5. Suppose that the change in the value of a portfolio over a one-day time period is normal with a mean of zero and a standard deviation of $2 million; what is (a) the one-day 97.5% VaR, (b) the five-day 97.5% VaR, and (c) the five-day 99% VaR? [a) 3.92 mil., b) 8.77 mil., c) 10.4 mil.]
6. What difference does it make to your answer to Problem 5 if there is first-order daily autocorrelation with a correlation parameter equal to 0.16? [a) the same, b) 9.96 mil., c) 11.82 mil.]

7. The change in the value of a portfolio in one month is normally distributed with a mean of zero and a standard deviation of \$2 million. Calculate the VaR and ES for a confidence level of 98% and a time horizon of three months. [VaR = 7.11 mil., ES = 8.39 mil.]
8. Suppose that each of two investments has a 4% chance of a loss of \$10 million, a 2% chance of a loss of \$1 million, and a 94% chance of a profit of \$1 million. They are independent of each other.

   - What is the VaR for one of the investments when the confidence level is 95%? [-1 mil.]
   - What is the expected shortfall when the confidence level is 95%? [8.2 mil.]
   - What is the VaR for a portfolio consisting of the two investments when the confidence level is 95%? [-9 mil.]
   - What is the expected shortfall for a portfolio consisting of the two investments when the confidence level is 95%? [-9.96 mil.]
   - Show that, in this example, VaR does not satisfy the subadditivity condition, whereas expected shortfall does. [-9 mil. > -2 mil.]

9. Suppose the first-order autocorrelation for daily changes in the value of a portfolio is 0.12. The 10-day VaR, calculated by multiplying the one-day VaR by $\sqrt{10}$, is \$2 million. What is a better estimate of the VaR that takes account of autocorrelation?
10. The change in the value of a portfolio in three months is normally distributed with a mean of \$500,000 and a standard deviation of \$3 million. Calculate the VaR and ES for a confidence level of 99.5% and a time horizon of three months.

# 7 Reading Week

Tip

Take some rest and revise the first six topics!

# 8 Market Risk II

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 12 - Historical Simulation and Extreme Value Theory
  - Chapter 13 - Model-Building Approach
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 6 - Risk Management

**Learning outcomes:**

- Understand the concept of Value at Risk (VaR) and Expected Shortfall (ES) as a measure of portfolio risk, and the importance of different estimation methods.
- Understand the Historical Simulation method VaR and ES estimation.
- Recognize the benefits and limitations of using historical data to estimate future portfolio risk.
- Learn Parametric Approach to VaR and ES estimation.
- Understand the limitations of assuming a normal distribution of returns.
- Acquire knowledge of the Monte Carlo Simulation method.
- Understand the computational intensity and the assumptions involved in the Monte Carlo simulation method.
- Develop the ability to critically evaluate and compare the advantages and disadvantages of each of these methods in different investment contexts and scenarios.

## 8.1 Introduction

There are several estimation methods for calculating Value at Risk (VaR):

1. **Historical Simulation**: This method involves using historical data to simulate the potential changes in the value of a portfolio. VaR is then calculated based on the worst losses observed in the historical data for a given confidence level.

2. **Parametric (Analytical) Approach**: This method assumes that returns on the assets in a portfolio are normally distributed, and uses the mean and standard deviation of the returns to calculate VaR. It can be used for individual assets as well as portfolios, taking into account the correlations between assets.
3. **Monte Carlo Simulation**: This method uses random number generation to simulate thousands of possible price paths for the assets in a portfolio. VaR is then calculated based on the distribution of simulated portfolio losses.
4. **Extreme Value Theory (EVT)**: This method focuses on modeling the tail of the loss distribution, which can be useful for calculating VaR at very high confidence levels.

## 8.2 Historical Simulation

- The historical simulation method for VaR uses the current portfolio and revalues it using actual historical changes in key factors experienced during a lookback period.
- The process starts by decomposing the portfolio into risk factors and collecting historical returns of each factor over the lookback period.
- The current portfolio is repriced using returns from each day of the lookback period, and results are sorted from largest loss to greatest gain.
- To estimate a one-day VaR at a 5% confidence interval, a point is chosen on the distribution beyond which 5% of the outcomes result in larger losses.
- To calculate **expected shortfall** using historical simulation, we average the losses that are worse than VaR.

### 8.2.1 Historical Simulation Example

**General Procedure:**

1. Collect historical price data for the asset.
2. Calculate the returns or the percentage change for each period.
3. Sort these returns in ascending order.
4. Identify the return at the (100 - confidence level)% quantile.
5. Identify VaR and ES

- Consider the last 500 daily returns for `SPY` ETF. What is daily VaR and ES with 99% confidence level? Below is the worst 20 returns during selected time priod.

```
import numpy as np
import pandas as pd
import yfinance as yf

# Step 1: Download historical data
```

```python
data = yf.download("SPY", period="3y", progress=False)
data.index = data.index.date

# Step 2: Calculate the log returns
data["Log Returns"] = np.log(data["Adj Close"] / data["Adj Close"].shift(1))

# Remove any NA values that might have been resulted from the log return calculation
data = data.dropna()

# Keep only the last 500 observations
data = data.iloc[-500:].copy()

# Step 3: Sort these returns
sorted_returns = data["Log Returns"].sort_values()

print(pd.DataFrame(sorted_returns).head(10))
```

```
            Log Returns
2024-08-05    -0.029556
2022-10-07    -0.028290
2022-11-02    -0.025417
2022-12-15    -0.024766
2022-10-14    -0.023052
2024-07-24    -0.022923
2022-09-29    -0.021110
2022-11-09    -0.020817
2024-09-03    -0.020794
2023-02-21    -0.020265
```

- 1% from 500 observations is 5.
- VaR is the 5th worst loss.
- ES is the average of the 4 worst losses.

```python
print("The 1-day 99% VaR is:", round(sorted_returns[4], 6))
print("The 1-day 99% ES is:", round(sorted_returns[:4].round(6).mean(), 6))
```

```
The 1-day 99% VaR is: -0.023052
The 1-day 99% ES is: -0.027007
```

### 8.2.2 Stressed VaR and Stressed ES

- Historical simulation for calculating VaR and ES typically uses the most recent data, referred to as current VaR and current ES.
- Stressed VaR and ES, introduced by regulators, use data from a chosen 251-day period that's particularly stressful for the current portfolio.
- The chosen period data creates 250 scenarios, each based on market variable changes between consecutive days.
- One-day 99% stressed VaR is the midpoint loss between the second and third worst scenarios, while one-day 99% ES is the average of the two worst losses.

### 8.2.3 Accuracy of VaR

- The historical simulation approach to Value-at-Risk (VaR) estimates the distribution of portfolio changes from a finite number of observations, making it subject to error.
- The standard error of the estimate for the $q$-percentile of the distribution, estimated as $x$, is given by the formula:

$$\text{Standard Error} = \frac{1}{f(x)}\sqrt{\frac{(1-q)q}{n}}$$

- where $n$ is the number of observations and $f(x)$ is an estimate of the probability density function of the loss evaluated at $x$.
- Standard error of a VaR estimated using historical simulation is typically high, decreasing with lower VaR confidence level or larger sample size.
- Losses typically have heavier tails than the normal distribution, so a Pareto distribution for $f(x)$ may provide better standard error estimates.

#### 8.2.3.1 Example

1. Suppose we want to estimate the 99th percentile of a loss distribution from 500 observations, so $n = 500$ and $q = 0.99$.
2. We assume a normal distribution with a mean of 0 and a standard deviation of \$10 million.
3. Using Excel, the 99th percentile is $\text{NORMINV}(0.99, 0, 10) = 23.26$.
4. The value of $f(x)$ is $\text{NORMDIST}(23.26, 0, 10, \text{FALSE}) = 0.0027$.
5. The standard error of the estimate is:

$$\frac{1}{0.0027} \times \sqrt{\frac{0.01 \times 0.99}{500}} = 1.67$$

6. If the estimate of the 99th percentile using historical simulation is \$25 million, a 95% confidence interval is from $25 - 1.96 \times 1.67$ to $25 + 1.96 \times 1.67$, or from \$21.7 million to \$28.3 million.

### 8.2.4 Extensions and Computational Issues

- VaR and ES are calculated using recent historical data, assuming that this is a good guide to future behavior.
- The basic historical simulation approach gives equal weight to each day in the past.
- **Weighting of Observations**: Recent observations could be weighted more heavily, reflecting current volatilities and macroeconomic conditions. Exponential weighting, where weights decline exponentially, is a natural choice.
- **Volatility Scaling for Market Variables**: It is possible incorporating volatility estimates into historical simulation. A variation is to use the Exponentially Weighted Moving Average (EWMA) to monitor the standard deviation of the simulated losses.
- **Bootstrap Method**: Creating many new similar data sets by sampling observations with replacement.
- Historical simulation is a method used to value the entire portfolio of a financial institution, often involving multiple simulations (e.g., 500 times).
- This approach can be computationally intensive, especially when certain instruments in the portfolio are valued using Monte Carlo simulation, leading to a "simulation within a simulation" problem.
- To reduce computation time, financial institutions sometimes use an approximation method based on the Taylor series expansion.

### 8.2.5 Extreme Value Theory

- Extreme Value Theory (EVT) is a scientific method used to estimate the tails of a distribution.
- EVT is based on the power law and can be used to improve Value at Risk (VaR) or Expected Shortfall (ES) estimates, particularly for very high confidence levels.
- EVT smooths and extrapolates the tails of an empirical distribution, providing better estimates of VaR and ES that reflect the entire shape of the tail.
- EVT can be applied to the probability distribution of portfolio daily changes derived from historical simulation, avoiding calculation uncertainties.
- Even with limited data (e.g., 500 days), EVT can provide estimates of VaR or ES for confidence levels as high as 99.9%.

### 8.2.6 Advantages of Historical Simulation Method

- The method is easy to understand and implement, as it relies on actual historical data to simulate possible future scenarios.
- Unlike other methods, historical simulation does not assume a specific parametric distribution (like normal distribution) for the returns, making it more robust in the presence of non-normal returns.
- By using actual historical data, this method naturally incorporates periods of high volatility, which can provide a more realistic estimate of VaR during turbulent times.
- The method is based on empirical evidence rather than theoretical assumptions, making it more reliable for estimating the risk of extreme events.
- Historical simulation can be easily modified to consider different time horizons, confidence levels, or other variations to suit specific risk management needs.
- As new historical data becomes available, it can be easily incorporated into the simulation, allowing the method to adapt to changing market conditions.
- It can accomodate options, unlike the parametric method.

## 8.3 Parametric Approach

- The parametric method, also known as the analytical, variance-covariance method or model-building approach, is an alternative to historical simulation for estimating risk measures like VaR and ES.
- The method assumes normal distribution for return distributions of portfolio risk factors.
- The normal distribution requires two parameters, expected value and standard deviation, to describe the distribution.
- Historical data is used to estimate these parameters, with adjustments to make them consistent with long-run values.
- It's ideal when the portfolio's value change is linearly dependent on market variable changes, and if these changes are assumed to be multivariate normal, it's computationally faster than historical simulation.
- This approach extends Harry Markowitz's portfolio theory.
- It is not suitable for portfolios with non-linear derivatives.
- Other distributions with additional parameters can be used, but they add complexity to VaR estimation.
- It is also possible to calculate Stressed Measures by selecting appropriate historical data for estimating volatilities and correlations.

### 8.3.1 The Basic Methodology

- Assuming a normal distribution of returns, VaR and ES are calculated using the equations from the previous chapter:

$$\text{VaR} = \mu + \sigma \times \sqrt{T} \times N^{-1}(X)$$

$$\text{ES} = \mu + \sigma \times \sqrt{T} \times \frac{e^{-(Y^2/2)}}{\sqrt{2\pi}(1-X)}$$

- Where:
  - $\mu$ is the mean of losses, usually assumed as zero.
  - $\sigma$ is the standard deviation of losses.
  - $X$ is the confidence level.
  - $N^{-1}(\cdot)$ is the inverse cumulative normal distribution.
  - $Y$ is the $X$th percentile point of the standard normal distribution.
  - $T$ is the time horizon in days, for longer time horizons it is better to use appriate return frequency directly.
- Standard deviation of multi-asset portfolio is calculated as follows:

$$\sigma_P = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} \rho_{ij}\delta_i\delta_j\sigma_i\sigma_j}$$

- This equation can also be written as:

$$\sigma_P = \sqrt{\sum_{i=1}^{n} \delta^2\sigma^2 + \sum_{i\neq j} \rho_{ij}\delta_i\delta_j\sigma_i\sigma_j}$$

#### 8.3.1.1 Asset A

- Consider one-share portfolio of asset A valued at \$10 million.
- Time horizon of 10 days and a 99% confidence level for VaR.
- Daily volatility is 2% (\$200,0000).
- The expected change in a market variable over the considered time period is assumed to be zero in the model-building approach.
- Change in value is assumed to be normally distributed, with a standard deviation of \$200,000 and a mean of zero.
- One-day 99% VaR is $2.326 \times \$200,000 = \$465,300$.
- 10-day 99% VaR is $2.326 \times \$200,000 \times \sqrt{10} = \$1,471,300$.

- 10-day 99% ES is:

$$\text{ES} = \$200,000 \times \sqrt{10} \times \frac{e^{-(2.326^2/2)}}{\sqrt{2\pi}(1-0.99)} = \$1,686,994$$

#### 8.3.1.2 Asset B

- Consider one-share portfolio of asset B valued at \$5 million.
- Time horizon of 10 days and a 99% confidence level for VaR.
- Daily volatility is 1% (\$50,0000).
- The expected change in a market variable over the considered time period is assumed to be zero in the model-building approach.
- Change in value is assumed to be normally distributed, with a standard deviation of \$50,000 and a mean of zero.
- One-day 99% VaR is $2.326 \times \$50,000 = \$116,300$.
- 10-day 99% VaR is $2.326 \times \$50,000 \times \sqrt{10} = \$367,800$.
- 10-day 99% ES is:

$$\text{ES} = \$50,000 \times \sqrt{10} \times \frac{e^{-(2.326^2/2)}}{\sqrt{2\pi}(1-0.99)} = \$421,400$$

#### 8.3.1.3 Two-Asset Case

- Assume a correlation of 0.3 between asset A and asset B.
- The standard deviation of the sum of two variables $X$ and $Y$, given their standard deviations ($\sigma_X$ and $\sigma_Y$) and the coefficient of correlation between them ($\rho$):

$$\sigma_{X+Y} = \sqrt{\sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y}$$

- Using values from the previous example for asset A and asset B:

$$\sigma_{A+B} = \sqrt{200,000^2 + 50,000^2 + 2 \times 0.3 \times 200,000 \times 50,000} = \$220,227$$

- One-day 99% VaR is $\$2.326 \times \$220,227 = \$512,300$.
- 10-day 99% VaR is $\$512,300 \times \sqrt{10} = \$1,620,100$.

- 10-day 99% ES is:

$$\text{ES} = \$220,227 \times \sqrt{10} \times \frac{e^{-(2.326^2/2)}}{\sqrt{2\pi}(1-0.99)} = \$1,856,100$$

#### 8.3.1.4 Benefits of Diversification

- The 10-day 99% VaR for the portfolio of asset A is $1,471,300.
- The 10-day 99% VaR for the portfolio of asset B is $367,800.
- The 10-day 99% VaR for the portfolio of both assets is $1,620,100.

$$(\$1,471,300 + \$367,800) - \$1,620,100 = \$219,000$$

- The difference of $219,000 represents the benefits of diversification.
- If both assets were perfectly correlated, the VaR for the combined portfolio would equal the sum of the individual VaRs.
- Less than perfect correlation results in some risk being diversified away, true for both ES and VaR.

  For non-normal distributions, the VaR of two portfolios considered jointly can be greater than the sum of their VaRs. ES does not have this disadvantage.

### 8.3.2 Parametric Approach vs. Historical Simulation

- **Advantages of Parametric Approach**:
  - Results can be produced quickly.
  - Can be used easily with volatility and correlation updating procedures.
  - Volatility changes can be incorporated into the historical simulation approach, but in a more artificial way.
- **Disadvantages of Parametric Approach**:
  - The change in the value of the portfolio is linearly related to the proportional or actual changes in the risk factors.
  - The daily changes in the risk factors are assumed to have a multivariate normal distribution.
  - Daily changes in risk factors often have distributions that differ from normal distribution.
  - Parametric method is very sensitive to parameter estimates, particularly covariances (when assuming a portfolio).
  - It is difficult to use when dealing with portfolios containing options due to their non-linear pay-off structure.

## 8.4 Monte Carlo Simulation

- This method lets users develop their own assumptions about distribution statistics to generate random hypothetical returns for a portfolio.
- Widely used in various fields to estimate statistical distributions, especially for complex investments with uncertain variables or pricing complex options.
- Can handle portfolios with multiple assets, complex risk interactions, and doesn't require normal distribution assumptions.
- Users must decide the number of random values to generate, with more values increasing reliability but taking more time.
- Monte Carlo simulation also requires accounting for the correlation between variables, so it can't simply simulate two random variables independently.
- The method involves simulating the returns of risk factors, repricing portfolio exposures under a range of simulated returns, sorting these results, and then selecting a percentile as the VaR.
- The method's flexibility allows it to accommodate any distribution, making it more accurate for option positions or bonds with embedded options compared to parametric methods.

### 8.4.1 Numerical Example in Python

**General Approach:**

1. Define the probability distribution of the returns of the asset. This is typically done by fitting the historical returns of the asset to a certain probability distribution (normal, lognormal, etc.).
2. Generate a large number of random outcomes from the defined distribution. Each random outcome represents a possible future return of the asset.
3. Reprice the asset using each random outcome and record the result. This will give a distribution of possible future asset values.
4. Sort the recorded results from worst to best.
5. Calculate VaR and ES.

```
import yfinance as yf
import numpy as np
import pandas as pd

# Download stock data
ticker = "SPY"
stock_data = yf.download(ticker, period="3y", progress=False)
price_data = stock_data["Adj Close"]
```

```python
## Calculate historical volatility and mean return
daily_returns = price_data.pct_change().dropna()
sigma = daily_returns.std()
mu = daily_returns.mean()

# Monte Carlo simulation
simulations = 10000
initial_price = price_data[-1]

random_shocks = np.random.normal(0, 1, simulations)

# Simplified Geomeetric Brownian motion (log-normal distribution)
simulated_prices = initial_price * np.exp(
    (mu - 0.5 * sigma**2) + sigma * random_shocks
)

# Calculate 1% VaR and ES
confidence_level = 1  # for 1% VaR and ES
percentile = np.percentile(simulated_prices, confidence_level)
VaR = initial_price - percentile

below_VaR = simulated_prices[simulated_prices <= percentile]
ES = np.mean(initial_price - below_VaR)

print(f"1-day 99% VaR is ${VaR:.2f}")
print(f"1-day 99% ES is ${ES:.2f}")
```

```
1-day 99% VaR is $14.28
1-day 99% ES is $16.31
```

## 8.5 Practice Questions and Problems

### 8.5.1 Historical Simulation

1. What assumption is being made when VaR is calculated using the historical simulation approach and 500 days of data?
2. Suppose we estimate the one-day 95% VaR from 1,000 observations (in millions of dollars) as 5. By fitting a standard distribution to the observations, the probability density function of the loss distribution at the 95% point is estimated to be 0.01. What is the standard error of the VaR estimate?

3. Download the lates data for selected ETFs (e.g. S&P 500, NASDAQ, Bonds, Gold) and calculate:
    - (a) the 95%, 97%, and 99% one-day VaR,
    - (b) the 95%, 97%, and 99% one-day ES,
    - (c) repeat the (a) and (b) for 5-day VaR and ES
    - (d) repeat (a), (b), (c) for a stock-bond portfolio (choose your allocation)

### 8.5.2 Parametric Approach

1. Consider a position consisting of a $100,000 investment in asset A and a $100,000 investment in asset B. Assume that the daily volatilities of both assets are 1% and that the coefficient of correlation between their returns is 0.3. What are the five-day 97% VaR and ES for the portfolio?
2. A financial institution owns a portfolio of instruments dependent on the U.S. dollar–sterling exchange rate. The delta of the portfolio with respect to percentage changes in the exchange rate is 3.9. If the daily volatility of the exchange rate is 0.7% and a linear model is assumed, estimate the 10-day 99% VaR.
3. Explain why the linear model can provide only approximate estimates of VaR for a portfolio containing options.

### 8.5.3 Summary

Use excel file 'daily_returns.xlsx' containing daily percentage returns in % for selected ETFs and calculate/show:

1. Show distribution of daily returns for each asset for the whole period.
2. Use historical simulation approach:
    - Calculate 95% and 99% one-day VaR for the whole period.
    - Calculate 95% and 99% one-day ES for the whole period.
    - Convert VaR and ES to 10-day estimates.
    - Repeat the calculations for the last 500 days.
3. Use variance-covariance (parametric) approach:
    - Calculate 95% and 99% one-day VaR for the whole period.
    - Calculate 95% and 99% one-day ES for the whole period.
    - Convert VaR and ES to 10-day estimates.
    - Repeat the calculations for the last 500 days.
4. Create your own portfolio (combination of given assets):
    - Calculate 95% and 99% one-day VaR for the last 500 days.

- Calculate 95% and 99% one-day ES for the last 500 days.

5. ***Bonus.*** Try to find 95% stressed VaR and ES (251-days) for each asset.

# 9 Credit Risk

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 17 - Estimating Default Probabilities
  - Chapter 18 - xVAs
  - Chapter 19 - Credit Value at Risk

**Learning outcomes:**

- Understand the importance of estimating default probabilities and recovery rates in risk management.
- Recognize credit ratings and statistics from rating agencies as sources for estimating default rates.
- Learn about the credit default swap (CDS) market and its role in estimating credit spreads.
- Distinguish between real-world and risk-neutral default probabilities.
- Grasp the concept of recovery rate.
- Identify different methods for inferring default probabilities from credit spreads.
- Understand the concept of Credit Value Adjustment (CVA) and how it measures the expected future loss from counterparty default.
- Recognize the computational complexities in estimating CVA.
- Grasp the idea of wrong-way and right-way risk in CVA calculations.
- Learn about the risks associated with CVA.
- Understand Debit Value Adjustment (DVA) and its impact on the dealer.
- Understand the definition of Credit Value at Risk (VaR) and its similarity to market risk VaR.
- Familiarize with three key models for calculating Credit VaR.
- Learn methods to calculate VaR or Expected Shortfall for credit-sensitive instruments.

## 9.1 Estimating Default Probabilities

### 9.1.1 Credit Ratings

- **Rating Agencies (e.g., Moody's, S&P, Fitch):**

  - Provide credit ratings for corporate bonds, indicating creditworthiness.
  - Focus on stability, so ratings change infrequently to avoid frequent reversals.
  - May "rate through the cycle," ignoring short-term economic changes.

- **Other Companies (e.g., Moody's KMV, Kamakura):**
  - Provide estimates of default probabilities based on equity price and other variables.
  - More responsive to market information, not focusing on stability.

- **Internal Credit Ratings in Banks:**
  - Used for small and medium-sized companies without publicly traded debt.
  - Utilize profitability ratios and balance sheet ratios, like return on assets and debt to equity.
  - Estimate cash flow statements to analyze a company's ability to service its debt.

- **Altman's Z-Score:**
  - A statistical technique to predict default using accounting ratios.

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$$

  - **Variables:**
    * $X_1$: Working capital/Total assets
    * $X_2$: Retained earnings/Total assets
    * $X_3$: Earnings before interest and taxes/Total assets
    * $X_4$: Market value of equity/Book value of total liabilities
    * $X_5$: Sales/Total assets
  - **Interpretation:**
    * $Z > 3.0$: Unlikely to default
    * $2.7 < Z < 3.0$: On alert
    * $1.8 < Z < 2.7$: Good chance of default
    * $Z < 1.8$: Very high probability of financial embarrassment.

### 9.1.2 Historical Default Probabilities

- Ratings agencies produce data to show default experience over time based on credit ratings.
- For investment-grade bonds (BBB and above), default probability tends to increase over time. For poor-rated bonds, it may decrease.
- Both unconditional and conditional default probabilities can be computed from the data.
- Table presents the historical default probabilities for bonds of various credit ratings from 1981 to 2020.

- The table assists in understanding the risk profiles associated with bonds of different credit ratings, reflecting the perceived creditworthiness of issuers over time and allowing risk management in financial decisions.
- Investment-grade bonds have rating BBB and above.
- **Example**: A bond initially rated BBB has a 0.16% chance of defaulting in the first year, 0.43% in the second year, etc.
- **Yearly Probability Calculation**: The probability of defaulting in a specific year can be calculated by subtraction. E.g., for the second year, $0.43\% - 0.16\% = 0.27\%$.

| Time (years) | 1 | 2 | 3 | 4 | 5 | 7 | 10 | 15 |
|---|---|---|---|---|---|---|---|---|
| AAA | 0.00 | 0.03 | 0.13 | 0.24 | 0.34 | 0.51 | 0.70 | 0.90 |
| AA | 0.02 | 0.06 | 0.11 | 0.21 | 0.30 | 0.49 | 0.70 | 0.99 |
| A | 0.05 | 0.13 | 0.22 | 0.33 | 0.46 | 0.76 | 1.20 | 1.84 |
| BBB | 0.16 | 0.43 | 0.75 | 1.14 | 1.54 | 2.27 | 3.24 | 4.54 |
| BB | 0.63 | 1.93 | 3.46 | 4.99 | 6.43 | 8.89 | 11.64 | 14.65 |
| B | 3.34 | 7.80 | 11.75 | 14.89 | 17.35 | 20.99 | 24.62 | 28.24 |
| CCC/C | 28.30 | 38.33 | 43.42 | 46.36 | 48.58 | 50.75 | 52.76 | 54.76 |

- **Unconditional Default Probability**:
  - The probability of default during a specific year as seen at the initial time (time zero).
  - The unconditional default probability of a CCC/C bond during the third year is calculated as $43.42\% - 38.33\% = 5.09\%$.
- **Conditional Default Probability**:
  - The probability that a bond will default during a specific year, given that no earlier default has occurred.
  - **Calculation**:
    1. Probability of surviving until the end of year two: $100\% - 38.33\% = 61.67\%$.
    2. Conditional probability of default during the third year: $\frac{5.09\%}{61.67\%} = 8.25\%$.
  - The conditional default probability considers the probability of survival until a specific point, then calculates the probability of default from that point onwards.
- Conditional probabilities are often more relevant for ongoing risk management as they reflect updated information and context, whereas unconditional probabilities provide a broader, initial view of risk.

#### 9.1.2.1 Hazard Rates

- **Hazard Rate ( (t))** is a measure of default intensity over a short time period, conditional on no prior default.

- The hazard rate is a useful tool for understanding the risk of default over time, taking into consideration both conditional and unconditional probabilities.
- It accounts for the possibility of survival and provides insights into the company's creditworthiness.
- **Formula**:
  - Conditional Probability:

$$\frac{V(t) - V(t + \Delta t)}{V(t)} = \lambda(t)\Delta t$$

  - Cumulative Survival:

$$V(t) = e^{-\int_0^t \lambda(\tau)d\tau}$$

  - Cumulative Default Probability:

$$Q(t) = 1 - e^{-\int_0^t \lambda(\tau)d\tau} = 1 - e^{-\lambda(t)t}$$

- **Example**:
  - Constant hazard rate 1.5% per year.
  - Probabilities:
    * 1st Year: $1 - e^{-0.015\times 1} = 0.0149$
    * 2nd Year: $1 - e^{-0.015\times 2} = 0.0296$
    * 3rd Year: 0.0440
    * 4th Year: 0.0582
    * 5th Year: 0.0723
  - Unconditional Probability in 4th Year: $0.0582 - 0.0440 = 0.0142$
  - Conditional Probability in 4th Year: $\frac{0.0142}{1-0.0440} = 0.0149$

### 9.1.3 Recovery Rates

- Recovery rate for a bond refers to the price at which it trades approximately one month after default as a percentage of its principal amount.
- Creditors may agree to partial payments, or assets are sold to meet claims. Some claims have priorities over others.
- Recovery rates vary among different bond categories, ranging from 54.6% for first lien bonds to 22.3% for junior subordinated bonds.
- The average recovery rate is negatively related to the default rate.
- Higher recovery rates during good economic conditions, and lower during poor conditions.
- A bad year for defaults can lead to both higher defaults and lower recovery rates, intensifying the negative impact for lenders.

### 9.1.4 Credit Default Swaps

- A CDS is a financial contract that functions as a form of insurance against the risk of credit default.
- Involves two parties - the buyer pays periodic premiums to the seller (often quarterly), and the seller agrees to compensate the buyer if a specified credit event, such as default, occurs.
- Can be linked to corporate bonds, government bonds, or other credit instruments.
- Provides the buyer with protection against the risk that the issuer of the underlying credit instrument might default.
- Pricing is determined by factors such as the creditworthiness of the reference entity, market conditions, and the maturity of the swap.
- Plays a role in price discovery for credit risk and can influence overall market liquidity and stability.

#### 9.1.4.1 CDS Spread

- The CDS spread is the annual amount, usually expressed in basis points (bps), that the protection buyer pays to the protection seller over the length of the contract.
- Typically calculated as a percentage of the notional amount of the underlying credit instrument paid per year.
- A wider spread often indicates a higher perceived risk of default for the underlying credit entity, whereas a narrower spread suggests a lower perceived risk.
- **Example**: If the CDS spread for a particular corporate bond is 200 bps, the buyer would pay the seller 2% of the notional amount annually for the protection.
- The CDS spread serves as a valuable indicator of credit risk and market sentiment for the referenced credit entity.
- Credit indices are benchmarks that track the performance of a basket of credit instruments such as bonds or credit default swaps.
- They include a diversified portfolio of credit instruments, often representing various sectors, credit ratings, or geographic regions.
- Used to gauge the overall health and trends in the credit market, providing insights into credit risk, liquidity, and investor sentiment.
- **Examples**: Popular credit indices include the Markit iTraxx in Europe, and the CDX in North America, tracking credit default swap spreads across various categories of issuers.
- They often enhance liquidity in the credit market by creating standardized products that can be traded more easily.

### 9.1.5 Credit Spreads

- The credit spread is the extra rate of interest per annum required by investors for bearing a particular credit risk.
- It can be measured by CDS spreads or bond yield spreads (Bond yield - risk-free bond).
- A CDS can hedge a corporate bond, essentially converting it into a risk-free bond.
- N-year CDS spread should approximate the yield difference between an N-year corporate bond and an N-year risk-free bond.
- Liquidity constraints can limit the effectiveness of arbitrage between CDS and bonds.
- CDSs provide a direct estimate of credit spread.
- Risk-free rates are generally derived from Treasury bonds.
- Implied risk-free rates often found to be higher than Treasury rates.
- Asset swaps give direct estimates of excess bond yields over a floating reference rate.

#### 9.1.5.1 CDS–Bond Basis

- The excess of the CDS spread over the bond yield spread.

$$\text{CDS-Bond Basis} = \text{CDS Spread} - \text{Bond Yield Spread}$$

- Various factors like counterparty default risk and bond prices can cause deviations from zero.
- The CDS-bond basis was positive before the 2007 market turmoil but has since usually been small and negative.

### 9.1.6 Estimating Default Probabilities from Credit Spreads

- Objective is to estimate default probabilities using credit spreads like CDS spread, bond yield spread, or asset swap spread.

#### 9.1.6.1 Approximate Calculation

- **Scenario**: 5-year credit spread is 240 basis points; recovery rate is 40%.
- **Formula**:

$$\overline{\lambda} = \frac{s(T)}{1-R}$$

- where $s(T)$ is the credit spread for a maturity $T$, $R$ is the recovery rate, and $\overline{\lambda}$ is the average hazard rate between time zero and time $T$.

$$\overline{\lambda} = \frac{0.024}{1 - 0.4} = 4\%$$

- Works well for CDS, bond yield, and asset swap spreads.
- Assumes bond price close to par value.

#### 9.1.6.2 A More Exact Calculation

- Suitable for bonds with prices significantly deviating from par value.
- **General Idea**: Incorporates bond price, coupon rate, and yield for more precise estimates.
- **Calculation Steps**:
  - Compare prices of corporate and similar risk-free bonds.
  - Calculate expected loss from default.
  - Equate this to present value of future default losses.
  - Solve for default probability $Q$.
- **Advantages**: Higher degree of accuracy; accounts for complex market conditions.

### 9.1.7 Comparison of Default Probability Estimates

- Real world (historical data) vs. risk-neutral world (credit spreads).
- Hazard rates from credit spreads are generally higher than those from historical data.
- Differences in rates increase as credit quality declines.
- Multiple reasons for these differences:
  - Illiquidity of corporate bonds.
  - Systematic risk due to correlated defaults.
  - Non-systematic risk harder to diversify in bonds.
- When **valuing** credit derivatives or estimating the impact of default risk on the pricing of instruments, we should use **risk-neutral default probabilities**.
- When carrying out **scenario analyses** to calculate potential future losses from defaults, we should use **real-world default probabilities**.
  - Merton 1974, model for calculating default probabilities.
  - Equity of a company is a call option on the value of the assets of the company with a strike price equal to the repayment required on the debt.

## 9.2 xVAs

- Over-the-counter derivatives are cleared either through Central Clearing Parties (CCPs) or bilateral agreements.
- Regulatory changes mandate standardized derivatives to be cleared via CCPs.
- Bilaterally cleared transactions involve more collateral to mitigate credit risk.
- In both cases, there is very little credit risk.
- However, there are transactions to which regulations do not apply (e.g., OTC transactions with non-financial end users).
- Credit risk assessment for derivatives is more complicated than for loans.
- Unlike fixed loan exposures, the future exposure in derivatives is uncertain, and the potential loss could be theoretically infinite.
- Credit Value Adjustment (CVA) and Debit Value Adjustment (DVA) are risk management tools used in the valuation of counterparty risk.
- **xVAs**: Credit Value Adjustment (CVA) and Debit Value Adjustment (DVA) are key metrics among xVAs, which also include Funding Value Adjustment (FVA), Margin Value Adjustment (MVA), and Capital Value Adjustment (KVA). These quantify different aspects of credit risk costs or benefits in derivatives.
- The chapter concentrates on CVA as a primary tool for quantifying the dealer's credit risk in bilaterally cleared transactions.

### 9.2.1 Credit Exposure on Derivatives

- Example: one transaction, cleared bilaterally and no collateral is posted.
- **Three Credit Exposure Scenarios**:

  1. The derivative is a liability (e.g., short option): No credit exposure for the dealer.
  2. The derivative is an asset (e.g., long option): Dealer has credit exposure.
  3. The derivative can be either(e.g., forward, swap): Exposure varies over time.

- In multiple transactions between a dealer and a counterparty, exposures are netted.
- The dealer's net exposure is $\max(V, 0)$, where $V$ is the net market value of all transactions.
- In calculating its credit risk, the dealer is interested in both its credit exposure today and what that exposure might be in the future.

### 9.2.2 Credit Value Adjustment (CVA)

- Credit Value Adjustment (CVA) estimates expected loss from counterparty default in OTC derivatives.
- CVA impacts balance sheet and profits (increase CVA = decrease profits).
- Dealer calculates it for each counterparty.

#### 9.2.2.1 Calculations

- Time $T$ is the life of the longest derivative.
- Divide $[0, T]$ into intervals $[t_{i-1}, t_i]$.
- Variables:
  - $q_i$: Risk-neutral probability of default during $i$th interval.
  - $v_i$: Present value of net exposure at $i$th interval's midpoint.
  - $R$: Estimated recovery rate.
  - $s_i$: An estimate of the counterparty's credit spread for a maturity of $t_i$.
- CVA Equation, total expected loss from a default:

$$\text{CVA} = \sum_{i=1}^{n} (1 - R) q_i v_i$$

$$q_i = \exp\left(\frac{-s_{i-1} t_{i-1}}{1 - R}\right) - \exp\left(\frac{-s_i t_i}{1 - R}\right)$$

- $v_i$ usually calculated using Monte Carlo simulation.

#### 9.2.2.2 Collateral and Cure Periods

- Collateral agreements are integrated into the calculation of $v_i$.
- **Exposure,** $E$: Defined as $E = \max(V - C, 0)$, where $V$ is the market value of outstanding transactions and $C$ is the collateral posted at the time of default.
- **Cure Period**: Usually 10 days. Time lapse between a counterparty ceasing to post collateral and the transaction close-out. Affects the collateral value at the time of default.
- It must be taken into account in the Monte Carlo simulation.

#### 9.2.2.3 Peak Exposure and Downgrade Triggers

- **Peak Exposure**: - Calculated at the midpoint of each interval, representing a high percentile (e.g., 97.5%) of exposures in Monte Carlo simulations.
- It should use real-world, not risk-neutral world, simulations, but for short time periods the difference is small.
- **Downgrade Triggers**: Clauses in ISDA master agreements that require more collateral or termination if one party's credit rating falls below a specified level.
- Downgrade triggers can lead to significant liquidity crises if a company cannot meet collateral demands, potentially resulting in bankruptcy (e.g., if it is overused).
- They offer limited protection against drastic changes in credit rating.
- Examples include AIG, Enron, Xerox, and Ambac.

### 9.2.3 The Impact of a New Transaction

- Positive correlation with existing transactions increases CVA; negative correlation reduces it.
- Companies may get favorable quotes from dealers with whom they already have transactions due to CVA/netting considerations.
- Directly recalculating CVA for each new transaction is impractical due to computational intensity.
- Stored values from previous Monte Carlo simulations can be used for efficient incremental CVA calculation.

### 9.2.4 CVA Risk and Wrong-Way Risk

- CVA can be considered as a complex derivative.
- A rise in CVA decreases dealer's income, hence the need to hedge CVAs in the same way as other derivatives (using Greek letters).
- Computing Greek letters for CVA is computationally intensive.
- Incorporating risks from changes in credit spreads into market risk capital calculations is a required by regulators.
- **Wrong-way risk** arises when probability of default and exposure are positively correlated.
- **Right-way risk** is the opposite.
- Evaluating wrong-way or right-way risk requires insights into counterparty's business and their other transactions.
- Regulatory mechanism to adjust CVA for wrong-way risk using alpha multiplier (min. 1.2).

### 9.2.5 Debit Value Adjustment (DVA)

- Debit Value Adjustment (DVA) is the counterparty's expected cost due to the dealer's possible default.
- It is essentially the counterparty's CVA.
- DVA is a benefit to the dealer, offsetting the dealer's own CVA. This is in line with the zero-sum nature of derivatives.
- The book value of derivatives is calculated as $f_{nd} - CVA + DVA$, where $f_{nd}$ is the value of the derivatives assuming neither side will default.
- An increase in a dealer's credit spread leads to an increase in DVA and consequently boosts reported profits.
- Due to the potential for inflated profits, regulators have excluded DVA gains and losses from the definition of common equity in regulatory capital calculations.

## 9.3 Credit Value at Risk

- Value at Risk (VaR) is crucial for regulatory capital determination and internal credit risk management in both financial and nonfinancial firms.
- Credit risk VaR is the potential credit loss over a specific time, not to be exceeded at a given confidence level.
- It may consider default losses alone or include downgrade and credit spread changes.
- Banks use Credit VaR for both regulatory and economic capital.
- Longer horizon for credit risk VaR (often one year) compared to market risk VaR (usually one day).
- Credit risk VaR requires more elaborate models than market risk VaR, which often uses historical simulation.
- **Credit Correlation**: A key variable that accounts for the interdependence of defaults among different companies, important especially in economic downturns.
- Increased credit correlation during stressed conditions raises the risk for financial institutions with credit portfolios.

### 9.3.1 Ratings Transition Matrices

- Tools showing the likelihood of a company moving from one credit rating to another over a specified time, based on historical data.
- **Example**: A firm with an 'A' rating has a 92.61% chance of maintaining that rating after one year, according to S&P's one-year transition matrix.
- One-year matrices can be extended to longer periods by matrix multiplication (e.g., for five-year horizon take fifth power of the one-year matrix).
- The model assumes rating changes are independent across periods, though this is not entirely accurate.
- Recent downgrades make further downgrades more likely.
- One-Year Ratings Transition Matrix, 1981–2020 (Source: S&P)
- Probabilities Expressed as Percentages and Adjustments for Transition to the NR (not rated) Category

| Initial Rating | AAA | AA | A | BBB | BB | B | CCC/C | D |
|---|---|---|---|---|---|---|---|---|
| AAA | 89.85 | 9.35 | 0.55 | 0.05 | 0.11 | 0.03 | 0.05 | 0.00 |
| AA | 0.50 | 90.76 | 8.08 | 0.49 | 0.05 | 0.06 | 0.02 | 0.02 |
| A | 0.03 | 1.67 | 92.61 | 5.23 | 0.27 | 0.12 | 0.02 | 0.05 |
| BBB | 0.00 | 0.10 | 3.45 | 91.93 | 3.78 | 0.46 | 0.11 | 0.17 |
| BB | 0.01 | 0.03 | 0.12 | 5.03 | 86.00 | 7.51 | 0.61 | 0.70 |
| B | 0.00 | 0.02 | 0.08 | 0.17 | 5.18 | 85.09 | 5.66 | 3.81 |
| CCC/C | 0.00 | 0.00 | 0.12 | 0.20 | 0.65 | 14.72 | 50.90 | 33.42 |
| D | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 100.00 |

### 9.3.2 Methods for Calculating Credit VaR:

- **Vasicek's Model**:
  - Utilizes a one-factor Gaussian copula model.
  - Focuses on time to default for companies in a portfolio.
  - Commonly used for regulatory capital determination.
- **Credit Risk Plus**:
  - Proposed by Credit Suisse in 1997.
  - Similar to insurance industry approaches.
  - Calculates default losses based on probabilities of individual company defaults.
  - Incorporates uncertainty in default rates.
- **CreditMetrics**:
  - Proposed by JPMorgan in 1997.
  - Uses a Gaussian copula model with a ratings transition matrix.
  - Considers losses from both defaults and credit rating downgrades.
- **CreditMetrics vs. Credit Risk Plus**:
  - Should predict the same long-term loss distribution if based on identical assumptions.
  - Differences arise in the timing, not the magnitude, of predicted losses.
  - **Example Scenario**: Loan downgraded from A to BBB in Year 1, BBB to B in Year 2, and defaults in Year 3.
  - **Credit Risk Plus Approach**: Assumes no losses in Year 1 and Year 2; calculates the loss only in Year 3.
  - **CreditMetrics Approach**: Calculates separate revaluation losses in each of the three years.
  - The total losses calculated using the CreditMetrics approach should theoretically equal those calculated using the Credit Risk Plus approach.

### 9.3.3 Credit Spread Risk

- Value of credit-sensitive products highly dependent on credit spreads.
- VaR or ES calculated considering potential credit spread changes.
- **Historical Simulation Approach**:
  - Collect last 500 days (or other period) of credit spread changes.
  - Limitations: Assumes no probability of default, may lack quality data.
- **Alternative: CreditMetrics Approach**:
  - Utilizes 10-day rating transitions matrix and Monte Carlo simulation.
  - Samples from matrix and credit spread probability distribution.

## 9.4 Practice Questions and Problems

### 9.4.1 Estimating Default Probabilities

1. What is a rating?
2. What rating agencies exist and how many categories do they use for investment-grade bonds?
3. What are the disadvantages of ratings with respect to credit risk for financial institutions?
4. What are alternative methods for estimating credit risk?
5. What is meant by a hazard rate?
6. Calculate the average hazard rate for a B-rated company during the first year from the data in Table in the text.
7. Calculate the average hazard rate for a BB-rated company during the third year from the data in Table in the text.
8. A credit default swap requires a semiannual payment at the rate of 60 basis points per year. The principal is $300 million and the credit default swap is settled in cash. A default occurs after four years and two months, and the calculation agent estimates that the price of the cheapest deliverable bond is 40% of its face value shortly after the default. List the cash flows and their timing for the seller of the credit default swap.
9. Explain the two ways a credit default swap can be settled.
10. Explain the difference between the default probabilities calculated from historical default experience and those calculated from credit spreads.
11. The spread between the yield on a three-year corporate bond and the yield on a similar risk-free bond is 50 basis points. The recovery rate is 30%. Estimate the implied average hazard rate per year over the three-year period.
12. The spread between the yield on a five-year bond issued by a company and the yield on a similar risk-free bond is 80 basis points. Assuming a recovery rate of 40%, estimate the implied average hazard rate per year over the five-year period. If the spread is 70 basis points for a three-year bond, what do your results indicate about the implied average hazard rate in years 4 and 5?
13. How are recovery rates usually defined?
14. "The position of a buyer of a credit default swap is similar to the position of someone who is long a risk-free bond and short a corporate bond." Explain this statement.
15. Why is there a potential asymmetric information problem in credit default swaps?

### 9.4.2 xVAs

1. Explain why a new transaction by a bank with a counterparty can have the effect of increasing or reducing the bank's credit exposure to the counterparty.
2. A company offers to post its own equity as collateral. How would you respond?
3. Can the existence of default triggers increase default risk? Explain your answer.

4. Give two examples of when (a) wrong-way risk and (b) right-way risk can be expected to be observed.
5. Explain the term "cure period."
6. "Netting means that CVA cannot be calculated on a transaction-by-transaction basis." Explain this statement.
7. "DVA can improve the bottom line when a bank is experiencing financial difficulties." Explain why this is so.
8. What part of CVA risk is considered a component of market risk by regulators?

### 9.4.3 Credit Value at Risk

1. Explain the difference between Vasicek's model, the Credit Risk Plus model, and CreditMetrics as far as the following are concerned: (a) when a credit loss is recognized and (b) the way in which default correlation is modeled.
2. Explain what is meant by the constant level of risk assumption.
3. Use the transition matrix from the text to calculate a transition matrix for three years. What is the probability of a company rated AAA being AAA at the end of two years? What is the probability of it moving to AA?
4. How can historical simulation be used to calculate a one-day 99% VaR for the credit risk of bonds in the trading book? What are the disadvantages?

# 10 Operational Risk, Liquidity Risk, and Model Risk

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 20 - Operational Risk
  - Chapter 21 - Liquidity Risk
  - Chapter 22 - Model Risk Management

**Learning outcomes:**

- Recognize the evolution of capital charge for operational risk.
- Comprehend the need for key risk indicators in operational risk management.
- Learn how to allocate operational risk capital to business units.
- Be aware of insurance options for mitigating operational risk and their limitations.
- Acknowledge the evolving nature of operational risk management procedures.
- Understand two types of liquidity risk: trading and funding.
- Recognize factors affecting liquidity trading risk.
- Know how to measure liquidity using bid–ask spread methods.
- Understand the importance of forecasting cash needs in normal and stressed conditions.
- Identify "liquidity black holes" and the need for diverse trading strategies.
- Understand model risk as a form of operational risk.
- Identify the unique challenges posed by financial models compared to physical science models.

## 10.1 Operational Risk

- Regulators introduced operational risk capital requirements in 1999.
- Cyber risk is a growing form of operational risk, necessitating advanced security measures and ongoing education.
- Operational risk is harder to quantify and manage compared to credit or market risk.
- Unlike insurance companies that can raise premiums to cover large losses, banks do not have this option as they operate in a competitive environment.

### 10.1.1 Defining Operational Risk

- Basel Committee's 2001 definition: "The risk of loss resulting from inadequate or failed internal processes, people, and systems or from external events."
- Legal risk is included, but reputation and strategic decision risks are excluded.
- Operational risks affect bank costs and revenue, and can interact with credit and market risks.
- E.g., natural disasters, political and regulatory risk, security breaches, and so on.

### 10.1.2 Types of Operational Risk

- **Internal Fraud**: Intentional acts to defraud or circumvent law by internal parties. Examples include insider trading and rogue trading.
- **External Fraud**: Fraudulent acts by third parties such as robbery and computer hacking.
- **Employment Practices and Workplace Safety**: Violations related to employment laws, health and safety, including discrimination claims.
- **Clients, Products, and Business Practices**: Failures to meet professional obligations or misuse of customer information. Examples include fiduciary breaches and money laundering.
- **Damage to Physical Assets**: Losses from natural disasters or vandalism. Examples include fires and earthquakes.
- **Business Disruption and System Failures**: Issues like hardware failures and utility outages disrupting business.
- **Execution, Delivery, and Process Management**: Transaction failures, documentation issues, and vendor disputes.

### 10.1.3 Loss Severity and Loss Frequency

- Loss frequency and loss severity are two fundamental distributions to estimate potential size of operational risks.
- **Loss Frequency**: Distribution of the number of losses observed in one year. Commonly modeled using the Poisson distribution.

$$P(n) = e^{-\lambda T} \frac{(\lambda T)^n}{n!}$$

- Where $\lambda$ is the average number of losses per year.
- **Loss Severity**: Distribution of the size of a loss, given that a loss occurs. Often modeled using the lognormal distribution.

- **Monte Carlo Simulation**: Used to combine loss frequency and severity to determine a total loss distribution.
- Banks often have limited internal data but may utilize data from other banks or vendors.
- Losses from other banks must be adjusted to reflect size of the institution.
- Publicly reported data, like that from vendors, is often skewed towards larger losses.
- Scenario analysis is used to assess operational risks that have not yet been observed.
- Committee estimates for loss frequency and severity are subjective.
- Scenarios can be developed to think actively about potential adverse events and formulate responses.

### 10.1.4 The Standardized Measurement Approach

- Defined by regulators in December 2017, see www.bis.org/bcbs/publ/d424.pdf.
- Implementation date: January 2023
- Its purpose was to standardize the calculation of capital for operational risk in banks.
- **Business Indicator (BI)** to measures the bank's size and therefore its potential operational risk exposure:
  - Interest Leases and Dividend Component (ILDC)
  - Services Component (SC)
  - Financial Component (FC)
  - **Formula**: $\text{BI} = \text{ILDC} + \text{SC} + \text{FC}$
- **Business Indicator Component (BIC)**:
  - Small Banks: $BIC = 0.12 \times BI$
  - BI between 1 and 30 billion euros: $BIC = 0.12 + 0.15 \times (BI - 1)$
  - BI above 30 billion euros**: $BIC = 4.47 + 0.18 \times (BI - 30)$
- **Internal Loss Multiplier (ILM)**:
  - **Loss Component (LC)** is equal to 15 times average annual operational risk losses over previous 10 years.

$$ILM = \ln\left[e - 1 + \left(\frac{LC}{BIC}\right)^{0.8}\right]$$

- **Operational Risk Capital**:
- For Small Banks (BI $<$ 1 billion euros): Equal to BIC.

- For Other Banks: Equal to $BIC \times ILM$.
- National Supervisors' Discretion: May set capital equal to BIC for all banks or use loss data for small banks.
- If $LC = BIC$, then $ILM = 1$ and required capital is BIC.
- Banks with operational risk losses higher than average will have $ILM > 1$.
- Banks with losses lower than average will have $ILM < 1$.

### 10.1.5 Preventing Operational Risk Losses

- Operational risk managers aim to prevent losses by studying failures at other banks.
- Managers explore causal relationships between decisions and operational risk to identify preventive measures, such as modifying employee qualifications or upgrading systems.
- Business unit managers identify operational risks, often using questionnaires and scorecards (risk control and self-assesment).
- Key risk indicators act as early warning systems, tracking metrics like:
  1. Staff turnover
  2. Number of failed transactions
  3. Number of positions filled by temps
  4. Ratio of supervisors to staff
  5. Number of open positions
  6. Percentage of staff who did not take 10 days consecutive leave in the last 12 months

### 10.1.6 Allocation of Operational Risk Capital

- Capital should be allocated in a way that motivates business units to manage operational risks effectively.
- Reduction in operational risk should lead to less allocated capital, thereby increasing return on capital for the unit.
- However, costs of reducing certain risks may outweigh the benefits.
- Support from senior management and board oversight is critical.
- Basel Committee on Banking Supervision recommends regular board reviews of risk management programs.

### 10.1.7 Insurance

- **Moral Hazard**: Insuring against operational risks may encourage lax behavior in risk management.

- **Adverse Selection**: Insurance companies may attract higher-risk clients, requiring rigorous assessment of bank controls.
- Deductibles, coinsurance, and policy limits are used to manage these insurance risks.

## 10.2 Liquidity Risk

- Elevated by the Global Financial Crisis of 2007.
- **Solvency** refers to a company having more assets than liabilities, so that the value of its equity is positive.
- **Liquidity** refers to the ability of a company to make cash payments as they become due.
- Financial institutions that are solvent can fail because of liquidity problems.
- **Liquidity Funding Risk**: Risk associated with inability to meet short-term obligations.
- **Liquidity Trading Risk**: Concerns bid-ask spreads and asset marketability.
- Regulatory liquidity requirements aim to ensure worst-case scenario survival.
- "Liquidity Black Holes": Extreme situations where market liquidity evaporates suddenly.

### 10.2.1 Liquidity Trading Risk

- Factors Affecting Liquidity:
  - Asset type (e.g., IBM shares vs. non-investment-grade bonds)
  - Trade size
  - Speed of liquidation
  - Economic environment
- Market makers may widen bid-ask spreads for large trades.
- Without market makers, implicit bid-ask spreads exist, varying from 0.01% to 10% of mid-market price.
- Market participants may anticipate large position unwinds, affecting liquidity.
- Importance of maintaining confidentiality to avoid predatory trading.
- Lack of transparency can lead to illiquid markets.
- During the Global Financial Crisis, complex derivatives became illiquid due to lack of understanding.
- "Liquidity Black Hole": liquidity evaporates when multiple institutions try to unwind similar positions simultaneously.

### 10.2.2 Measuring Market Liquidity

- **Bid-Ask Spread**: Measures liquidity either as a cash amount or as a proportion of the asset price.

- **Cash Bid-Ask Spread**: $p = \text{Ask price} - \text{Bid price}$
- **Proportional Bid-Ask Spread**: $s = \frac{\text{Ask price} - \text{Bid price}}{\text{Mid-market price}}$
- One measure of the liquidity of a book is how much it would cost to liquidate the book in normal market conditions within a certain time.
- **Cost of Liquidation**: Estimates the cost to liquidate an asset or book of assets.
- **Cost of Liquidation (Normal Market)**:

$$\text{Cost} = \sum_{i=1}^{n} \frac{s_i \alpha_i}{2}$$

- where $s_i$ is the proportional bid-ask spread and $\alpha_i$ is the value of the position.
- **Cost of Liquidation (Stressed Market)**:

$$\text{Cost} = \sum_{i=1}^{n} \frac{(\mu_i + \lambda \sigma_i)\alpha_i}{2}$$

- $\mu_i$ is the mean of the proportional bid-offer spread for the $i$th financial instrument.
- $\sigma_i$ is standard deviation of the proportional bid-offer spread for the $i$th financial instrument.
- $\lambda$ gives the required confidence level for the spread.
- Diversification doesn't necessarily reduce liquidity risk.
- Monitoring liquidity regularly is crucial.
- The measure is sensitive to the time period assumed for liquidation.
- Holding many small positions rather than a few large positions therefore tends to entail less liquidity risk.
- Setting limits to the size of any one position can therefore be one way of reducing liquidity trading risk.

#### 10.2.2.1 Additional Notes

- **Liquidity-Adjusted VaR**:
  - It incorporates both market risk and liquidity risk into a single metric (VaR plus Cost of liquidation).
  - It can be calculated using either normal market conditions or stressed market conditions.
- **Unwinding a Position Optimally**:
  - A dilemma exists between liquidating quickly (higher bid-ask spread, less market risk) and liquidating slowly (lower bid-ask spread, higher market risk).
  - Almgren and Chriss (2001) provide a mathematical approach to optimize this process.
- **Other Measures of Market Liquidity**:
  - Besides bid-ask spread, trading volume and price impact are also considered.
  - A metric proposed by Amihud relates daily return to daily dollar volume, suggesting that higher expected returns compensate for lower liquidity.
  - **Amihud's Measure**: Average of

$$\frac{\text{Absolute value of daily return}}{\text{Daily dollar volume}}$$

  - This is a simpler but similar measure to price impact.
  - Amihud (2002) demonstrates that as liquidity decreases, an asset's expected return increases, implying that investors are compensated for taking on liquidity risk.

### 10.2.3 Liquidity Funding Risk

- Focuses on a financial institution's capacity to meet cash needs.
- Institutions can be solvent yet fail due to liquidity issues.
- Causes of Liquidity Funding Problems:
  1. **Economic Liquidity Stresses**: Reluctance from investors to fund any credit-risk bearing activities.
  2. **Aggressive Funding Decisions**: Tendency to fund long-term needs with short-term instruments, causing liquidity mismatch.
  3. **Poor Financial Performance**: Leads to loss of confidence, deposit withdrawal, and funding difficulties.
- Often, all three causes may contribute to severe liquidity problems.
- **Managing Risk**:

1. Predicting Cash Needs: Some are predictable (e.g., bond coupons), others less so (e.g., deposit withdrawals).
2. Complex Financial Instruments: Increase unpredictability of cash needs due to factors like downgrade triggers and counterparty defaults.

- The key is to ensure cash needs can be met even in adverse scenarios.

#### 10.2.3.1 Sources of Liquidity

1. **Cash and Treasury Securities**: Highly liquid but expensive, limiting the amount held.
2. **Liquidating Trading Book**: Requires monitoring liquidity under stressed market conditions.
3. **Ability to Borrow**: Easy during unstressed periods but difficult during market stress.
4. **Retail and Wholesale Deposits**: Volatile and dependent on market-wide conditions.
5. **Securitization**: Used to be reliable but became risky after 2007.
6. **Central Bank Borrowing**: Lender of last resort, subject to collateral and haircut.

- **Additional Considerations**:
  1. **Stressed vs Normal Conditions**: Different liquidity sources function differently under stress.
  2. **Collateral and Haircut**: Important when borrowing from central banks.
  3. **Hedging Issues**: May result in liquidity problems due to margin requirements.

### 10.2.4 Reserve Requirements and Regulation

- A set percentage of a bank's deposits must be kept in reserve.
- Affects the money supply; varies by country.
- Basel III Liquidity Risk Requirements:
  - **LCR (Liquidity Coverage Ratio)**: High-quality liquid assets to net cash outflows in a 30-day stress period should be $\geq 100\%$.
  - **NSFR (Net Stable Funding Ratio)**: Amount of stable funding to required stable funding should be $\geq 100\%$.

### 10.2.5 Liquidity Black Holes

- A situation where market liquidity dries up due to everyone wanting to sell and no one wanting to buy, or vice versa.
- It is still possible to buy/sell assets but the bid-ask spread is huge.
- Negative Feedback Traders: Buy when prices fall, sell when prices rise.

- Positive Feedback Traders: Do the opposite, exacerbating price movements.
- Dominance of positive feedback traders leads to unstable, illiquid markets.
- Reasons for positive feedback trading:

  1. **Trend Trading**: Buy or sell based on asset price trends.
  2. **Stop-Loss Rules**: Automatic selling below certain price levels.
  3. **Dynamic Hedging**: Can reduce market liquidity when large short option positions are hedged.
  4. **Synthetic Options**: Commercial programs for synthetic put options can cause liquidity black holes.
  5. **Margins**: Big market moves lead to margin calls, further destabilizing prices.
  6. **Predatory Trading**: Knowing an investor must sell large quantities can drive prices further down.
  7. **LTCM Failure**: Example of how leveraging and margin calls can exacerbate liquidity issues.

- **Leveraging and Deleveraging**:
  - Banks awash with liquidity make credit easily available, leading to increased asset prices ("leveraging").
  - Deleveraging is the opposite: reduced liquidity in banks results in tightened credit and decreased asset prices.
  - Hedge funds are particularly affected by leveraging-deleveraging cycles.

- **Irrational Exuberance**:
  - Term coined by Alan Greenspan in 1996 to describe overvaluation of asset prices.
  - Often self-reinforcing due to high short-term profits.
  - Leads to liquidity black holes and market crashes.

- **The Impact of Regulation**:
  - Uniform regulatory environment can lead to synchronous behavior among banks, exacerbating market issues.
  - Different financial cycles necessitate different capital requirements; uniformity can be problematic.

- **The Importance of Diversity**:
  - Market participants often do not act independently, causing liquidity black holes.
  - Different types of financial institutions should be regulated differently to introduce market diversity.
  - Contrarian investment strategies may offer benefits but are hard to implement under short-term VaR-based risk management.

## 10.3 Model Risk Management

- Models increasingly used for various applications like credit decisions, liquidity management, and risk assessment.
- Advanced analytic techniques, including machine learning, automate tasks previously done by humans.
- Large institutions require a model risk management function for appropriate use and validation.
- Models are approximations, dependent on assumptions about the phenomena they represent.
- Importance of understanding these assumptions and their limitations.
- Model risk is considered a component of operational risk.
- Banks must develop, validate, and use models in accordance with regulatory systems.

### 10.3.1 Regulatory Guidance

- SR 11-7 by the U.S. Board of Governors of the Federal Reserve System outlines principles for effective model risk management.
- Model risk defined as adverse consequences from incorrect or misused models.
- Model risk occurs primarily for two reasons:
  1. The model may have fundamental errors (data, calculation)
  2. The model may be used incorrectly or inappropriately (limitations and assumptions of the model)
- **Model Development**:
  - Importance of thorough documentation for model continuity, regulatory scrutiny, and validation.
  - Testing should be rigorous, including out-of-sample tests and performance under extreme conditions.
- **Model Validation**:
  - Model validation groups provide an independent check.
  - Three core elements: conceptual soundness, ongoing monitoring, and outcomes analysis.
  - Importance of independence and corporate culture.
- **Vendor Models**:
  - Vendor models should undergo similar validation processes.
  - Require documentation and ongoing monitoring from vendors.
  - Contingency plans for vendor unavailability.

### 10.3.2 Models in Physics and Finance

- Physical models are precise because they describe fundamental laws of nature.
- Financial models are approximate, influenced by human behavior.
- Parameters like gravitational pull in physics are constant.
- Financial parameters like volatility change frequently.
- Financial models can result in significant errors and losses.
- Incorrect assumptions and lack of validation can be costly.
- To manage model risk, monitoring trading patterns and competitiveness in various products is crucial.

### 10.3.3 Models for Pricing Actively Traded Products

- A situation that is common in the over-the-counter market is one where a financial instrument that has to be valued is a standard product such as an option, but not exactly the same as one that trades in the market.
- Actively traded financial instruments have market-determined prices.
- Models like Black–Scholes–Merton are used for similar but non-standard products to ensure pricing consistency.
- Models are primarily interpolation tools; changes in model usually don't affect prices significantly.
- Model risk exists in extreme market conditions and must be validated.
- Hedging focuses on both within-model and outside-model variables.
- Portfolio value changes can be attributed to unhedged risks, model imperfections, and new trades.
- Risk is often diversified away in large financial institutions, minimizing the impact of inaccurate hedging.

### 10.3.4 Models for Less Actively Traded Products

- For actively traded products, calibrating models to market prices minimizes model risk.
- For less-traded products, model choice significantly affects valuation and hedging.
- Institutions often aim for relative, not absolute, valuation based on similar market-traded instruments.
- Multiple models should be used for non-standard products to understand model risk.
- Using multiple models can provide a range of possible prices.
- Institutions should charge at the higher end of this range and be conservative about recognizing profits.
- Accounting standards categorize valuations as Level 1, 2, or 3 based on reliability and need for modeling assumptions.

### 10.3.5 Successful Pricing Models and Common Mistakes

- The goal is to create models that aid in market understanding and decision-making without unnecessary complexity.
- **Successful Pricing Models**:
  1. Consistency: Aligns with observed market prices, useful for interpolation.
  2. Communication: Facilitates stable, easier-to-understand metrics like implied volatilities.
  3. Simplicity: Intuitive enough for traders to develop a nuanced understanding.
- **Model-Building Considerations**:
  1. Avoid Over-fitting: Complex models may capture data well but lack utility and might produce unrealistic outcomes.
  2. Caution on Over-parameterization: Adding too many parameters complicates the model and could fail during market regime changes.
  3. Preference for Simplicity: Simple models are often preferred as they are easier to understand and less of a 'black box.'

## 10.4 Practice Questions and Problems

### 10.4.1 Operational Risk

1. What risks are included by regulators in their definition of operational risks? What risks are not included?
2. Discuss how moral hazard and adverse selection are handled in car insurance.
3. When is a trading loss classified as a market risk and when is it classified as an operational risk?
4. Discuss whether there is (a) moral hazard and (b) adverse selection in life insurance contracts.
5. Give two examples of key risk indicators that might be monitored by a central operational risk management group within a bank.
6. Consider two banks, each with a BIC of 200 million euros. Bank A has suffered 10 operational risk losses of 20 million euros in the past 10 years. Bank B has suffered one 200 million euro loss in the past 10 years. What is the operational risk regulatory capital for each bank under SMA?

### 10.4.2 Liquidity Risk

1. An asset is quoted bid 50, offer 55. What does this mean? What is the proportional bid–offer spread?
2. Suppose that an investor has shorted shares worth $5,000 of Company A and bought shares worth $3,000 of Company B. The proportional bid–offer spread for Company A is 0.01 and the proportional bid–offer spread for Company B is 0.02. What does it cost the investor to unwind the portfolio?
3. Suppose that in Problem 2. the bid–offer spreads for the two companies are normally distributed. For Company A the bid–offer spread has a mean of 0.01 and a standard deviation of 0.01. For Company B the bid–offer spread has a mean of 0.02 and a standard deviation of 0.03. What is the cost of unwinding that the investor is 95% confident will not be exceeded?
4. Explain the difference between the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR).
5. Why is it risky to rely on wholesale deposits for funding?
6. What is meant by (a) positive feedback trading and (b) negative feedback trading? Which is liable to lead to liquidity problems?
7. What is meant by liquidity-adjusted VaR?
8. Explain how liquidity black holes occur. How can regulation lead to liquidity black holes?
9. Why is it beneficial to the liquidity of markets for traders to follow diverse trading strategies?

### 10.4.3 Model Risk Management

1. How is a financial institution likely to find that it is using a model different from its competitors for valuing a particular type of derivatives product?
2. Distinguish between within-model and outside-model hedging.
3. What is meant by P&L decomposition?
4. What is meant by over-fitting and over-parameterization in model building?
5. "For structured products, traders mark to model. They do not mark to market." Explain this remark.

# 11 Climate Risk and Enterprise Risk Management

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 23 - Climate Risk, ESG, and Sustainability
  - Chapter 24 - Enterprise Risk Management

**Learning outcomes:**

- Understand the growing importance of environmental risks in risk management practices.
- Recognize the terms ESG, sustainability, greenwashing.
- Understand the holistic approach of Enterprise Risk Management (ERM).
- Recognize the importance of integrating business units into an enterprise-wide risk management system.
- Distinguish between quantitative and qualitative aspects of risk appetite.
- Emphasize the significance of a strong risk culture, especially in medium- to long-term risk assessment.
- Be aware of the potential pitfalls in focusing solely on short-term profits, as evident from historical crises.

## 11.1 Climate Risk, ESG, and Sustainability

- Environmental risks pose a threat to Earth's habitability for future generations.
- Companies and financial institutions are under increasing societal pressure to address these risks.
- The prominence of environmental risks is expected to rise in the coming years.
- Climate risk is identified as the most significant environmental risk.

### 11.1.1 Climate Risk

- Climate risk is one of the most pressing environmental concern.
- Human activities emit tens of billions of tons of greenhouse gases annually.

- Greenhouse gases, primarily CO2, persist for hundreds of years, exacerbating global warming.
- Paris Agreement (2015) and Glasgow Climate Pact (2021) aim for net-zero emissions by

2050.

- Uncertainties exist in achieving these goals due to varying global policy responses.
- Difficulty in balancing current emissions with future environmental damage (trade-off between emissions today and damage in the future).
- Major sectors contributing to greenhouse gas emissions in the US are transportation, electricity, and industry.
- China, U.S., and India as major emitters, with per capita disparities. https://www.worldometers.info/co2-emissions/co2-emissions-by-country/
- Climate change causes rising sea levels due to ice melt and water expansion.
- Increase in extreme weather events like storms, heatwaves, and hurricanes.
- Extreme events already cost billions; e.g., Hurricane Ida cost $75 billion in 2021.

#### 11.1.1.1 Physical and Transition Risks

- Analysts differentiate between physical and transition climate risks.
- **Physical risks** involve costs from extreme climate events like floods and hurricanes.
- **Transition risks** are linked to policies for lower emissions, such as carbon taxes.
- Companies may need to shut down fossil fuel plants earlier due to transition risks.
- Lower emissions also present new business opportunities in green technologies.
- Estimated $100–150 trillion needed for economic transition over next 30 years.
- **Carbon tax** and **cap-and-trade** systems as tools to regulate emissions, varying adoption globally.

#### 11.1.1.2 Insurance Companies

- Climate change significantly impacts the insurance industry by affecting risk correlations.
- Traditional "law of large numbers" less applicable due to correlated climate-related disasters.
- Short-term contracts allow pricing adjustments.
- Life insurance less adaptable to climate risks.
- Resource allocation needed for better climate risk understanding and regulatory compliance.
- Companies increasingly reluctant to insure high-risk areas for natural disasters.

#### 11.1.1.3 Banks

- Banks actively quantify climate change effects through forums like PCAF (Partnership for Carbon Accounting Financials).
- Regulatory bodies launch climate stress tests for banks.
- Climate risks impact multiple traditional risk categories:
  - **Credit Risk**: Climate risk can impair debt service and collateral value.
  - **Market Risk**: Asset prices may decline due to climate effects.
  - **Liquidity Risk**: Market changes may limit access to stable funding.
  - **Operational Risk**: Legal and compliance risks may rise.
  - **Reputational Risk**: Public sentiment can heighten reputational risks.
- Mortgages vulnerable to long-term climate change impacts on real estate values.
- Severe climate events can reduce corporate profitability, raising credit risk.
- Mass migrations and agricultural changes expected, affecting economic stability.
- Sovereign debt increasingly risky, particularly in less-developed countries.
- Transition to low-carbon economy creates winners and losers, affecting lending.
- Reputational risks make banks cautious in lending to carbon emitters.

#### 11.1.1.4 Institutional Investors

- Increasing focus on incorporating climate risk in portfolios.
- Companies incentivized to align with emission targets to attract institutional investments.
- **Green Bonds**: Innovative financial instruments designated for climate-friendly investments, underwritten by investment banks.
- Multiple organizations aim to standardize climate-related financial reporting, but efforts are still in progress.
- Asset managers like BlackRock are proactive, while others like Berkshire Hathaway are resistant.
- **Political Counterpressures**: Legislation proposals and requests to ignore environmental factors in state pension investments.
- Despite political setbacks, focus on emissions and climate risk disclosure is growing.

### 11.1.2 ESG

- Shift from profit-centric approach to incorporating environmental, social, and governance (ESG) factors.
- **Social Aspect**: Addresses diversity, equity, inclusion, and worker well-being. Metrics still under development.

- **Governance Aspect**: Focuses on transparent accounting and accountability to shareholders. Avoids conflicts of interest and unethical behavior.
- Multiple ESG data providers exist, often yielding conflicting reports.
- Absence of regulated ESG reporting standards allows companies to shop for favorable scores, creating risk for investors.

### 11.1.3 Sustainability

- **Sustainability Defined**: Meets current needs without jeopardizing future generations.
- **Examples of Nonsustainability**: Greenhouse gas emissions, ocean pollution, soil and water contamination.
- Sustainability Accounting Standards Board (SASB) and Global Reporting Initiative provide guidelines.
- **Three Pillars**:
  - Environmental Sustainability: Resource conservation.
  - Economic Sustainability: Long-term stability with social and environmental care.
  - Social Sustainability: Supports healthy, livable communities for current and future generations.
- Sustainable investment funds aim to invest in companies with good ESG or sustainability practices.
- No significant outperformance compared to traditional funds, despite increased demand. https://onlinelibrary.wiley.com/doi/abs/10.1111/jofi.12841
- Some reports overstate benefits, suggest high returns with low risks contrary to finance theory.

### 11.1.4 Greenwashing

- **Greenwashing Concept**: Companies misleadingly claim ecological responsibility to attract consumers.
- **Examples**:
  - Innocent drinks: False claims about recyclable plastics.
  - Keurig: Misleading recycling claims led to a $3 million fine.
  - IKEA: Used illegally sourced wood despite good sustainability credentials.
  - Ryanair: Unsubstantiated claim of being Europe's "lowest emissions airline."
- **Financial Institutions**: Deutsche Bank's DWS exaggerated green credentials, leading to CEO's resignation.
- **Implications**: Considered a serious offense, impacting consumer trust and corporate credibility.

## 11.2 Enterprise Risk Management

- Risk managers must evaluate risks and adopt a holistic view through Enterprise Risk Management (ERM).
- Top-down and bottom-up approaches essential for effective risk management.
  - Top-down sets overall risk appetite.
  - Bottom-up assesses risks at the business unit level.
- Shift from compliance orientation to strategic orientation in risk management is noted.

### 11.2.1 Risk Appetite

- How much risk is the organization prepared to take in order to achieve its strategic objectives and business plans?
- Risk appetite is an essential part of Enterprise Risk Management (ERM).
- For fund managers, risk appetite is more straightforward to quantify using beta, expected return, and risk.
- For fund managers it is possible to use $\beta$ that is consistent with a particular VaR objective.

$$\beta = \frac{R - R_F}{R_M - R_F + N^{-1}(1-p)\sigma_M}$$

- $R$ is the return that has a probability $p$ of being exceeded in one year.
- $R_M$ is the expected market return and $R_F$ is the risk-free rate.
- Unfortunately, the available risk-return alternatives for a financial institution such as a bank or an insurance company cannot usually be quantified as easily as for a fund manager.
- VaR or expected shortfall often specify worst-case loss aspect of risk appetite.
- Other elements like reputational risk, credit rating risk, legal risks, etc., may be qualitative.
- Risk quantification can be:
  1. Bottom-Up: Calculated by business units, then aggregated.
  2. Top-Down: Set by the board, guiding risk limits for individual business lines.
- Risk appetite should address **risk Concentration**: Limit exposure to sectors, market variables, companies.

- Monitoring and adherence to set limits crucial for effectiveness.

### 11.2.2 Risk Culture

- It is all about how decisions are made.
- If the risk culture is good, all decisions are made in a disciplined way with a careful consideration of possible outcomes, weighing risks and rewards.
- A strong risk culture should align with an institution's risk appetite and long-term objectives.
- Short-term profits should not outweigh long-term risks.
- The role of compensation in influencing decisions is crucial and is being revised to mitigate short-term focus.
- E.g., spreading bonuses over years, clawback provisions for adverse outcomes.
- Regulatory guidelines (Financial Stability Board) can assist in establishing a robust risk culture.

### 11.2.3 Identifying Major Risks

- Core to ERM, similar to stress testing; consider historical and emerging trends.
- Regulatory Changes: Banks face tighter regulations; shift of activities to less-regulated shadow banks.
- Cognitive Biases:
  - Wishful Thinking: Confusing desires with likely outcomes.
  - Anchoring: Fixation on initial estimates, neglecting full range of outcomes.
  - Availability: Overemphasis on recent information.
  - Representativeness: Stereotyping situations based on past experiences.
  - Sunk Costs Bias: Ignoring sunk costs critical for rational decision-making.
- Underestimating tail risks due to cognitive biases; requires methods to improve estimations.
- Inclusion of diverse views mitigates bias, improves risk identification.

### 11.2.4 Strategic Risk Management

- Focuses on assessing assumptions and potential failures in a company's strategy.
  - Key Questions: Impact of competitors, customer reactions, and external events.
  - Risk is tied to the poor performance under 'wrong' scenarios.
- Courses of Action Once Risk Identified:
  - Avoid: Terminate the business or investment.

  - Reduce: Mitigate likelihood or impact of adverse events.
  - Adjust: Modify plans to lower risks.
  - Share/Insure: Transfer or share some risk.
  - Accept: No action taken.

- Choice of action varies by context but requires thorough consideration.

## 11.3 Practice Questions and Problems

### 11.3.1 Climate Risk, ESG, and Sustainability

1. Why does the burning of fossil fuels lead to temperature rises?
2. What is the goal that was agreed to in Paris in 2015?
3. What is the difference between a carbon tax and a cap and trade system?
4. What is the difference between physical risks and transition risks?
5. What are the risks of climate change to insurance companies?
6. What does ESG stand for?
7. What is greenwashing?
8. How can pension plans and mutual funds help solve the global warming problem?
9. Give examples of activities covered by ESG and sustainability that are not climate risk issues.

### 11.3.2 Enterprise Risk Management

1. Explain the difference between top-down and bottom-up approaches to risk management. Why are both necessary in ERM?
2. "The risk culture at a bank should lead it to consider the longer-term implications of decisions as well as short-term profits." Discuss.
3. What steps can a bank take to encourage employees to consider more than short-term profits?
4. What distinguishes enterprise risk management from more traditional approaches to risk management?
5. A fund's risk appetite is such that it wants to be 90% certain that it will not lose more than 20% in any one year. Using the performance of the S&P 500 between 1970 and 2016 (mean return 11.74%, standard deviation 18.86%), determine the beta the fund should have. Assume a risk-free rate of 3% per annum.
6. Give three examples of cognitive biases in finance.

# 12 Regulation, Economic Capital, and RAROC

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 25 - Basel I, Basel II, and Solvency II
  - Chapter 26 - Basel II.5, Basel III, and Other Post-Crisis Changes
  - Chapter 27 - Fundamental Review of the Trading Book
  - Chapter 28 - Economic Capital and RAROC

**Learning outcomes:**

- Trace bank capital requirements from the '80s to 2007.
- Note Basel Accord's shift to risk-weighted assets in 1988.
- Identify Basel II features: credit and operational risk.
- Understand Solvency II for insurance; relate to Basel II.
- Post-2007: grasp reasons for regulatory change.
- Uncover Basel II.5 and Basel III changes.
- Acknowledge Dodd–Frank's impact in the U.S.
- Understand "unintended consequences" in regulations.
- Note shift from VaR to ES in FRTB.
- Identify Basel's two approaches: standardized, internal.
- Grasp economic capital concept.
- Identify risks considered in economic capital.
- Understand enterprise-wide aggregation.
- Explain RAROC's role in performance, strategy.

## 12.1 Basel I, Basel II, and Solvency II

- The Basel Accord of 1988 initiated international standards for bank regulation.
- Regulatory evolution has seen new rules modify old ones while retaining certain approaches.
- Understanding history is crucial for grasping the current regulatory landscape.
- The chapter outlines Basel I, netting provisions, and the 1996 Amendment.
- Basel II, introduced around 2007, significantly reformed the regulatory framework.

- Solvency II, akin to Basel II, was adopted by the European Union in 2016 for insurance companies.

### 12.1.1 The Reasons for Regulating Banks

- The primary aim of bank regulation is to ensure sufficient capital against risks.
- Governments aim to minimize the probability of bank defaults for economic stability.
- Self-regulation by banks is not historically supported; formal regulation enhances capital and risk awareness.
- Deposit insurance programs, while protecting depositors, can inadvertently encourage risk-taking ("moral hazard").
- Systemic risk is a major concern, as a default by one large bank can trigger a financial system collapse.
- Governments face a dilemma between bailing out struggling institutions and risking systemic collapse.
- The 2008 market turmoil involved both bailouts and allowed failures, exemplified by Lehman Brothers, to send market signals.

### 12.1.2 Bank Regulation Pre-1988

- Pre-1988 bank regulation focused on setting minimum capital-to-asset ratios within individual countries.
- Regulatory stringency and definitions varied globally, leading to competitive imbalances.
- Major international loans and complex transactions, like OTC derivatives, exposed the inadequacy of existing capital requirements.
- Basel Committee formed in 1974 to standardize regulations.
- 1988 BIS Accord, later Basel I, was the Committee's first major output.

### 12.1.3 The 1988 BIS Accord

- First international risk-based standards for capital adequacy.
- Signed by all 12 Basel Committee members.
- Increased resources for risk measurement and management in banks.
- The Accord set the groundwork for modern risk management but faced criticisms for being too simplistic.
- **Cooke Ratio**: Focused on total risk-weighted assets (credit risk exposures).
- Categories of Credit Risk Exposures:

    1. **On-Balance Sheet Assets**: Excludes derivatives.
    2. **Off-Balance Sheet Items**: Excludes derivatives.
    3. **Over-the-Counter Derivatives**: Includes swaps, forwards, etc.

#### 12.1.3.1 Key Equations:

- **Risk-Weighted Assets (On-Balance)**:

$$\sum_{i=1}^{N} w_i L_i$$

- where $L_i$ is the principal amount and $w_i$ is its risk weight.
- **Credit Equivalent Amount (Off-Balance)**:
- Conversion factors applied to principal amounts of instruments like bankers' acceptances, guarantees, and loan commitments.
- **Credit Equivalent Amount (Derivatives)**:

$$\max(V, 0) + aL$$

- where $V$ is current value, $a$ is an add-on factor, and $L$ is the principal amount.
- **Total Risk-Weighted Assets**:

$$\sum_{i=1}^{N} w_i L_i + \sum_{j=1}^{M} w_j^* C_j$$

- where $C_j$ is the credit equivalent amount for off-balance-sheet items, and $w_j^*$ is the risk weight of the counterparty.

#### 12.1.3.2 Capital Requirements

- Required banks to maintain capital equal to at least 8% of total risk-weighted assets.
- **Tier 1 Capital**:
  - Includes equity and noncumulative perpetual preferred stock.
  - Goodwill is subtracted from equity.
  - Must constitute at least 50% of required capital (i.e., 4% of risk-weighted assets).
  - Must include 2% of risk-weighted assets as common equity.
- **Tier 2 Capital**:
  - Also known as Supplementary Capital.
  - Includes cumulative perpetual preferred stock, certain 99-year debentures, and subordinated debt with original life over five years.
  - Provides cushion for depositors in case of insolvency.
- Some countries and banks maintain capital ratios above the Basel Committee's minimum requirements.

### 12.1.4 Netting

- Standard in over-the-counter (OTC) derivatives markets.
- Netting clause stating that all transactions are a single transaction in case of default.
- Netting effectively reduces credit risk and modifies regulatory capital requirements.
- **Basel Accords**: Modified in 1995 to include netting.
- **Without Netting**: Loss to the bank = Sum of positive value transactions.
  - Example: Bank's transactions worth +\$24M, -\$17M, +\$8M. Loss = \$24M + \$8M = \$32M.
- **With Netting**: Loss to the bank = Sum of all transactions.
  - Example: Loss = \$24M - \$17M + \$8M = \$15M.
- **General Formula**:
  - Without Netting: $\sum_{i=1}^{N} \max(V_i, 0)$
  - With Netting: $\max(\sum_{i=1}^{N} V_i, 0)$

### 12.1.5 The 1996 Amendment

- The 1996 Amendment to the 1988 Basel Accord added capital charges for market risks in trading activities.
- It mandated fair value accounting, also known as marking to market, for a bank's trading book.
- Two approaches for calculating market risk: standardized and internal model-based.
- Value-at-risk (VaR) metrics and Specific Risk Charge (SRC) introduced.
- Total capital combines credit and market risk, with Tier options.
- Back-testing required; multiplier adjusted based on model performance.

### 12.1.6 Basel II

- Follow-up to the 1988 Basel Accord to address its weaknesses.
- **Overall Goal**: More sophisticated and comprehensive regulatory framework to manage risks in banking.
- Initial proposal in June 1999, revised in January 2001 and April 2003.
- Final rules published in June 2004, updated in November 2005.
- Implementation started in 2007.
- **U.S.**: Applies to internationally active banks; small banks follow Basel IA.
- **EU**: Applies to all banks and securities companies.

#### 12.1.6.1 Three Pillars of Basel II

1. **Minimum Capital Requirements**:

- More risk-sensitive approach to credit risk.
- Operational risk included.
- Market risk remains unchanged.
- 8% of risk-weighted assets (RWA) still the base requirement.

2. **Supervisory Review**:

- Emphasizes qualitative and quantitative risk management.
- Regulatory flexibility but seeks global consistency.
- Focuses on early intervention.

3. **Market Discipline**:

- Greater disclosure requirements.
- Aims to improve risk management through transparency.

$$\text{Total Capital} = 0.08 \times (\text{credit risk RWA} + \text{market risk RWA} + \text{operational risk RWA})$$

### 12.1.7 Solvency II

- Solvency II is the regulatory framework for insurance companies in the European Union, replacing Solvency I in 2016.
- Unlike Solvency I, which focused solely on underwriting risks, Solvency II also considers investment and operational risks.
- Structure resembles Basel II with three pillars:

  1. Pillar 1: Minimum Capital Requirements (MCR) and Solvency Capital Requirement (SCR) calculated. Breach triggers supervisory intervention.
  2. Pillar 2: Supervisory review process for internal capital adequacy.
  3. Pillar 3: Disclosure of risk management information to the market.

- SCR can be calculated using the standardized approach or internal models involving Value-at-Risk (VaR) with a one-year time horizon and 99.5% confidence limit.
- SCR includes charges for investment risk (market and credit), underwriting risk (life, non-life, health), and operational risk.
- Capital adequacy tests include various adverse event scenarios (e.g., 32% stock market decrease, 20% real estate price decrease).
- Internal models must pass three tests: statistical quality, calibration, and use.
- Types of capital:

1. Tier 1: Equity capital, retained earnings.
2. Tier 2: Liabilities subordinated to policyholders, available in wind-down scenarios.
3. Tier 3: Liabilities subordinated to policyholders, not meeting Tier 2 criteria.

- Capital composition rules are specified for Tier 1, Tier 1 + Tier 2, and Tier 1 + Tier 2 + Tier 3.

## 12.2 Basel II.5, Basel III, and Other Post-Crisis Changes

- Basel II's implementation roughly coincided with the onset of the Global Financial Crisis.
- Critics argue Basel II encouraged self-regulation, allowing banks to use their own model inputs for regulatory capital.
- Basel II may not be solely to blame:
  - The origins of the crisis predate Basel II.
  - U.S. was late in implementing Basel II, affecting capital levels.
- Basel II.5 was introduced post-crisis to amend market risk capital calculations.
  - Implementation date: December 31, 2011.
- Basel III was a more comprehensive post-crisis regulatory overhaul.
  - First set of regulations: December 2010.
  - Finalized: December 2017.
  - Implementation phased over several years.
- Other post-2008 regulations:
  - Dodd–Frank Act in the U.S., signed July 21, 2010.
  - New rules in the European Union and the United Kingdom.

### 12.2.1 Basel II.5

- Implemented on December 31, 2011, it introduced changes to the calculation of market risk capital.
- **Three Main Changes**:
  1. Stressed Value-at-Risk (VaR)
  2. Incremental Risk Charge (IRC)
  3. Comprehensive Risk Measure (CRM)
- **Impact**: Significantly increased the capital large banks need to hold for market risk.

#### 12.2.1.1 Details:

- **Stressed VaR**:
  - Originally based on 10-day, 99% VaR.
  - Introduces a stressed VaR, based on 250-day stressed market conditions.

    $$\text{Total Capital Charge} = \max(VaR_{t-1}, m_c \times VaR_{\text{avg}}) + \max(sVaR_{t-1}, m_s \times sVaR_{\text{avg}})$$

  - $VaR_{t-1}$ and $sVaR_{t-1}$ are the Value-at-Risk and stressed Value-at-Risk calculated on the previous day with a 10-day time horizon and a 99% confidence level.
  - $VaR_{\text{avg}}$ and $sVaR_{\text{avg}}$ are the averages of Value-at-Risk and stressed Value-at-Risk calculated over the previous 60 days with the same confidence level and time horizon.
  - $m_c$ and $m_s$ are multiplicative factors determined by bank supervisors, with a minimum value of three.
  - Effect: At least doubles the previous capital requirement.
- **Incremental Risk Charge (IRC)**:
  - Addresses discrepancy between trading and banking book capital requirements.
  - Calculates a one-year 99.9% VaR, considering both credit rating changes and defaults.
  - Includes a "liquidity horizon" for each instrument.
  - Employs the "constant level of risk assumption."
- **Comprehensive Risk Measure (CRM)**:
  - Focuses on correlation-dependent instruments like ABSs and CDOs.
  - Replaces the Incremental Risk Charge and specific risk charge for these instruments.
  - Includes both standardized and internal model-based approaches.

### 12.2.2 Basel III

- Introduced post-2007–2009 Global Financial Crisis to overhaul Basel II.
- First set of regulations published in December 2010, finalized in 2017.
- Implementation delayed to January 2023.
- **Six Key Components**:
  1. Capital Definition and Requirements
  2. Capital Conservation Buffer
  3. Countercyclical Buffer
  4. Leverage Ratio
  5. Liquidity Risk
  6. Counterparty Credit Risk

#### 12.2.2.1 Capital Definition and Requirements

- Consists of Tier 1 equity capital, Additional Tier 1 capital, and Tier 2 capital. No Tier 3 capital.
- Each tier has different characteristics and is intended to absorb losses to varying degrees, thereby contributing to a bank's overall financial stability.

1. **Tier 1 Equity Capital (Common Equity Tier 1 or CET1)**
   - Includes share capital and retained earnings.
   - Excludes goodwill, deferred tax assets, and defined benefit pension plan deficits.
2. **Additional Tier 1 (AT1) Capital**
   - Items like non-cumulative preferred stock.
   - Previously considered Tier 1 but not common equity.
3. **Tier 2 Capital**
   - Includes subordinated debt with an original maturity of at least five years.

- **Capital Requirements**:
  - Tier 1 equity capital: At least 4.5% of risk-weighted assets.
  - Total Tier 1 capital: At least 6% of risk-weighted assets.
  - Total capital (Tier 1 + Tier 2): At least 8% of risk-weighted assets.

#### 12.2.2.2 Capital Conservation Buffer

- Further CET1 capital equal to 2.5% of risk-weighted assets.
- Dividend restrictions based on capital ratios.
- To ensure banks accumulate additional common equity Tier 1 (CET1) capital during normal economic conditions.
- This buffer serves as a reserve for absorbing losses during financial stress.
- Banks must maintain a CET1 capital conservation buffer equal to 2.5% of risk-weighted assets.
- **Dividend Restrictions**: If the buffer is partially or wholly used, dividend payouts are constrained until the capital is replenished.

#### 12.2.2.3 Countercyclical Buffer

- To counteract the cyclicality of bank earnings and protect the financial system during economic downturns.
- The buffer can range from 0% to 2.5% of a bank's total risk-weighted assets and must be met with Tier 1 equity capital (CET1).
- The implementation level is determined by each country's regulatory authorities, allowing for regional economic conditions.

#### 12.2.2.4 Leverage Ratio

- **Purpose**: To limit banks' discretion in calculating risk-weighted assets and to provide additional information to regulators.
- **Potential Issue**: Could encourage banks to hold riskier assets if this becomes the constraining ratio.
- The leverage ratio is the ratio of Tier 1 capital to the exposure measure.
- Basel III specifies a minimum leverage ratio of 3%.
- **Capital Measure**: Total Tier 1 capital.
- **Exposure Measure**: Sum of (a) on-balance-sheet exposures, (b) derivatives exposures, (c) securities financing transaction exposures, and (d) off-balance-sheet items.
- No risk-weighting adjustments are made.
- Some countries have proposed or implemented higher minimums.

#### 12.2.2.5 Liquidity Risk

- Basel III introduces the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR).
- **LCR**: Focuses on a bank's ability to survive a 30-day liquidity disruption.

$$\text{LCR} = \frac{\text{High-Quality Liquid Assets}}{\text{Net Cash Outflows in a 30-Day Period}}$$

- **NSFR**: Focuses on liquidity management over one year.

$$\text{NSFR} = \frac{\text{Amount of Stable Funding}}{\text{Required Amount of Stable Funding}}$$

- **Liquidity Management**: Designed to ensure that banks can survive liquidity pressures.

#### 12.2.2.6 Counterparty Credit Risk

- Banks calculate a Credit Value Adjustment (CVA) for each derivatives counterparty.
- CVA measures expected loss from counterparty default.
- Changes in CVA can be due to shifts in risk factors or credit spreads.
- Basel III includes CVA risk in market risk capital requirements.

#### 12.2.2.7 G-SIBs, SIFIs, and D-SIBs

- Regulators aim to ensure that large, systemically important institutions hold sufficient capital.
- G-SIBs: Global Systemically Important Banks
- SIFIs: Systemically Important Financial Institutions (includes both banks and non-banks).
- Importance is determined through scoring methodologies and other approaches like network theory.
- G-SIBs required to hold extra Tier 1 equity capital, based on categorization.
- Financial Stability Board publishes annual G-SIB list.
- Total Loss-Absorbing Capacity (TLAC) rules have been set for G-SIBs.
- Non-compliance with TLAC can limit a bank's dividend issuance.
- Some countries impose stricter capital requirements (e.g., Switzerland).
- Domestic Systemically Important Banks (D-SIBs) have their own sets of rules and capital requirements.

### 12.2.3 Contingent Convertible Bonds (CoCos)

- CoCos are bonds that automatically convert into equity under certain conditions, generally when a bank faces financial difficulties.
- Offer higher interest rates than regular debt.
- Benefits for banks: High return on equity in normal times, and an equity cushion during financial stress.
- Benefits for regulators: Reduces the need for public bailouts.
- Specifying the trigger for conversion, often based on Tier 1 equity capital ratios.

### 12.2.4 Dodd–Frank Act

- The Dodd-Frank Wall Street Reform and Consumer Protection Act was signed into U.S. law in July 2010.
- Aimed to prevent future bailouts and protect consumers.
- Its primary objectives are to:

  - Enhance regulation and oversight of financial markets and institutions.
  - Reduce systemic risks in the financial sector.
  - Protect consumers through the creation of the Consumer Financial Protection Bureau (CFPB).
  - Increase transparency in financial products and services.

## 12.3 Fundamental Review of the Trading Book

- FRTB was introduced by the Basel Committee in May 2012, final rules published in January 2019, implementation slated for January 2023, but delays are expected.
- **Previous Approaches**:
  - Basel I used Value at Risk (VaR) with a 99% confidence level.
  - Basel II.5 added "stressed VaR" based on historical periods of financial stress.
- **Key Features of FRTB**:
- Shifts from VaR to Expected Shortfall (ES) with a 97.5% confidence level (and stressed ES).
- Liquidity horizons introduced, varying from 10 to 120 days, based on the risk factor considered.
- Stricter definitions to counter regulatory arbitrage between the trading book and the banking book.
- The Basel Committee outlines two approaches for banks: Standardized Approach and Internal Models Approach.
- Banks approved for Internal Models Approach must also implement the Standardized Approach.
- Standardized Approach calculates regulatory capital using delta, vega, and gamma exposures in the trading book.
- Internal Models Approach uses stressed expected shortfall for regulatory capital calculation.
- Calculations are performed separately for each trading desk.

### 12.3.1 Standardized Approach

- Capital requirement is a sum of risk charge, default risk charge, and residual risk add-on.
- **Risk Charge**:
  - Seven Risk Classes: General interest rate, foreign exchange, commodity, equity, and three credit spread risks.
  - Types of Risk Charges: Delta, Vega, and Curvature.
- **Default Risk Charge**:

  - Calculated by Loss Given Default (LGD) and a default risk weight.

- **Residual Risk Add-On**:
  - For risks not captured by delta/vega/curvature approach, e.g., exotic options.
- **Simplified Approach**:
  - For smaller banks; excludes vega and gamma risks.

### 12.3.2 Trading Book vs. Banking Book

- **Trading Book**: Contains financial instruments intended for trading. Subject to market risk capital.
- **Banking Book**: Consists of instruments intended to be held to maturity. Subject to credit risk capital.
- Regulatory Arbitrage: Past practices allowed banks to choose book placement based on advantageous capital requirements.
- FRTB aims to clarify and standardize criteria for book classification.
- 'Intent to trade' insufficient; must be manageable on a trading desk and pose risks to solvency.
- Instruments are initially assigned to a specific book with limited ability to transfer later.
- Allow for book transfers, but any capital benefits from such moves are disallowed.

## 12.4 Economic Capital and RAROC

- **Economic Capital**: Internal estimate of the capital a financial institution requires for its risk exposure.
- It varies from regulatory capital, which follows standard rules.
- Acts as a "currency" for risk-taking within an institution.

### 12.4.1 Definition of Economic Capital

- **Definition**: Amount of capital needed to absorb losses over one year at a specific confidence level, tailored to maintain credit ratings like AA or BBB.
- Economic capital fills the gap between expected loss and unexpected loss.
- **Case Example**: An AA-rated bank with expected annual losses of 1% and a one-year 99.98% VaR of 5% would require $4 in economic capital per $100 of loans.
- **Approaches to Measure Economic Capital**:
  - **Top-Down**: Uses asset volatility to estimate risk, often using Merton's model.
  - **Bottom-Up**: Aggregates loss distributions by risk types and business units.

- **Risk-Adjusted Return on Capital (RAROC)**:
  - Measures profitability relative to allocated economic capital.
  - Used for assessing past and forecasting future performance, also aids in capital allocation decisions.
- **Types of Risks Considered**:
  - Regulatory Capital covers Credit, Market, and Operational risks.
  - Some institutions also consider Business risk, which is not mandated for regulatory capital.

### 12.4.2 Components of Economic Capital

- **Market Risk Economic Capital**:
  - Time horizon is usually one year; confidence level can go up to 99.98%.
  - Standard deviation for one-year loss or gain derived from annualized daily metrics.
- **Credit Risk Economic Capital**:
  - Banks have freedom to choose their credit correlation models.
  - Popular approaches include CreditMetrics and Credit Risk Plus.
  - Can be conditional (cycle-specific) or unconditional (cycle-neutral) based on economic conditions.
- **Operational Risk Economic Capital**:
  - Regulatory procedure often adopted; some banks use more advanced models.
  - Aims to create a complete distribution of losses.
- **Business Risk Economic Capital**:
  - Involves strategic and reputational risks.
  - Difficult to quantify, largely subjective.
  - Quantification can inform the impact of new strategic initiatives on total risk.
- Characteristics of Loss Distributions for Different Risk Types

| | Standard Deviation | Skewness | Kurtosis |
|---|---|---|---|
| Market risk | High | Zero | Low |
| Credit risk | Moderate | Moderate | Moderate |
| Operational risk | Low | High | High |

### 12.4.3 Aggregating Economic Capital

- To calculate the total economic capital for a financial institution by aggregating various types of risks such as market, credit, and operational risks.

1. **Simple Summation (Basel Committee Method)**
   - Formula: $E_{\text{total}} = \sum_{i=1}^{n} E_i$
   - Assumes perfect correlation between risks.
   - Overestimates capital requirement by approx. 40%.
2. **Normal Distributions**
   - Formula: $\sigma_{\text{total}} = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} \sigma_i \sigma_j \rho_{ij}}$
   - Underestimates capital due to disregard for skewness and kurtosis.
3. **Using Copulas**
   - Maps loss distributions to a standard distribution.
   - Gaussian copula as the simplest case.
4. **Hybrid Approach**
   - Formula: $E_{\text{total}} = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} E_i E_j \rho_{ij}}$
   - Offers a balance between underestimation and overestimation of capital.

### 12.4.4 RAROC

- Risk-adjusted performance measurement (RAPM) is crucial for assessing business units.
- RAROC (Risk-Adjusted Return on Capital) is a widely-used approach.
- Compares expected return with economic capital, adjusted for risk.

$$\text{RAROC} = \frac{\text{Revenues} - \text{Costs} - \text{Expected losses}}{\text{Economic capital}}$$

- Numerator can be pre-tax or post-tax.
- An alternative includes adding a risk-free rate of return to the numerator.
- Ex-ante calculations: For business expansion/contraction decisions.
- Ex-post calculations: For performance evaluation and bonuses.
- Decisions should not solely be based on ex-post results; consider long-term expectations.

## 12.5 Practice Questions and Problems

### 12.5.1 Basel I, Basel II, and Solvency II

1. "When a steel company goes bankrupt, other companies in the same industry benefit because they have one less competitor. But when a bank goes bankrupt other banks do not necessarily benefit." Explain this statement.
2. "The existence of deposit insurance makes it particularly important for there to be regulations on the amount of capital banks hold." Explain this statement.
3. An interest rate swap involves the exchange of a fixed rate of interest for a floating rate of interest with both being applied to the same principal. The principals are not exchanged. What is the nature of the credit risk for a bank when it enters into a five-year interest rate swap with a notional principal of $100 million? Assume the swap is worth zero initially.
4. In a currency swap, interest on a principal in one currency is exchanged for interest on a principal in another currency. The principals in the two currencies are exchanged at the end of the life of the swap. Why is the credit risk on a currency swap greater than that on an interest rate swap?
5. A four-year interest rate swap currently has a negative value to a financial institution. Is the financial institution exposed to credit risk on the transaction? Explain your answer.
6. All the derivatives transactions a bank has with a corporate client have a positive value to the bank. What is the value to the bank of netting provisions in its master agreement with the client?
7. What is the difference between the trading book and the banking book for a bank? A bank currently has a loan of $10 million to a corporate client. At the end of the life of the loan, the client would like to sell debt securities to the bank instead of borrowing. How does this potentially affect the nature of the bank's regulatory capital calculations?
8. Under Basel I, banks do not like lending to highly creditworthy companies and prefer to help them issue debt securities. Why is this? Do you think this changed as a result of Basel II?
9. Banks sometimes use what is referred to as regulatory arbitrage to reduce their capital. What do you think this means?

### 12.5.2 Basel II.5, Basel III, and Oter Post-Crisis Changes

1. What are the three major components of Basel II.5?
2. What are the six major components of Basel III?
3. What is the difference between VaR as it has been traditionally measured and stressed VaR?
4. Explain how the incremental risk charge is calculated. Why was it introduced by the Basel Committee?
5. Explain how CoCo bonds work. Why are they attractive to (a) banks and (b) regulators?

### 12.5.3 Fundamental Review of the Trading Book

1. Outline the differences between the way market risk capital is calculated in (a) Basel I, (b) Basel II.5, and (c) the FRTB.
2. What are the advantages of expected shortfall over value at risk?
3. What is the difference between the trading book and the banking book? Why are regulators concerned about specifying whether an instrument should be one or the other in the FRTB?

### 12.5.4 Economic Capital and RAROC

1. What is the difference between economic capital and regulatory capital?
2. What determines the confidence level used by a AA-rated financial institution in its economic capital calculations?
3. What types of risk are included in business risk?
4. A bank is considering expanding its asset management operations. The main risk is operational risk. It estimates the expected operational risk loss from the new venture in one year to be \$2 million and the 99.97% VaR (arising from a small chance of an large investor lawsuit) to be \$40 million. The expected fees it will receive from investors for the funds under administration are \$12 million per year, and administrative costs are expected to be \$3 million per year. Estimate the before-tax RAROC.
5. RAROC can be used in two different ways. What are they?

# 13 Financial Innovation and Risk Management Mistakes

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 29 - Financial Innovation
  - Chapter 30 - Risk Management Mistakes to Avoid

**Learning outcomes:**

- Understand tech's impact on financial services by 2040.
- Recognize shift to digital payments.
- Learn machine learning's role in credit and fraud.
- Know blockchain's impact on record-keeping.
- Assess changes in the insurance industry.
- Analyze trends in investing and borrowing.
- Evaluate banks' adaptation to tech changes.
- Formulate risk management mistakes to avoid.
- Understand common misconceptions about risk.

## 13.1 Financial Innovation

- Technological changes have led to disintermediation in finance.
- New tech-based intermediaries emerge (reintermediation).
- Financial firms, like banks, face disruption from FinTech.
- Future FinTech will impact digitization, customer behavior, and firm adaptation.
- These changes are of particular relevance to risk management professionals.

### 13.1.1 Technological Advances

- Dramatic increase in computer processing power since the 1960s.
- Introduction of Robotics Process Automation (RPA) for tasks such as customer account creation and employee hiring.

#### 13.1.1.1 Machine Learning

- Differentiates from third industrial revolution through data-driven decision-making.
- Applied in areas like credit decision automation, improving upon traditional human-led methods.
- **Types of Machine Learning**:
  1. Unsupervised learning: Identifies patterns and clusters, e.g., customer segmentation.
  2. Supervised learning: Prediction and classification, e.g., house valuation or loan default prediction.
  3. Reinforcement learning: Sequential decision-making, e.g., hedging strategies in derivatives trading.
- **Challenges**:
  - Overfitting: Model may not generalize to new data.
  - Solution: Use training, validation, and test datasets to estimate model accuracy.

#### 13.1.1.2 Blockchain Summary

- **Challenges with Traditional Ledgers**: Risks include fraud, lack of trust, hacking, and inconsistencies across multiple record-keepers.
- **Blockchain as a Solution**: A distributed ledger technology allowing multiple entries controlled by a community, aiming to negate the aforementioned risks.
- Each block is sealed using a cryptographic hash, ensuring data integrity and tamper-proofing.
- In blockchain systems like Bitcoin, miners are rewarded for validating and adding new blocks, thereby ensuring ongoing, decentralized maintenance.
- Types of Blockchains:
  - Permissionless: Open to all, like Bitcoin.
  - Permissioned: Restricted access; often used by financial institutions for efficient transaction settlements.
- **Potential in Financial Sector**: Blockchain is increasingly being adopted for trading securities, settlements, and back-office functions, aiming for near real-time operations.
- Blockchain poses both threats and opportunities—potentially lowering bank fees while opening new service avenues.
- May also lower operational risks and regulatory capital requirements.
- Adoption and utility in various sectors are still under evaluation; however, it holds significant potential for operational improvements.

### 13.1.2 Payment Systems

- Evolution from cash to digital payments, with rapid adoption in developing countries.
- Key attributes: speed, convenience, security, simplicity, and cost.
- Innovations: Instant fund transfer (PayPal), digital wallets (Apple Pay), and use of cryptocurrencies.
- Security: Focus on biometric authentication to reduce fraud and costs.
- Privacy concerns: Data collection and biometric information.
- Credit facilities: High-interest credit cards versus FinTech solutions like PayPal's free 14-day credit.
- Future services: Easier foreign exchange and data analytics.
- Data-driven credit decisions: Risks around privacy and data selling.
- Future trends: Decline of cash, political resistance to full digitization.
- Cryptocurrencies: Volatile but secure, growing acceptance.
- Central Bank Digital Currencies (CBDCs): Emergence and implications for financial control.

### 13.1.3 Lending

#### 13.1.3.1 P2P Lending

- P2P lending disintermediates traditional banks but adds new intermediaries for risk assessment and transaction facilitation.
- P2P often serves borrowers rejected by traditional banks, potentially charging higher interest rates than conventional loans but lower than credit cards.
- Diversification and secondary markets are crucial for lenders in P2P platforms to manage risks.

#### 13.1.3.2 Crowdfunding

- Crowdfunding democratizes equity investment but faces valuation complexities and regulatory hindrances.
- JOBS Act in the U.S. provides some regulatory leeway, but limitations on funding exist.
- ICOs represent a speculative, risky, and largely unregulated form of crowdfunding with potential for both fraud and success.

### 13.1.4 Wealth Management

- **Traditional Wealth Management**:
  - Highly profitable for banks, with fees ranging from 1% to 1.5% per annum.

  - Involves client risk assessment and investment recommendations based on risk profiles.

- Emergence of **index funds** in 1975 by John Bogle, offering low fees (0.15%) and no need for human managers.
- **Robo-Advisers**:
  - Appeared around 2010, subject to regulatory oversight.
  - Utilize digital platforms for risk assessment and automated portfolio rebalancing.
  - Lower fees, typically around 0.25% per annum.
  - Can address cognitive biases.
  - Popularity may be cyclical, depending on market performance.

### 13.1.5 InsurTech

- **Nature of Insurance Industry**:
  - Resistant to disruption due to need for stability and strict regulations.
  - Difficult for newcomers to enter without significant capital.
  - Brokers and agents facilitate customer-insurer interactions.
  - Trend towards disintermediation; websites for direct customer dealings.
- **Technological Innovations**:
  - Online quote comparison made easy.
  - Data collection for risk assessment.
  - Enables precise risk profiling for premium adjustments.
  - Potentially simplifies claims in case of accidents.
- InsurTech unlikely to disrupt, but adoption of technology is advantageous.

### 13.1.6 Regulation and Compliance

- Financial institutions face a high regulatory barrier, affecting small FinTech startups.
- Regulators aim to balance innovation with client safety.
- "Sandbox" programs provide a testing environment for new financial products.
- Challenges include jurisdiction issues and evaluating FinTech's primarily code-based assets.
- Technology simplifies and streamlines regulatory compliance.
- Real-time monitoring helps avoid fines and ensures international compliance.

### 13.1.7 How Should Financial Institutions Respond?

- FinTech poses strategic risk to established financial entities.
- Institutions should adapt business models to changing consumer behavior.
- Various strategies include in-house development, acquisitions, and partnerships with startups.
- **Flexibility and Adaptability**: Key for surviving in the changing landscape.

## 13.2 Risk Management Mistakes to Avoid

- Clear and unambiguous risk limits essential for both financial and non-financial companies.
- Limits should be established at the board level and trickled down to individual risk managers.
- Daily reports must align with actual gains and losses for accurate risk assessment.
- Special Attention to Derivatives:
  - Close monitoring of derivatives required to differentiate between hedging, speculation, or arbitrage.
- Issue of Exceeding Risk Limits for Profit:
  - Profiting from exceeding risk limits creates a risky culture.
  - Penalties should apply equally for both profitable and unprofitable violations.
- Misconception of Market Predictability:
  - Success in trading often attributed to luck rather than skill.
  - Risk limits should not be increased just because a trader shows consistent profits.
- Importance of Diversification:
  - Risks can be halved by diversifying across 20 stocks, making it a poor choice to limit to one market variable.
- Necessity of Scenario Analyses and Stress Tests:
  - Risk measures like VaR should be complemented by stress tests.
  - Results of stress testing should inform strategic decision-making, often ignored during good times.

### 13.2.1 Managing the Trading Room

- All traders should be scrutinized, even high-performing ones, to ensure they are not taking undue risks or manipulating systems.

- Separation of Front, Middle, and Back Office to prevent internal mishandling of trades.
- Financial models are not foolproof and should not be blindly trusted.
- Model failures have led to significant losses for financial institutions.
- Profits from selling exotic instruments should be recognized conservatively over the life of the deal to mitigate the risk of aggressive trading behavior.
- Financial institutions should avoid selling inappropriate high-risk products to clients, as it undermines long-term trust and can lead to legal repercussions.
- Situations offering seemingly easy profits should be scrutinized for potential hidden risks.

### 13.2.2 Liquidity Risk

- Traders use actively traded instruments to price infrequently traded ones.
- Risk arises when assuming these can be traded close to theoretical prices, especially in market shocks.
- The need for scenario analyses and stress testing.
- Dangers escalate when multiple participants follow the same trading strategy, causing market volatility.
- Excessive use of short-term funding for long-term needs increases liquidity risks.
- Basel III introduced liquidity requirements.
- Lack of transparency in structured products exacerbated the 2007 credit crisis.
- Advocates for more information about underlying assets to minimize market disruptions (flight to quality during crisis).

### 13.2.3 Lessons for Nonfinancial Corporations

#### 13.2.3.1 Understand Your Trades

- Corporations should fully understand the trade or trading strategy before execution.
- If senior management doesn't understand a trade, it shouldn't be approved.
- Valuing the financial instrument is a way to ensure understanding.
- Relying solely on investment bankers for valuation can be risky.

#### 13.2.3.2 Hedgers vs Speculators

- Initially, a trader may focus on hedging but might gradually transition to speculation.
- A loss in speculation can lead to risky attempts to recover, potentially causing disaster.
- Senior management should set clear risk limits and controls.

#### 13.2.3.3 Treasury as Profit Center: Be Cautious

- Making the treasury department a profit center can be tempting but is fraught with risks.
- The treasurer might take additional risks to improve the bottom line.
- The primary goal of a hedging program is to reduce risks, not to increase profits.
- Treating the treasury as a profit center could lead to speculation and financial disasters.

### 13.2.4 A Final Point

- Known Risks: Quantifiable risks like market and credit risks.
- Unknown Risks: Known events with indeterminable probability, e.g., operational risks.
- Unknowable Risks: Risks where even the event is unknown; often termed as "black swans."
- Known risks are called "risk," while unknown risks are termed "uncertainty" (Knight, 1921).
- Understanding and preparing for various kinds of risks can significantly mitigate the potential for catastrophic losses.

#### 13.2.4.1 Managing Unknown Risks

- Subjective probabilities often used.
- Flexibility is key; avoid excessive leverage and aim for variable costs.
- Insurance products may be developed to cover some unknown risks.

#### 13.2.4.2 Managing Unknowable Risks

- Most challenging to handle due to their unpredictability.
- Diversification across products and markets can add a layer of protection.
- Insurance products for unknowable risks are an area of future development.

## 13.3 Practice Questions and Problems

### 13.3.1 Financial Innovation

1. Explain the terms disintermediation and reintermediation.
2. What is meant by (a) machine learning and (b) distributed ledger technologies?
3. Why might a central bank switch from fiat money to digital money?
4. Explain how P2P lending and equity crowdfunding work.

5. “When thinking about the impact of FinTech, financial institutions should be thankful for regulation.” Explain.
6. “Index funds are a great idea, but if everyone chose them there would be no price discovery.” Do you agree with this argument?

### 13.3.2 Risk Management Mistakes to Avoid

1. Discuss and name a few mistakes to avoid.